100% CONSTRUCTION DRAWINGS

Revision 1, Addendum 1 - 3 / 12 / 2026

Revision 2, Addendum 2 - 4 /   1 / 2026

B1146 - USDB TEXTILES WAREHOUSE EXPANSION & RENOVATION

FORT LEAVENWORTH
DIRECTORATE OF PUBLIC WORKS

Code Information
Code: UFC 1-200-01 with Change 1, 3-101-01 with Change 1
UFC 4-440-01
2021 International Building Code (IBC)
2021 International Mechanical Code (IMC)
2021 International Plumbing Code (IPC)
2021 International Fuel Gas Code (IFGC)
2021 International Fire Code (IFC)
2021 International Energy Conservation Code (IECC)
2021 International green Construction Code (IgCC)
2023 NFPA 70, National Electrical Code (NEC)
2023 NFPA 70E, Std. for Electrical Safety inWorkplace
2022 NFPA 72, National Fire Alarm & Signaling Code
NFPA 101 (2018) for Egress
Architectural Barriers Act (ABA)

2021 International Energy Conservation Code (IECC)
Contractor product submittals to highlight rule below and what the proposed product rating is for each product and assembly.

a. Window & Doors U-Factor 0.7 (BUT/h x ft2 x deg. F) Max
b. Opaque Thermal Envelope Insulation – Component Min. Requirements
   a. Metal Building R-19_R-11 LS
   b. Attic & Other R-49
   c. Metal building R-13 + R-13ci
   d. Metal Framed R-13 + R-7ci
   e. Unheated Slabs R-15 for 24” below
c. Opaque Thermal Envelope Assembly, Max. Req't U-Factor
   a. Metal Building: U-0.035
   b. Attic/Other: U-0.021
   c. Metal Framed Walls U-0.054
   d. Floors: U-0.057
   e. Unheated Slabs F-0.52
   f. Opaque Door-Swing U-0.37
   g. Opaque Dr.nonSwg U-0.31
   h. Garage Door U-0.31
d. Follow Effective R-Values for Steel Stud Wall Assemblies

Construction Must comply with the 2021 IECC and other codes noted below.

Construction Type: IIIB
Occupancy Type: S-1 (Warehouse)

Area Allowable: 60,000 SF (*UFC 3-600-01, 4-46.5.1*)

Area Actual:
Existing Building: 9,347 SF
New Addition: 1,236 SF
Total: 10,583 SF

Occupancy Count: 10,573 / 300 SF per person
= 35.2 (36 People) 18 Women, 18 Men

PATH OF EGRESS
Egress (*NFPA 101, Table 42.2.5*)
Dead End Corridor 100 Ft. Max., 5'-0" Ft. Actual
Common Path of Travel:100 Ft. Max., 5'-0" Ft.Actual
Maximum Path to Exit: 400 Ft. Max. 97'-0" Actual

No fire separation required B Office/S-1 Storage
(UFC 3-600-01; 4-46.5.2.2 states Re: IBC Chapt. 5)

Exits: 2 Required; 2 Provided

Plumbing Counts	Required	Provided
Toilets (1/100)	1/gender	2
Lavatories (1/100)	1/gender	2
Drinking Fountains (1/1000)	2	2
Janitor's Sink (1)	1	1

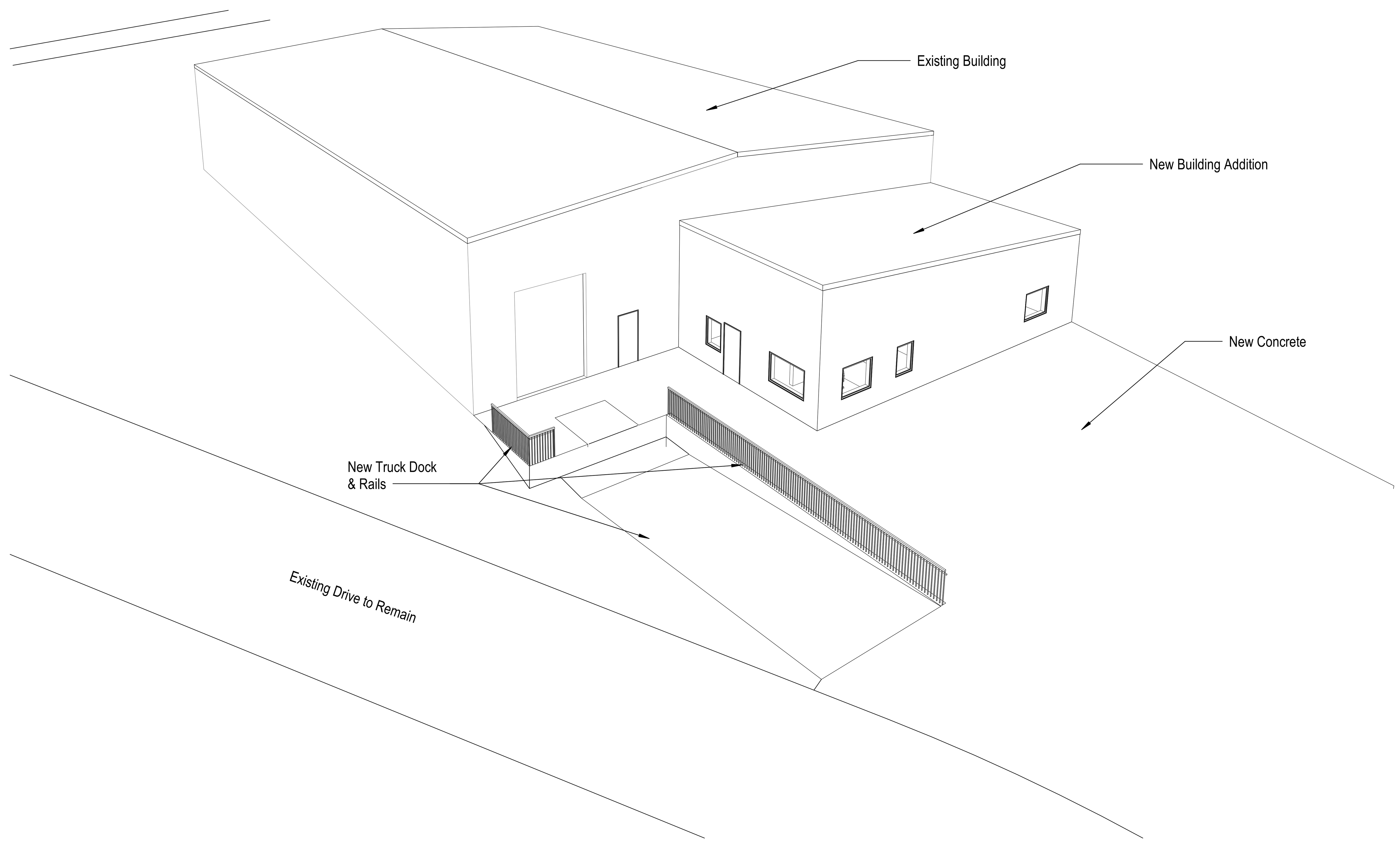
Plans were prepared in accordance with architect's understanding of the Architectural Barriers Act

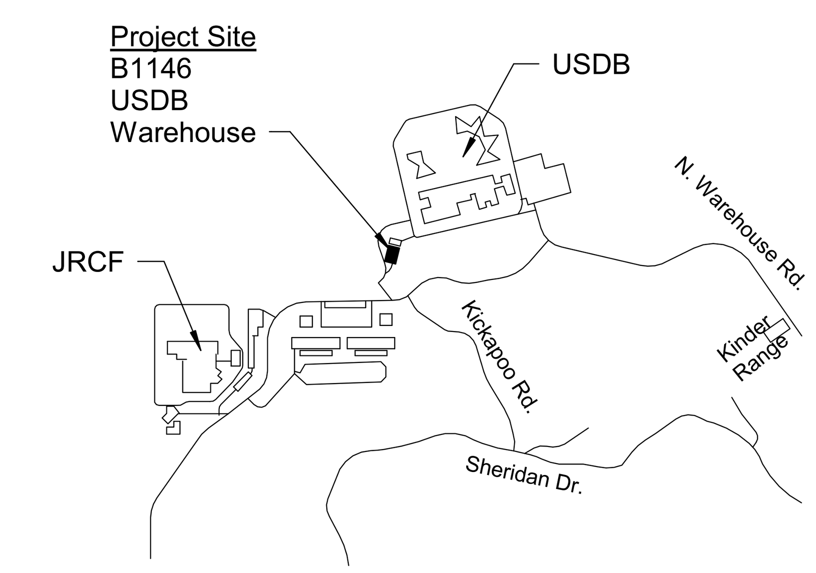
Building is sprinklered. Contractor to modify the existing sprinkler system to meet the UFC-3-600-01 and NFPA 13 codes within the new and existing building.

Contractor to provide in rack sprinkler heads with guards between all new and existing racks such that they do not obstruct rack storage space. Contractor to calculate the required fire pump size and replace the existing pump if required. Contractor shall submit load, pump calculations and shop drawings for the fire sprinkler design.

Contractor to add and/or relocate smoke and fire detectors and alarms, and to reprogram the fire alarm system for relocated and new added devices as required by applicable codes and regulations. New work to conform to UFC 3-600-01 and NFPA 72.

A1 Code Information
1/4" = 1'-0"





Fort Leavenworth, KS

B5 Location Map
1 : 1

Drawing Index

A5 Drawing Index
1 : 1

FORT LEAVENWORTH
DPW

MARK	DESCRIPTION	DATE
2	Revision 2, Addendum 2	4 / 1 / 2026
1	Revision 1, Addendum 1	3 / 13 / 2026

DESIGNED BY: JCM
DRAWN BY: JCM
CHECKED BY: SER
SIZE: ANSI 'D'
ISSUE DATE: 2 / 21 / 2025
PROJECT NO.: 300000331343
100% Drawings

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

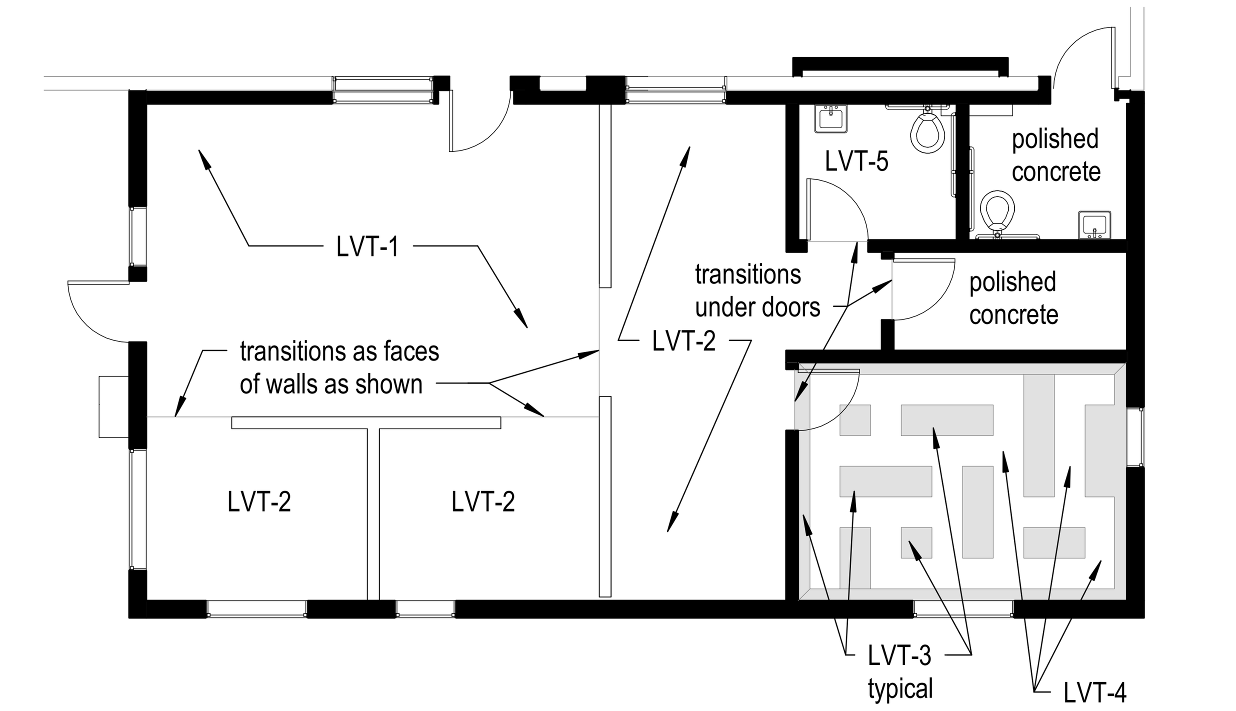
B1146 - Textiles Warehouse - Office Addition

Cover Sheet

SHEET ID
A0

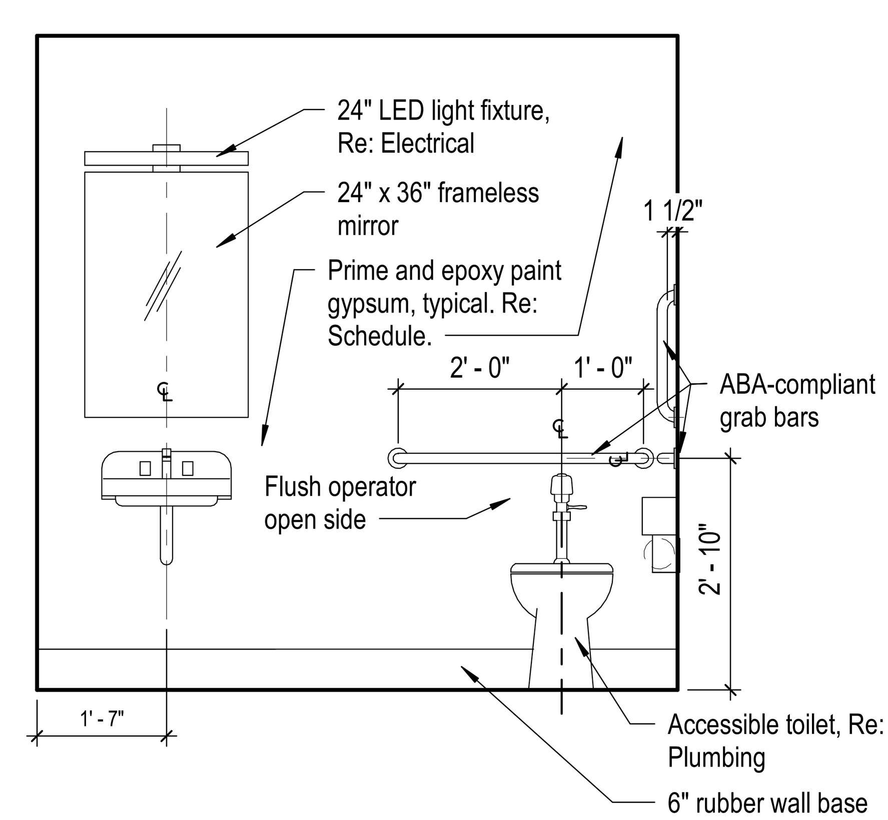
1 2 3 4 5

D C B A

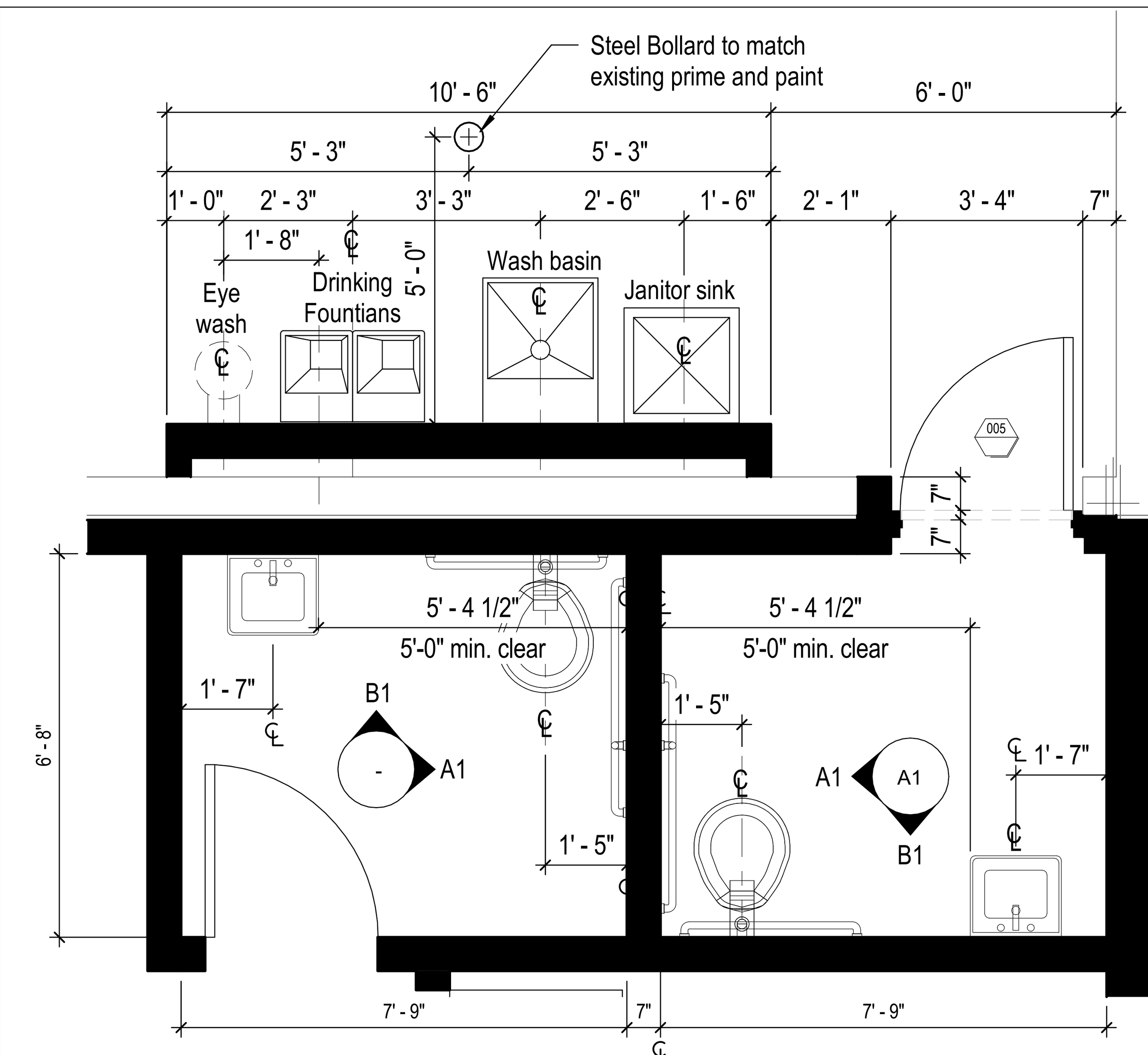


Note: All 18" LVT tile spacing unless noted otherwise. Orient grain of wood along long dimesion of patch of tile, miter at corners. Use north-south orientation for squares.

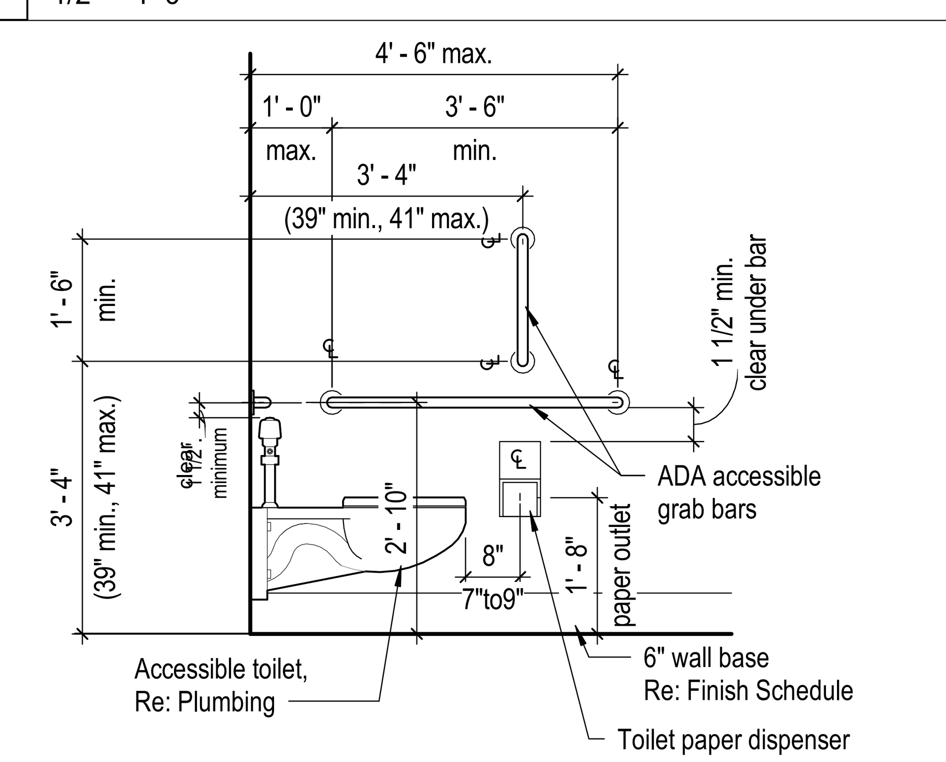
C4 | A2 — **Addition Floor Finish Plan**
1/8" = 1'-0"



B1 — **ABA Toilet - Flush Valve Wall Elev**
1/2" = 1'-0"

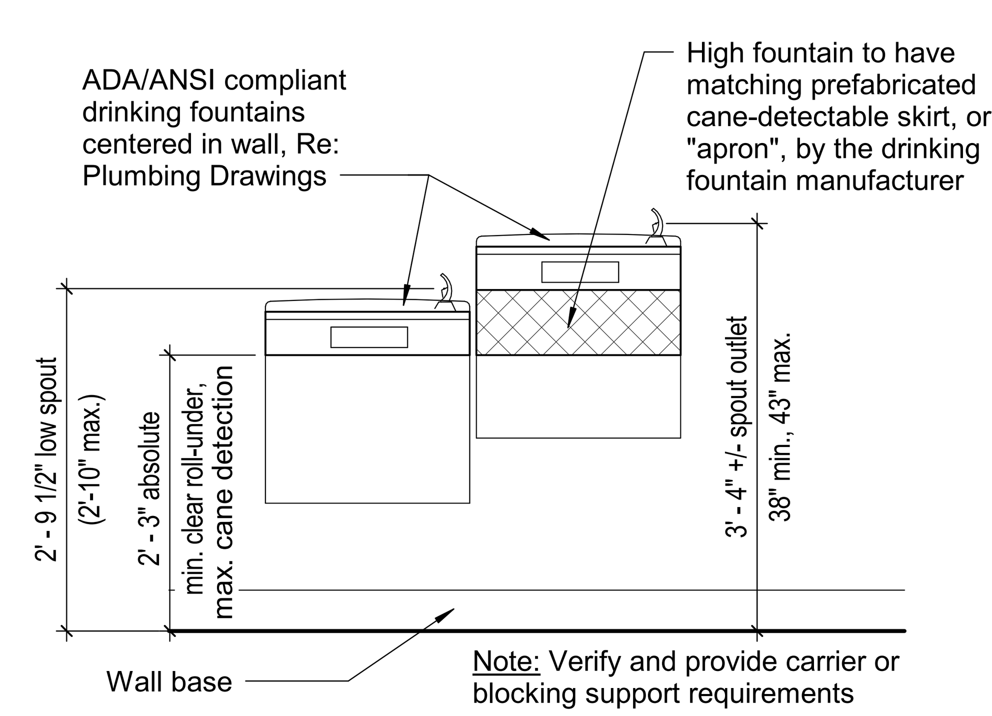


B2 | A3 — **Enlarged Latrine Plan**
3/8" = 1'-0"



Note: Any paint on walls or ceiling to be epoxy.

A1 — **ABA Toilet - Flush Valve Bars Rt**
1/2" = 1'-0"



Note: Verify and provide carrier or blocking support requirements

A2 — **ABA Drinking Fountains Elevation**
3/4" = 1'-0"

NOTE: CONTRACTOR HAS THE OPTION TO CONSTRUCT THE ROOF AND WALLS OF THE ADDITION IN EITHER TRADITIONAL METAL BUILDING METHODS TO MATCH THE EXISTING BUILDING, OR WITH METAL STUDS, BUT WITH EXTERIOR SIDING, ROOF WINDOWS, DOORS AND DETAILING TO MATCH EXISTING.

Demolition Notes

1. Demolish and remove all elements noted and/or shown as dashed. Coordinate the removal of associated MEP elements.
2. Contractor shall remove, haul off site and dispose of all existing construction not to be salvaged.
3. Remove existing flooring and wall finish. Repair and prepare substrate for new finishes.
4. Salvage existing doors, frames and hardware for reuse at new locations. Refinish for new appearance.
5. Contractor shall coordinate all schedules with owner prior to commencement of work, including demolition duration, working times, building access and after-hours work.
6. Provide exit lights, horn strobes and emergency lighting per code requirements.

General Construction Notes

1. All new exterior and walls, doors and frames to match existing.
2. All new interior doors and frames to match existing.
3. Prime new or reworked gypsum walls. Paint all gypsum walls with a minumum of (2) coats of paint. See Finish Schedule.
4. New switching to be located at 46" AFF to top of plate.
5. Raise existing window blinds and protect during the work.
6. Install additional fire protection devices, to include smoke detectors, fire alarms, alarm speakers, ADA strobes, etc. as required by building codes.
7. Add and relocate sprinkler heads as required. Contractor to provide design-build drawings for DPW review prior to work.
8. Notify Architect of any dimension changes per plan.

Wall Symbol Legend

- Existing Wall
- Wall to be Removed
- New Wall

Wall Type Legend

- 1 — Wall Type 1: Re: B4 / A5
- 1A — Wall Type 1A: Re: B4 / A5
- 1B — Wall Type 1B: Re: B4 / A5
- 2 — Wall Type 2: Exterior Walls of new addition **may be either** "stick-built" with metal studs and roof joists or with metal building construction similar to the existing building. Exterior building skin to match existing in all aspects. Interior of walls to have vapor barrier, insulation and 5/8" gypsum, painted.
- 3 — Wall Type 3: Same as "1", but with 3 5/8" studs, and walls up to 42" above floor with a 3/4" thick stained Birch hardwood cap. Wood cap to be same width as gypsum wall with 1/4" x 1/4" reveal below cap to gypsum board and all exposed edges "eased".

Annotations and Symbols

- Detail Number / SIM / 1 / A101 / Sheet Number — Interior Elevation
- Detail Number / SIM / 1 / A101 / Sheet Number — Detail Reference
- Detail Number / SIM / 1 / A101 / Sheet Number — Section Reference
- Detail Number / 1 / A102 / 1/8" = 1'-0" / Origin Sheet Number / Scale — View Name
- 0 2' 4' 8' — Graphic Scale
- Name Elevation — Elevation Mark
- 101 — Door Number
- ROOM NAME 101 — Room Name Room Number
- 1 — Revision Reference
- 9' - 0" — Ceiling Height
- A — Wall Tag
- NORTH — North Arrow

Light Fixture Symbols

- E — Existing Fluorescent Light Fixture
- N — New Fluorescent Light Fixture
- R — Relocated Fluorescent Light Fixture
- E — Existing Recessed Can Light Fixture
- N — New Recessed Can Light Fixture
- R — Relocated Recessed Can Light Fixture
- Linear Strip Fluorescent Light Fixture
- Wall Sconce

A3 — **Notes and Symbols**
Not to Scale

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE

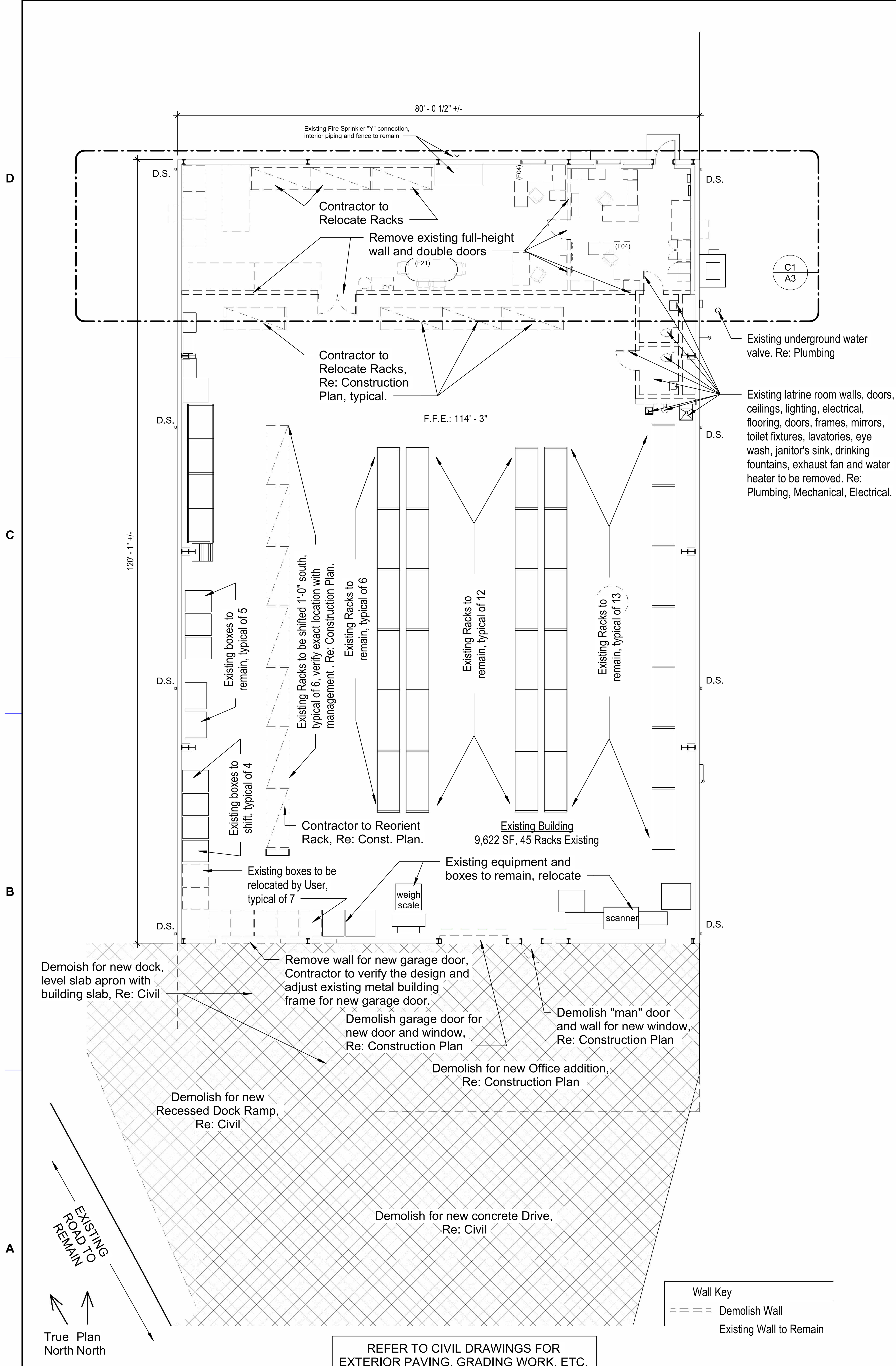
ISSUE DATE: 2 / 21 / 2025
PROJECT NO.: 30000331343
100% Drawings

DESIGNED BY: JCM
DRAWN BY: JCM
CHECKED BY: SER
SIZE: ANSI 'D'

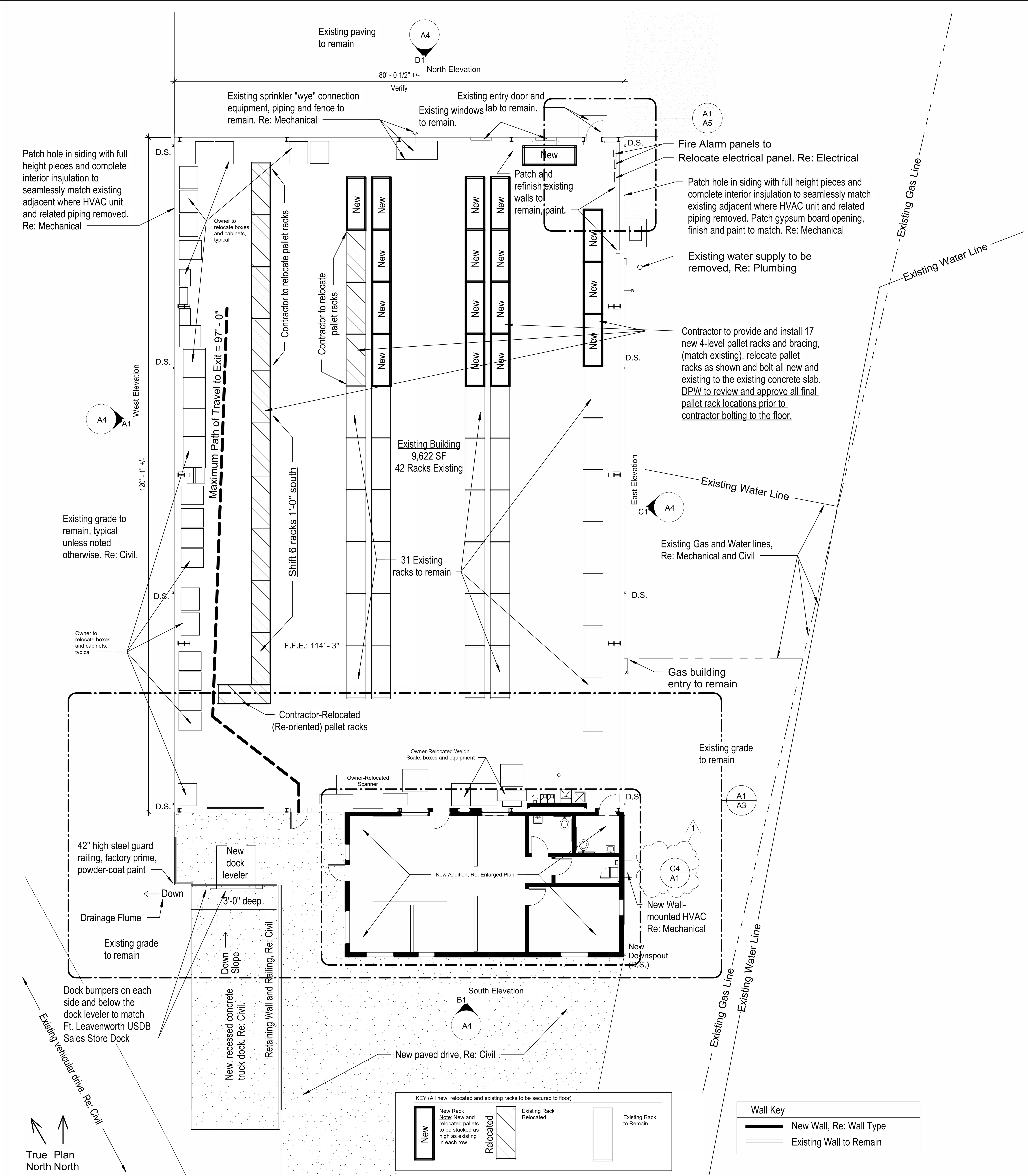
DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH
820 McCLELLAN AVE.
FORT LEAVENWORTH, KS 66027-1292

B1146 - Textiles Warehouse - Office Addition
General Notes, Code Information, Wall Types

SHEET ID
A1



A1	Floor Plan - Demolition
A4	3/32" = 1'-0"



A3	Floor Plan - Construction
A4	3/32" = 1'-0"

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE
1	Revision 1, Addendum 1	3 / 13 / 2026

DESIGNED BY: JCM
DRAWN BY: JCM
CHECKED BY: SER
ISSUE DATE: 2 / 21 / 2025
PROJECT NO.: 30000031343
SIZE: ANSI 'D'
100% Drawings

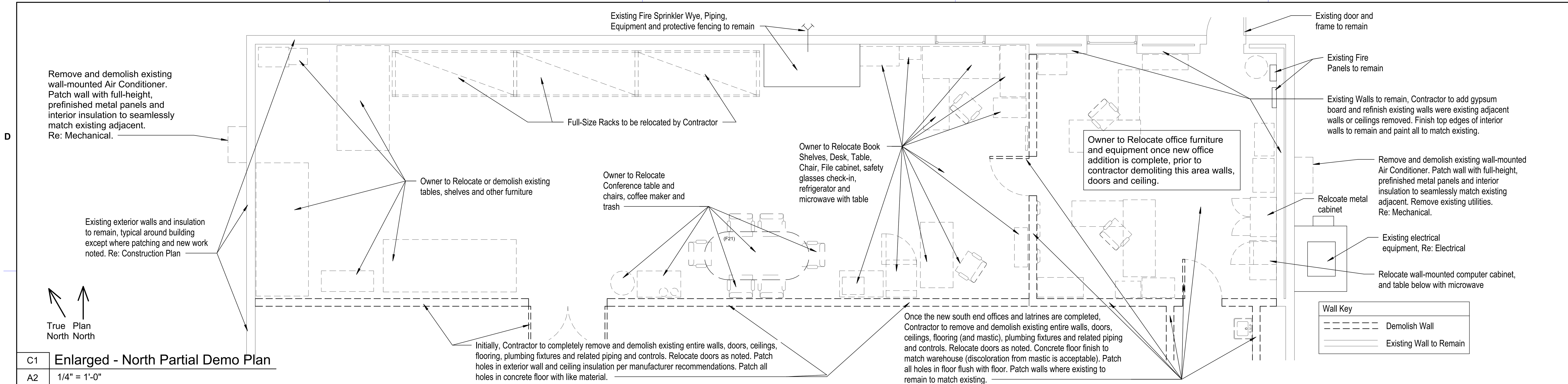
DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH
820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

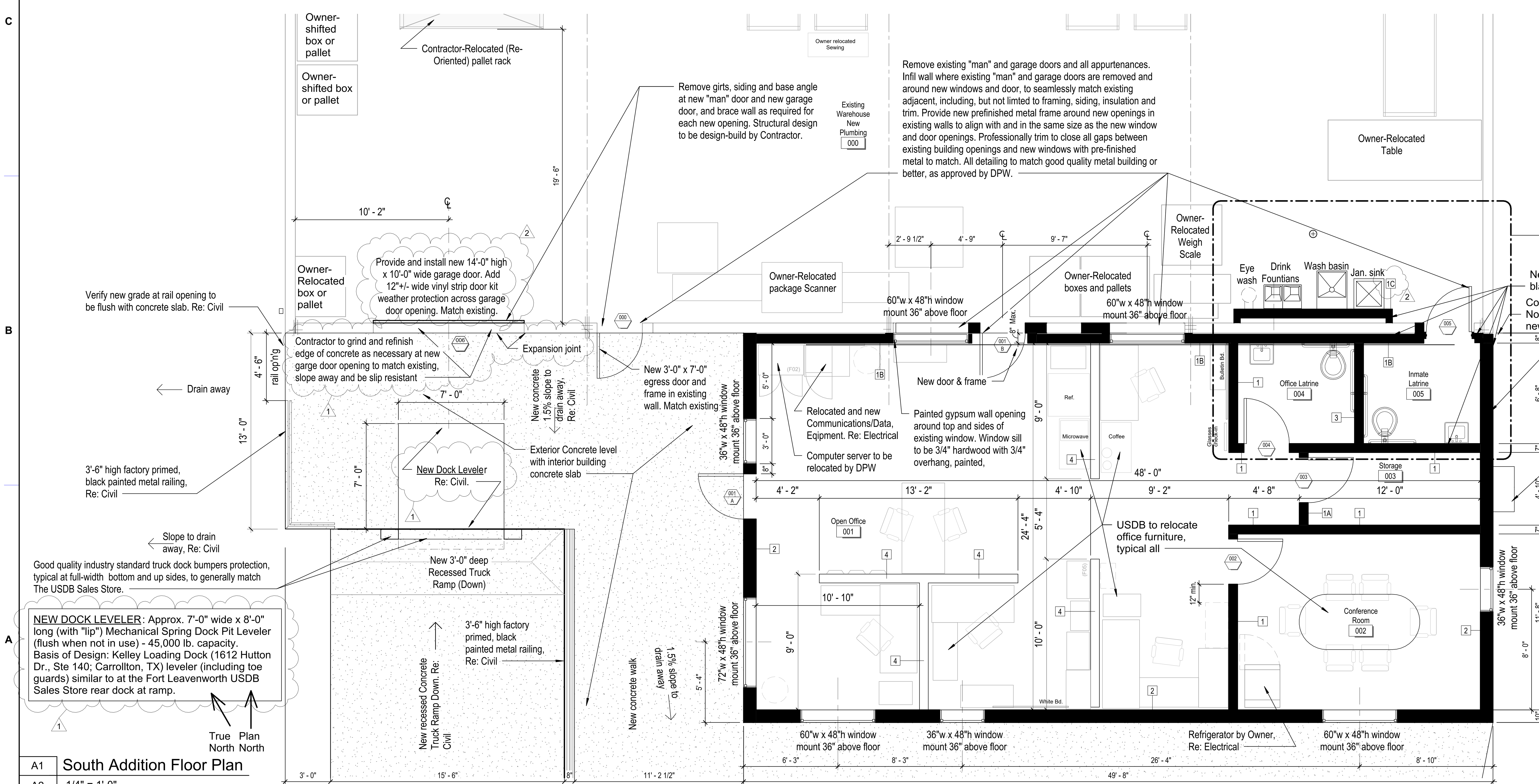
Overall Floor Plans

SHEET ID

A2



C1 Enlarged - North Partial Demo Plan
A2 1/4" = 1'-0"



A1 South Addition Floor Plan
A2 1/4" = 1'-0"

Wall Key
New Wall, Re: Wall Type
Existing Wall to Remain

PALLET RACKS:
All new and existing pallet racks to be drilled and bolted to the existing concrete floor by Contractor. Final locations as approved by DPW and User.

WINDOW NOTES:
1. All windows, including 2 interior and 6 exterior to be double pane insulated windows, Re Specification. Interior to be sound resistant.
2. Window frames to be "thermally broken."
3. Glazing to have both solar glazed, tinted, low-E glass with both panes laminated.
4. Windows to meet American Architectural Manuf. Assoc. (AAMA) type HC60 window specifications (AAMA WSG-11)
5. All windows to be as good or better quality than existing.
6. All exterior windows to be fixed, non-operational.
7. Both interior windows to be side-to-side slide windows.
8. New interior windows to have single frames which continuously close off the void between inside face of addition to opposite face of existing wall.
9. All windows to have interior 1" "plantation style", manually-operated blinds on addition side of wall.

FORT LEAVENWORTH DPW

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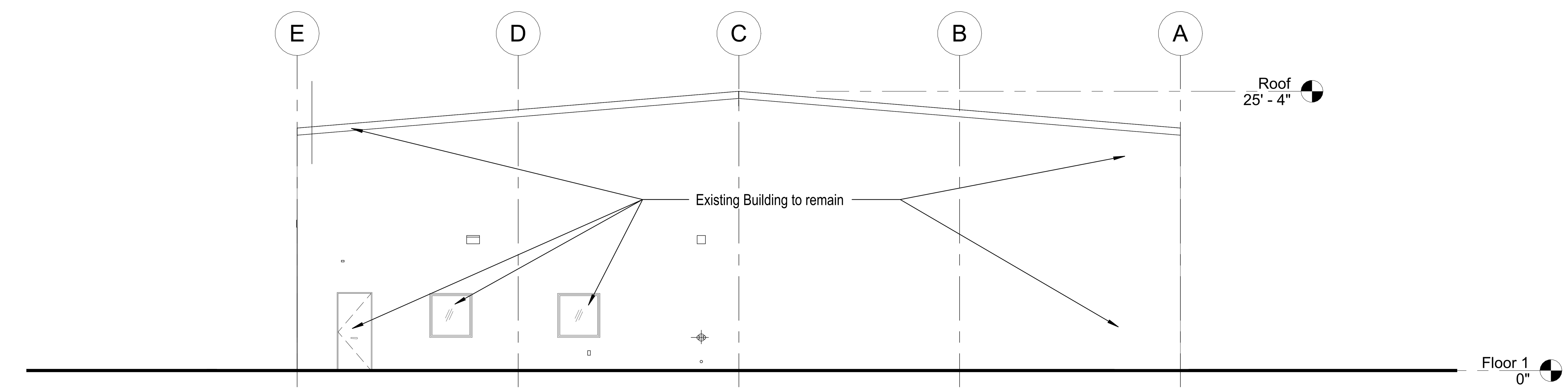
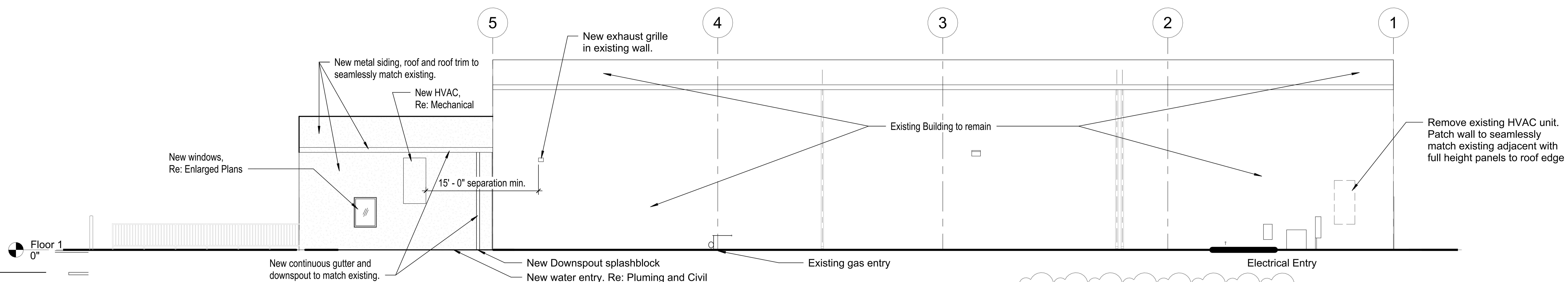
B1146 - Textiles Warehouse - Office Addition

Enlarged Plans

SHEET ID
A3

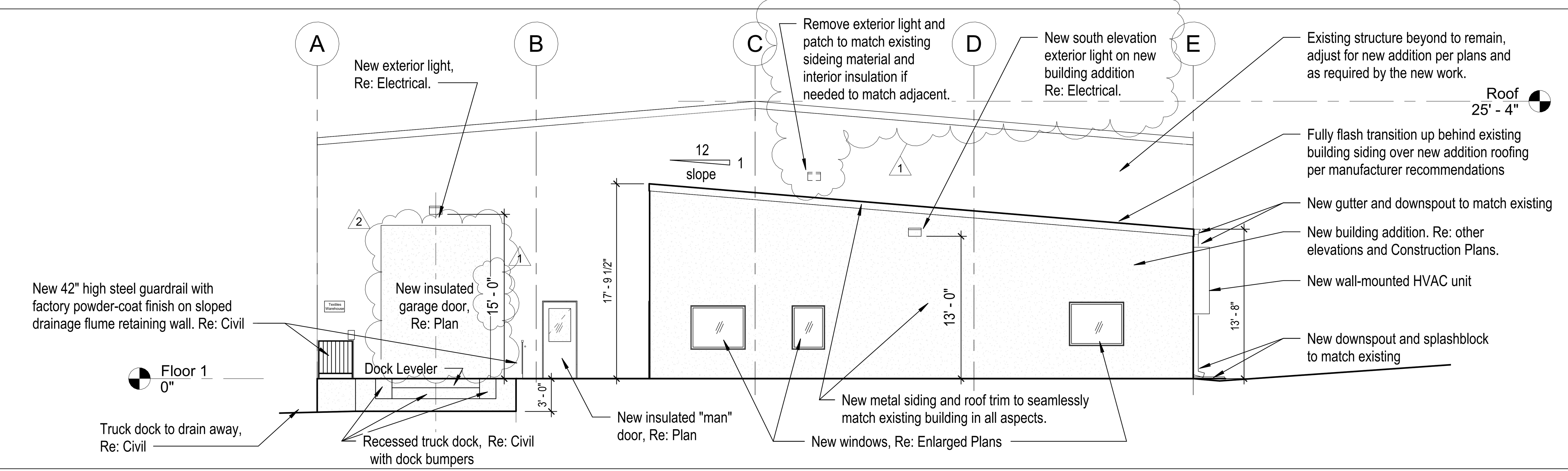


D1 North Elevation
A2 1/8" = 1'-0"

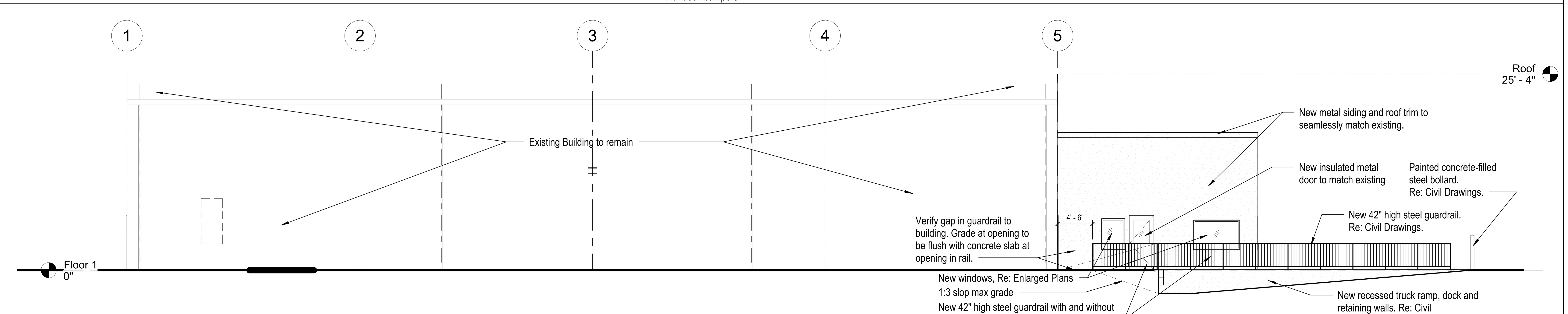


C1 East Elevation
A2 1/8" = 1'-0"

New Windows Notes:
1. All new door and window glazing to have be Category II safety glazing.
2. All new interior and exterior window glazing to double-pane insulated.
3. All new windows to be fixed, not operable, match existing details.
4. All new exterior windows to have exterior metal bars of type, spacing and building connections matching existing.



B1 South Elevation
A2 1/8" = 1'-0"



A1 West Elevation
A2 1/8" = 1'-0"

FORT LEAVENWORTH DPW

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1	Revision 1, Addendum 1	3 / 13 / 2026

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DRAWN BY: JCM
CHECKED BY: SER
ISSUE DATE: 2 / 21 / 2025
PROJECT NO.: 300000331343
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100% Drawings

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ENGINEERING DIVISION
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820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

Building Elevations

SHEET ID
A4

Door and Frame Schedule

Mark	Door Letter	Room Name	Door Type	Door Width	Door Height	Door Thickness	Frame Type	Frame Jamb	Frame Head	Rating	Hardware	Comments
000		Existing Building	A	3' - 0"	7' - 0"	2"	1				1	1, 2
001	A	Open Office	A	3' - 0"	7' - 0"	2"	1				2	1, 2
001	B	Open Office	B	3' - 0"	7' - 0"	2"	1				3	1, 2
002		Conference Room	C	3' - 0"	7' - 0"	2"	1				4	
003		Storage	C	3' - 0"	7' - 0"	2"	1				5	
004		Office Latrine	C	3' - 0"	7' - 0"	2"	1				6	3
005		Inmate Latrine	C	3' - 0"	7' - 0"	2"	1				7	2
006		Warehouse		8'-0"	14'-0"							4

1

Door Comments

1. Provide door thresholds below door, over concrete joint.
2. Provide 1/2" wide expansion joint flush with concrete slab.
3. Sound blocking sweep at bottom of door. Do not undercut excessively.

Re: Mechanical Plans for make-up air to exhaust fan thru transfer duct.

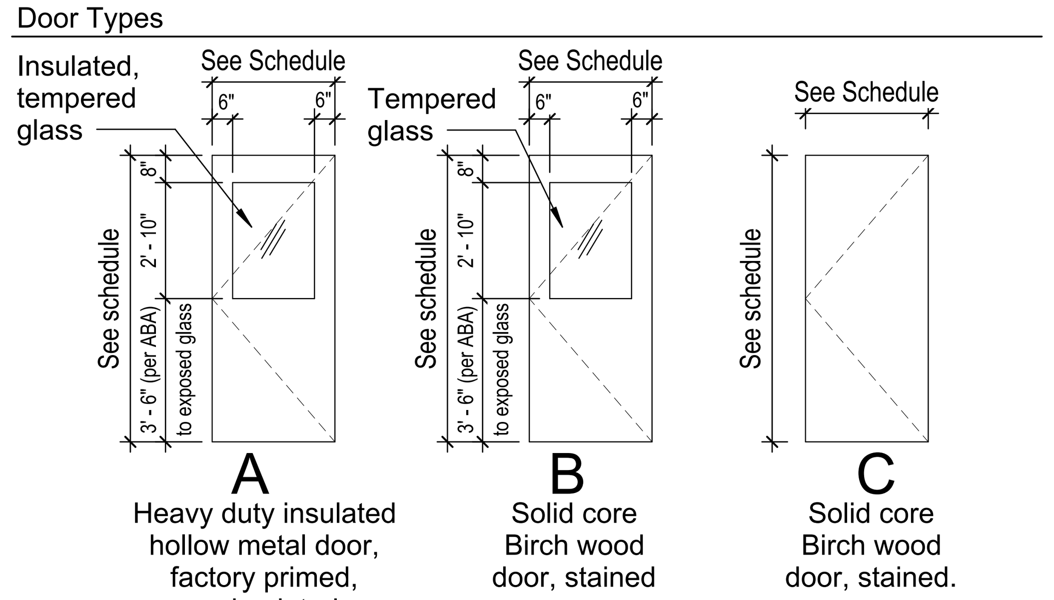
4. Coiling, motorized, insulated high performance commercial overhead door. Re: Electrical. Basis of Design: Overhead Door Company.

1

General Notes:

A. Doors to be keyed per Building standard.
B. All new hardware to match building standard.
C. Existing northeast building exterior door and frame to remain. Provide new lock hardware.
D. Verify all final hardware with DPW prior to order.
E. Final stain color to be selected by DPW from manufacturer's full range of colors
F. Provide door thresholds at Doors 000, 001A, 001B and 005, weather proof at 000, 001A.

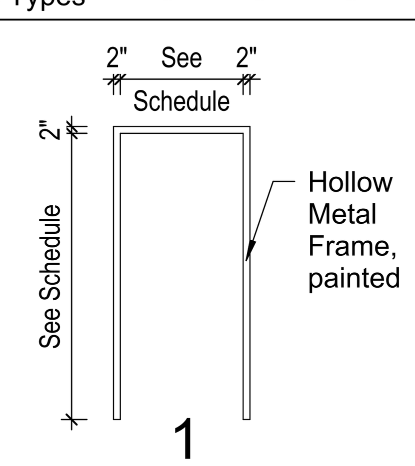
Door Types



Note: All stained wood doors to match and be factory finished.

2

Frame Types



B1 **Door and Frame Types**
1/4" = 1'-0"

Hardware Groups

Group 1 (Single Steel door, lock - 180 deg. swing)
- 3 Ea. Hinge
- 1 Ea. Interior Closer with limiting arm ("stop") and hold-open function
- 1 Ea. Lock Set
- 1 Set Ea. Silencers
- 1 Ea. 35" x 16" heavy duty, stainless-steel kickplate
- 1 Ea. Panic Hardware

Group 2 (Single Steel door, lock - 180 deg. swing)
- 3 Ea. Hinge
- 1 Ea. Interior Closer with limiting arm ("stop") and hold-open function
- 1 Ea. Push-Button "Cipher" Lock Set
- 1 Set Ea. Silencers
- 1 Ea. 35" x 16" heavy duty stainless-steel kickplate
- (*No Panic Hardware*)

Group 3 (WAREHOUSE - Single wood half-light door, lock)
- 3 Ea. Hinge
- 1 Ea. Closer with limiting arm ("stop") and hold-open function
- 1 Ea. Lock Set
- 1 Set Ea. Silencers
- 1 Ea. 35" x 16" stainless steel kickplate

Group 4 (CONF. RM. - Single wood door, no lock)
- 3 Ea. Hinge
- 1 Ea. Passage "Latch" Set
- 1 Set Ea. Silencers
- 1 Ea. Wall Stop
- 1 Ea. 35" x 16" stainless steel kickplate

Group 5 (STORAGE - Single door, storeroom lock)
- 3 Ea. Hinge
- 1 Ea. Storeroom Lock Set
- 1 Set Ea. Silencers
- 1 Ea. Wall Stop

Group 6 (OFFICE TOILET - Single door, privacy lock)
- 3 Ea. Hinge
- 1 Ea. Privacy Lock Set
- 1 Set Ea. Silencers
- 1 Set Smoke/Sound Gasketing
- 1 Ea. Wall Stop
- 1 Ea. 35" x 16" stainless steel kickplate
- 1 Ea. Sound blocking door sweep

Group 7 (INMATE TOILET - Single door, privacy lock)
- 3 Ea. Hinge
- 1 Ea. Privacy Lock Set
- 1 Set Ea. Silencers
- 1 Ea. Floor Stop
- 1 Ea. 35" x 16" stainless steel kickplate

Room Finish Schedule

Room Number	Room Name	Floor Finish	Base Finish	Walls North	Walls East	Walls South	Walls West	Ceiling Finish	Ceiling Height	Comments
000	Existing Warehouse New Plumbing	Existing Concrete	RB - 1	NA	P-1	P-1	P-1	N/A	Varies	1
001	Open Office	VCT-1, VCT-2	RB - 2	P-1, P-8	P-2, P-8	P-1, P-8	P-1, P-8	Gyp. P-8	Varies	2,4
002	Conference Room	VCT-3, VCT-4	RB - 2	P-4	P-3	P-3	P-3	CT-1	9' - 0"	
003	Storage	Polished Concrete	RB - 2	P-1	P-1	P-1	P-1	Gyp.P-7	Varies	2
004	Office Latrine	VCT-5	RB - 1	P-5	P-5	P-5	P-5	CT-1	9' - 0"	3
005	Inmate Latrine	Polished Concrete	RB - 1	P-6	P-2	P-6	P-6	Gyp. P-7	9' - 0"	3

Comments

1. Existing Warehouse Ceiling.
2. Sloped gypsum ceiling on framed roof.
3. Paint on walls and ceilings in Latrines to be Epoxy Paint.
4. Paint wall above 9'-4" AFF and gypsum ceiling matching darker color. Ductwork, electrical, and all allowed to be painted, paint to match ceiling.Verify all with DPW.

2

5. Paint full wall behind drinking fountains, janitor's sink, eye wash, etc. to be Epoxy Paint.

General Notes

1. All Epoxy Paint Walls and Ceiling in both latrines.

Finishes (all finishes are "Basis of Design"

Luxury Vinyl Floor Tile

LVT-1	Armstrong Flooring - Elmshorn J7208	(6" x 48")
LVT-2	Armstrong Flooring - Halden J7206	(6" x 36")
LVT-3	Armstrong Flooring - Palazzo Nero - ST550	(18" x 36")
LVT-4	Armstrong Flooring - Palazzo Scuro - ST551	(18" x 36")
LVT-5	Armstrong Flooring - Palazzo Grigio - ST556	(18" x 36")

Rubber Wall Base (Basis of Design - Tarkett)

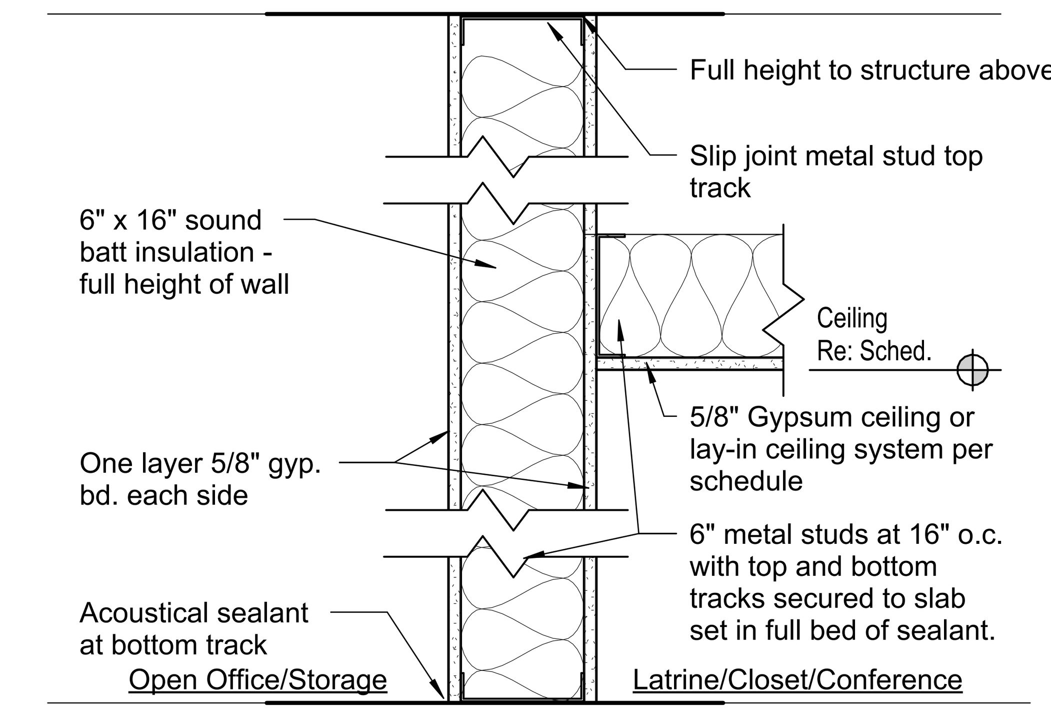
RB - 1	6" high cove base - 47 "Brown"
RB - 2	4" high cove base - 47 "Brown"

Paint (Basis of Design - Sherwin Williams)

PT-1	SW 9180 "Aged White"	Office Wall - Primary (below 9'-0")
PT-2	SW 7548 "Portico"	Office Wall - Accent (below 9'-0")
PT-3	SW 7548 "Portico"	Office Wall - Accent (below 9'-0")
PT-4	SW 7734 "Olive Grove"	Conf. Wall - Accent
PT-5	SW 6164 "Svelta Sage"	Office Latrine Walls
PT-6	SW 7011 "Natural Choice"	Inmate Latrine Walls
PT-7	SW 7103 "White Tail"	Gypsum Ceilings (Storage, Inmate Latrine)
PT-8	SW 7674 "Peppercorn"	Open Office - Upper Wall/Ceiling (above 9'-0")
PT-9	SW 7048 "Urbane Bronze"	Metal Doors and Frames

Ceiling Panels (Tile) and Grid

CT-1	Armstrong - Dune 2' x 2' panels with recessed tegular edge on 15/16" white aluminum grid.



Description:
6" metal stud (20 ga.) at 16" on center (o.c.) maximum to underside of structural deck. Four foot wide, 5/8" gypsum board on either side of stud. All exposed gypsum to be painted. Re: Room Finish Schedule. Acoustical batt insulation between studs, full height of wall. Use acoustical sealant to seal around openings and penetrations. Slip joint shall be installed at bottom of joist or to bottom of deck.

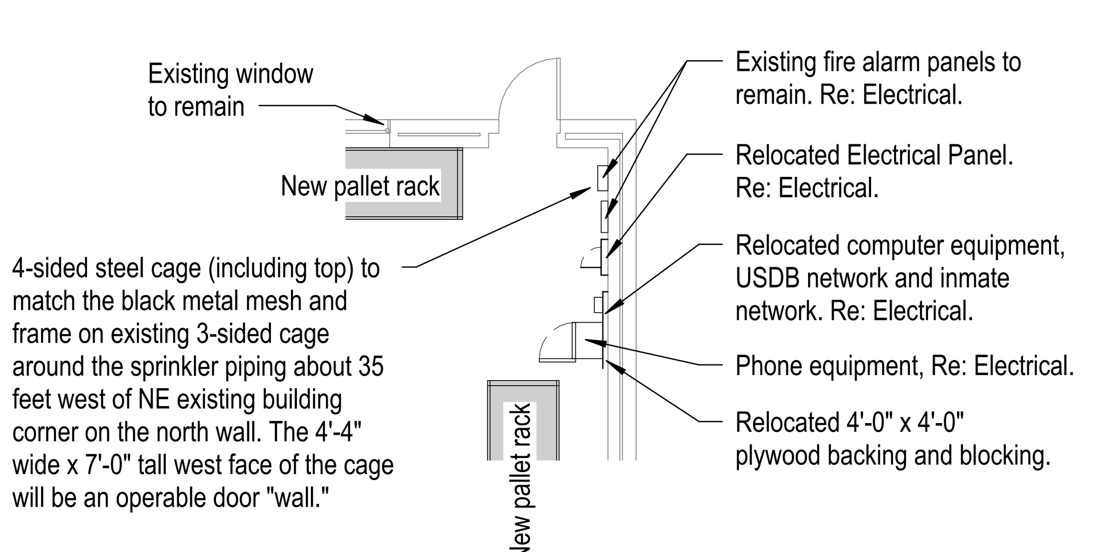
- 1 As stated above (except ceilings on both sides of wall between latrines)
- 1A As stated above, Except no low ceiling on either side of wall
- 1B As stated above, Except gypsum board only one side of wall
- 1C As stated above, Except gypsum board only one side of wall and a 45 deg, beveled framed gypsum cap at 9'-0" above floor.

2

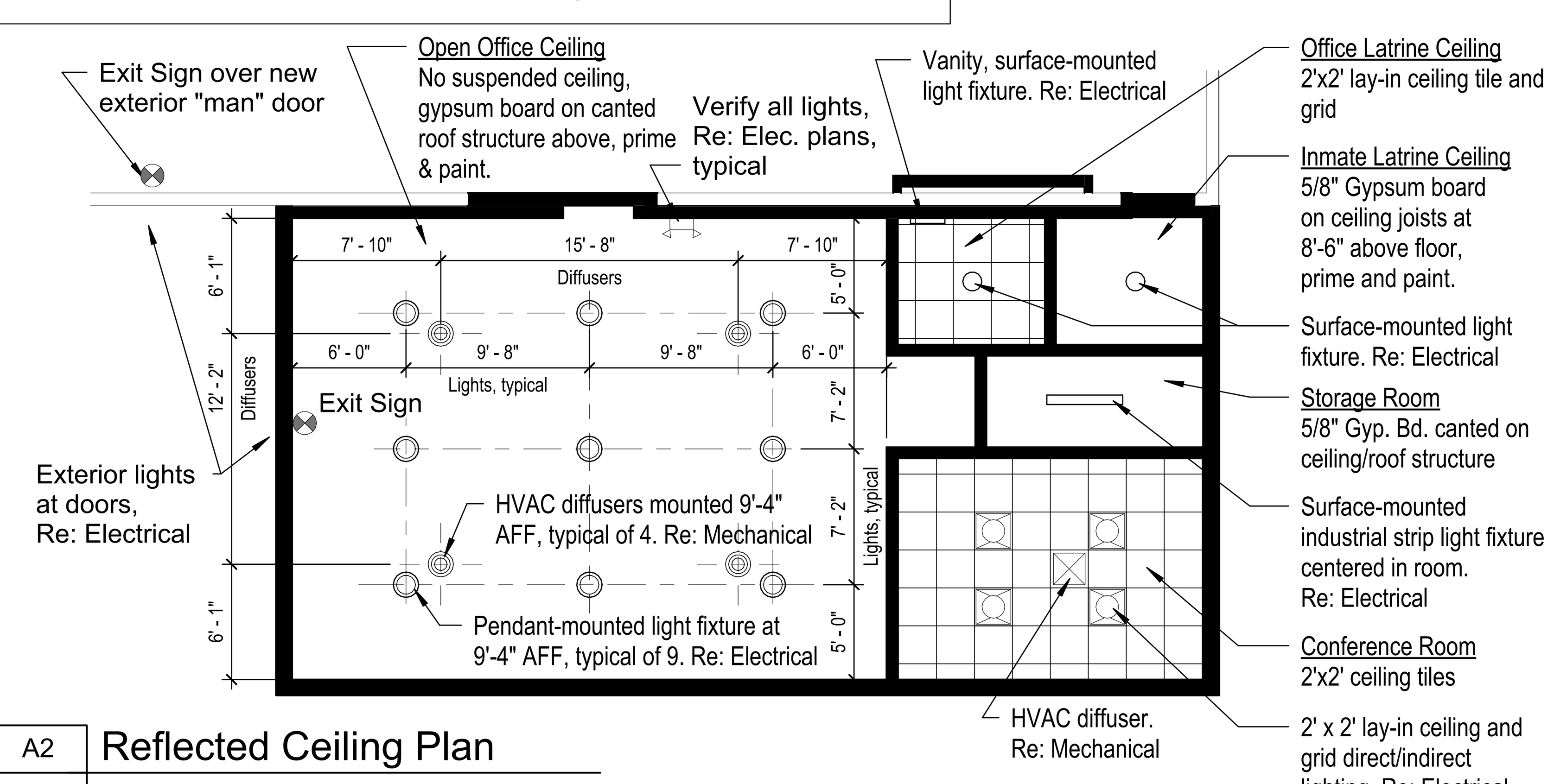
C4 **Wall Type 1**
1 1/2" = 1'-0"

Pre-Engineered Metal Building Construction exterior wall with gypsum board full height on interior. Re: Specifications for insulation and gypsum board. Re: Mechanical, Electrical, Plumbing and Fire Protection. Paint. Re: Finish Schedule.

B4 **Wall Type 2**
12" = 1'-0"



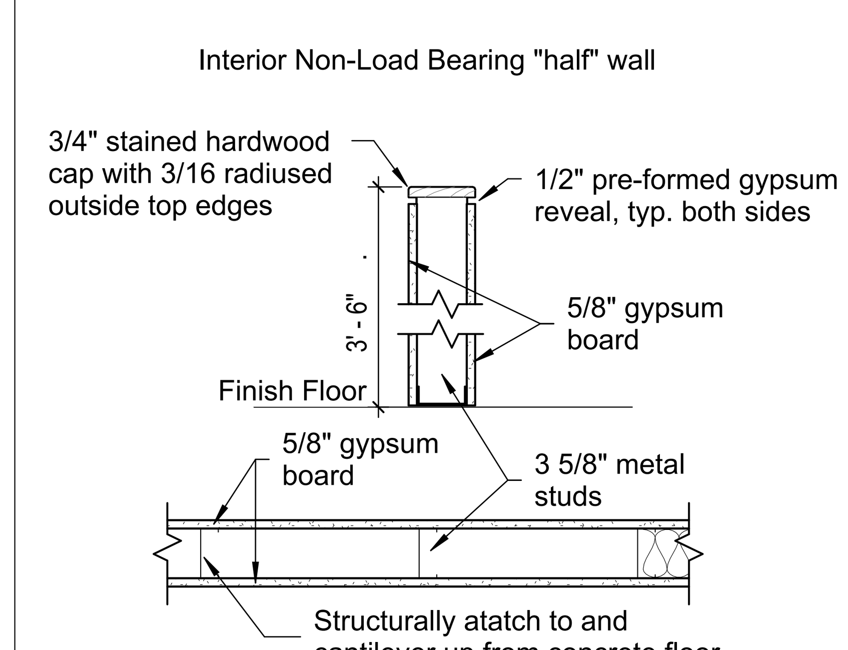
A1 **Enlarged Floor Plan** — Northeast Corner of Existing Building
A2 1/8" = 1'-0"



A2 **Reflected Ceiling Plan**
A4 1/8" = 1'-0"

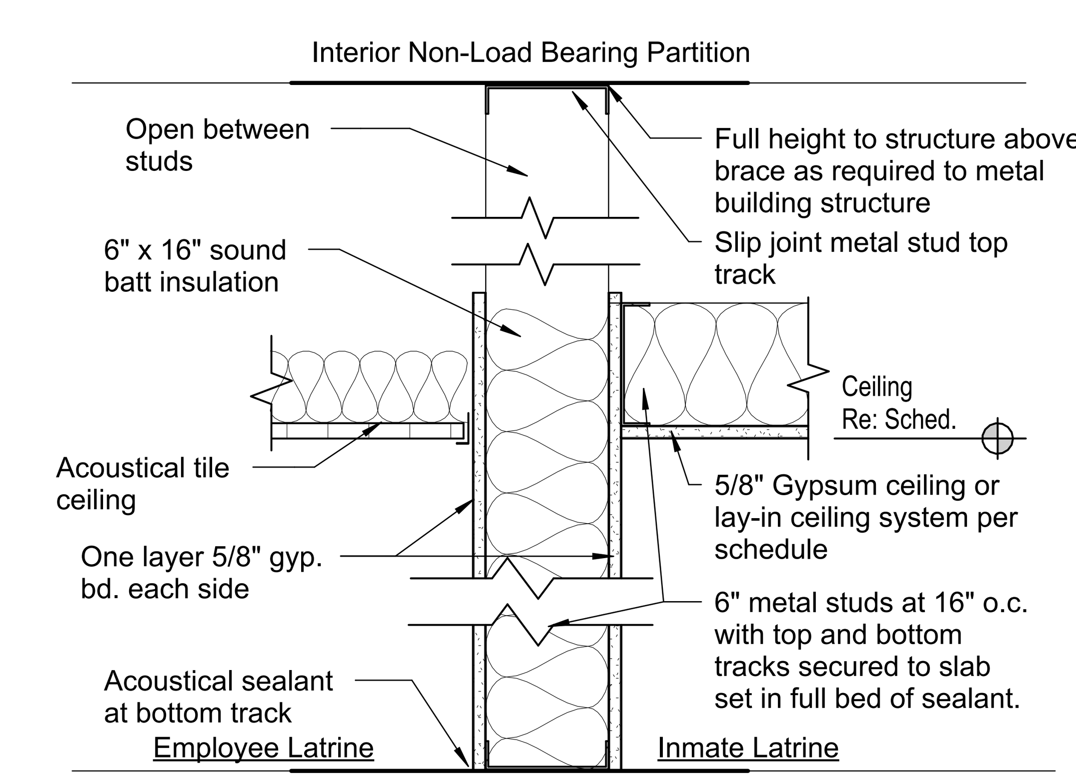
General Notes:

1. Re: Electrical for light schedule.
2. Re: Electrical for controls and electrical notes.
3. Re: Mechanical for HVAC
4. Coordinate all ceiling elements with Fire Protection (sensors, alarms, sprinkler heads, wiring, programming, emergency lighting, etc.) to be design-build by contractor, unless noted herein.



Description
3 5/8" metal stud, spaced max. 16" o.c. Four foot wide, 5/8" gypsum board on either side of stud.

A3 **Wall Type 4**
1" = 1'-0"



Description:
6" metal stud (20 ga.) at 16" on center (o.c.) maximum to underside of structural deck. Four foot wide, 5/8" gypsum board on either side of stud. All exposed gypsum to be painted. Re: Room Finish Schedule. Acoustical batt insulation between studs, full height of wall. Use acoustical sealant to seal around openings and penetrations. Slip joint shall be installed at bottom of joist or to bottom of deck.

- 3 As stated above

A4 **Wall Type 3**
1 1/2" = 1'-0"

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE
2	Revision 2, Addendum 2	4 / 1 / 2026
1	Revision 1, Addendum 1	3 / 13 / 2026

DESIGNED BY: JCM
DRAWN BY: JCM
CHECKED BY: SER
SIZE: ANSI 'D'
ISSUE DATE: 2 / 21 / 2025
PROJECT NO.: 300000331343
100% Drawings

DIRECTORATE OF PUBLIC WORKS ENGINEERING DIVISION DESIGN BRANCH
820 McCLELLAN AVE. FORT LEAVENWORTH, KS 66027-1292

B1146 - Textiles Warehouse - Office Addition

Schedules

SHEET ID
A5

The drawings include the General Conditions of the Contract for Construction AIA Document A201 1997 Edition and all sections of the specifications shown here.

DIVISION 1 – GENERAL REQUIREMENT

01010 – SUMMARY OF WORK

1. The drawings are generally design intent in nature and not detailed out. The drawings and specifications indicate the scope of the project in terms of architectural and engineering design concept, but do not necessarily indicate or describe all work required for the full performance of the work. Contractor's bid shall be based on these drawings with the understanding that the contractor and each subcontractor are to furnish and install all items required for the proper completion and full operation of the work. The Contractor shall make provisions for the necessary execution of the work. These include, but are not limited to, such items as demolition, new construction, hoisting, storage, freight access, debris removal, and items required for a complete project.
2. The intent is for a pre-engineered metal building (PEMB) addition to an existing PEMP. DPW will make available the shop drawings from the original (existing) PEMB. The Contractor must field verify all prior to any work.
3. The work will include general construction, plumbing, mechanical and electrical, and fire protection work. The work covers the construction and finishing of a new building addition to match existing and removing existing north end east-west full height wall, the NE building office-related items and the existing toilet rooms. The contractor will ensure coordination between the trades to provide for a complete project.
4. The contractor is responsible for the project to comply with all applicable codes and regulations.
5. No deviations or changes to the design intent shall be made without the approval of DPW.
6. Contractor shall re-execute any work that fails to conform to the construction documents and any defects due to faulty materials or workmanship.
7. Warranty shall be extended for all construction components and equipment and installations included in this work for a minimum of 12 months from the date of final completion (unless noted otherwise).
8. General contractor is responsible for securing the site work, on-site storage areas and equipment during construction for protection of the property, as well as, materials, tools, etc.
9. Safe construction practices shall be observed to provide workperson's safety on the job and safety of all on-site personnel including, but not limited to DPW representatives, architect and existing users who need to access or use the structure during construction
10. General contractor is responsible for coordination of all deliveries, access to, and protection of the building during construction.
11. General contractor is responsible for means and methods of construction, including verifying and coordinating movement of products and materials through the building.
12. General contractor shall facilitate observations of work by the Fort Leavenworth representatives at all times, as requested or needed.
13.General contractor shall cooperate, coordinate and meet with any Fort Leavenworth-retained contractors or vendors for the completion of the project.
14. Contractor is responsible for field measuring and verifying all existing conditions prior to the start of construction and shall notify architect immediately in writing of any required changes that may be necessary in order to not delay the project.
15. In the case of discrepancies between the contract documents, the contractor shall secure instruction from DPW's Contracting Officer Representative (COR), Architect or Engineer prior to proceeding. Always assume and maintain the greater value option until written instruction from DPW indicates otherwise.

01300 SUBMITTALS

General contractor is responsible for submitting the following items to the architect and owner for review: Submit shop drawings for DPW's review for all items requiring fabrication (millwork, glass, doors, etc.).

A. Shop drawings: All non-sample or mock-up submittals shall be transferred in a legible PDF format with a minimum 3" high x 4" wide white space for architect's review stamp, and another for contractor's review stamp. Submittals shall be clearly numbered. Re-submittals shall be marked with a sequential letter designation after the number. All should be noted in a submittal log.

B. Product Data: PDF.

C. Samples: A minimum of two (2) of each finish samples as listed herein, one for DPW, one for the Contractor. The Contractor shall not ask for or use the DPW Sample. If the Contractor requires more samples, they are to make more separately. Each sample is to be marked with the project name, date, contractor's name, specific name or designation corresponding to the construction drawings, numbered sequentially and logged.

D. Engineering Submittals: Submit final Structural design (including dock, foundations, retaining walls and new addition) mechanical, electrical, plumbing and fire protection construction drawings to DPW for review. DPW shall have up to ten (10) working days to review and approve/reject submittals, but in most cases likely strive to take much less time than this.

01400 PROJECT COORDINATION

1. A pre-construction meeting will be conducted to review all procedures, schedule and phasing of the project.
2. Contractor is to administer a weekly, on-site DPW Construction Meeting (DPW COR and Architect, MICC Contracting Officer, Contractor) and write meeting notes to be delivered/emailed within 2 days of the meeting, should DPW choose not to write them.
3. Contractor shall provide DPW a construction schedule that aligns with the contract requirements and provide critical path items. Weekly updates are to be provided.
4. A Project Schedule, Schedule of Values (at a minimum broken out per specification listing), Submittal Schedule, and a List of Subcontractors are to be submitted prior to the first regularly scheduled DPW construction meeting. The Project Schedule shall be updated if and when things change throughout construction.
5. The Contractor is to provide and complete work on its own internal punch list prior to DPW's punch list walk-thru. Contractor to send this pre-punch list prior to scheduling a Punch List walk-thru with DPW.
6. Pay applications will be submitted to the COR for review and will follow standard AIA conventions, including partial and final lien waivers.

01710 FINAL CLEANING

1. Prior to acceptance of the work by the owner, the general contractor shall clean once prior to arrival of furniture and once after it is installed:
   - A. Sweep and mop clean all work areas prior to Punch list walk-thru.
   - B. Dust all other than floors with clean rags.
   - C. Clean all new windows thoroughly.
   - D. All cleaning materials and methods shall be as recommended by each manufacturer.
   - E. Removed marks, stains, fingerprints, other soil and dirt from painted surfaces.
   - F. Clean and polish hardware for all trades as necessary to establish new condition.
   - G. Remove spots, soil, paint from flooring; wash and seal as Manuf. recommended.
   - H. Clean fixtures and equipment, remove stains, paint, dirt and dust.
   - I. Remove temporary floor protection for finished floor; clean and polish such floors.
   - J. Where resilient flooring is supplied, clean, and polish all resilient floor surfaces if and as recommended by flooring manufacturer and DPW.

DIVISION 2 - SITE WORK

02070 - DEMOLITION

1. Provide demolition related to the project work as necessary for full completion of the project, and left in very good move-in condition during the project as it will be a partially occupied building during construction and need pallet storage movements.
2. Return original building to building condition prior to the work, including, but not exclusive of where removing doors, door frames, hardware, finishes, existing partitions not noted to remain, existing floor-covering, acoustic ceiling tile and power/telephone receptacles where existing partitions are to be removed.
3. Patch base building with seamlessly matching and like materials where items removed.

DIVISION 3 - CONCRETE

1. Provide concrete per the Drawings and patching as required.
2. All trenching or floor cores should be coordinated with DPW. Re: Plumbing. Any damage caused during the process is the responsibility of the Contractor.

DIVISION 4 – MASONRY - N/A

DIVISION 5 – METALS:

All new hollow metal doors to be heavy duty. Exterior hollow metal door frames to be heavy duty. Interior door frames medium duty.

DIVISION 6 - WOOD & PLASTICS

06100 - ROUGH CARPENTRY

1. This work includes generally not exposed carpentry work. Rough carpentry includes:
   - A. Wood nailers, blocking or sleepers
   - B. Wood furring.
2. Provide seasoned lumber with 15% maximum moisture content at time of dressing.
4. Provide pressure treated blocking in all partitions where required to anchor wall standards, cabinetry, grab bars, toilet partitions, equipment, and other items.

06200 - FINISH CARPENTRY

1. Provide finish carpentry as shown on the drawings and in accordance with AWI quality standards established by the Architectural Woodwork Institute.
2. All hardwood for paint finish shall be natural birch, custom grade.
3. Contractor to include in bid, pricing for furring out walls for fire extinguisher cabinets.
4. Provide installation of welded hollow metal frames, wood doors, metal framing, and finish hardware as shown on drawings.
5. Adjust hardware for proper operation and code compliance of all doors in space as shown on drawings, including but not limited to minimum opening force allowable by the administrative authority for fire doors and 5 lbs. max for interior hinged doors and gates, sliding and folding doors.
6. The millwork contractor shall obtain and field verify all measurements and conditions at the building affecting his work and shall be responsible for all details and dimensions assuring precision fitting and proper assembly of his products.
7. All work shall be accurately set plumb, level and square, scribed tightly and accurately to adjacent surfaces, securely anchored in the position indicated on drawings and approved shop drawings.
8. Coordinate installation of in-wall steel anchorage and miscellaneous blocking with all trades for precise locations.
9. Make all necessary modifications in the field to accommodate conduit, piping, ductwork, outlets, boxes, sprinkler pipes, or other miscellaneous items.

DIVISION 7 - THERMAL & MOISTURE PROTECTION (See M/E/P drawings for additional information)

1. Provide acoustic insulation (6" sound attenuation batt) that is plenum approved above ceiling and in all walls of Latrines and Conference Room as well as exterior walls. Acoustic insulation shall be approved equal to Owens Corning's Thermafiber. Ceiling and floor metal tracks shall be bedded in two continuous beads of acoustic caulking.
2. For "Metal Building" addition: Insulation to have R-13 insulation between framing plus R-13 continous insulation outside the frame, per the IECC.
3. Roof - Metal Buildings R-19 + R-11 LS, Framed Building R-30 ci (continuous). Contractor to show/submit to DPW how insulation meets requirements prior to order.

DIVISION 8 - DOORS & WINDOWS (See door schedule)

08100 - METAL DOORS AND FRAMES

1. Submittals
   - A. Product Data: Submit manufacturer's technical product data substantiating that products comply with requirements.
   - B. Shop Drawings: Submit for fabrication and installation of steel frames. Include details of each frame type, conditions at openings, details of construction, location and installation requirements of finish hardware and reinforcements, and details of joints and connections. Show anchors and accessory items.
2. Provide schedule of frames using same reference opening numbers as those on the construction drawings.
3. Inspect metal work upon delivery for damage. Minor damages may be repaired provided refinished items are equal in all respect to new work and as acceptable to DPW; otherwise, remove and replace damaged items as described. Verify prior to work.
4.Prepare frames to receive finish hardware in accordance with final finish hardware schedule and templates provided by hardware supplier. Comply with specifications for door and frame preparation with applicable requirements of ANSI A115 series for hardware.
5. Provide for three (3) door mutes (silencers) per frame.
6. Provide for floor anchor clips for each jamb. Clips to be fourteen (14) gauge with two (2) anchor holes and welded frames.
7. Fire-rated door assemblies: Where fire-rated door assemblies are indicated or required, provide fire-rated door and frame assemblies that comply with NFPA 80 "Standard for Fire Doors and Windows" and have been tested, listed, and labeled in accordance with ASME 152 "Standard Methods of Fire Tests of Door Assemblies " by a nationally recognized independent testing and inspection agency acceptable to authorities having jurisdiction.
8. Metal "Swinging" Doors to be insulated with U-0.37 overall value for opaque portion
9. Garage Door to be insulated with U-0.31 overall value.
10. Provide shop-applied, rust-inhibitive primer to the more conservative of the paint manufacturer's recommendation or per industry standard. Feild paint, two coats.
11. Windows - Fixed: U-0.36, SHGC: 0.36; Operable U-0.45, SHGC - 0.33

08211 - FLUSH WOOD DOORS

1. Provide plain-sliced Birch solid core wood doors (book and running matched), unless noted on the drawings.
2. Provide underwriter lab markings for fire doors in accordance with National Fire Protection Association and applicable local codes. Moisture content of doors shall be 12% maximum at time of fabrication.
3. Maximum warp shall be 1/4" with allowable tolerances of 1/16" in overall dimension.
4. Flush doors shall be guaranteed for a period of two (2) years.
5. Solid core flush wood doors shall be stain grade, with block frame glued core with blocks of 2 1/2" maximum width bonded together with end joints staggered with adjacent rows.
6. Seal all door edges after fitting and undercutting.
7. Comply with WDMA's "How to Store, Handle, Finish, Install, and Maintain Wood Doors."

08710 - FINISH HARDWARE

1. Provide finish hardware for the project per as shown on drawings and agreed upon with DPW, unless otherwise noted. Verify all with DPW through shop drawings.
2. The hardware contractor shall be responsible for thoroughly detailing the entire project to insure the items specified will properly function in the indicated locations, this includes glazed doors and electronic strikes, magnetic locks, etc.
3. For labeled openings provide hardware in accordance with National Fire Protection Association's standards and local ordinances.
4. Provide keying and keying schedule with DPW and per DPW requirements.

08800 - GLASS & GLAZING

1. Provide each type of glass as shown on the drawings and listed in the item specification sheets, provide all components for a complete installation.
2. Glass shall conform with federal specifications DD G451-C.
3. Unless otherwise shown, confirm with details and procedures of the "Glazing Manual" (flat glass jobbers association).
4. All glass shall be tempered unless noted otherwise.
5. All tempered glass shall be tempered without tongmarks or labels.
6. All glass shall be 1/4" or greater per industry standard.
7. Each type of glass specified to be produced by one (1) manufacturer.
8. Openings and cut glass accurately to fit each opening with minimum edge. Measure all clearances and bit on glass as specified FGJA. If glass is to be cut to size at project site, deliver each piece to project at least 2" larger (in both dimensions) than required, so as to cut edges without the necessity of seaming or nipping.
9. Prime surfaces to receive glazing compounds in accordance with manufacturer's recommendations, using recommended primer.
10. Inspect each piece of glass before installation. Pieces which have significant impact damage at edges, scratches or abrasions of faces, or any other evidence of damage shall not be installed. Locate setting blocks at the quarter points of sill, but no closer than 6" to corners of glass.
11. Use blocks of proper size to support the glass in accordance with manufacturer's recommendations where continuous gaskets or provide spacers for all glass to separate glass from stops.
12.Locate spacers 36" O.C. maximum inside and out, with minimum of two (2) spacers per edge of glass.
13. Set glass in a manner which produces greatest possible uniform appearance. Face all glass which has dissimilar faces with matching faces in the same direction.
14.Glazing materials from different sources shall not be used in the same joint systems unless the manufacturer of each material has stated, in writing, that his material is fully compatible with the other materials.
15. Use masking tape or other suitable protection to limit coverage of glazing materials to the surface intended for sealants.
16. Butt lap ends of sealant tape in accordance with the manufacturer's recommendations.
17. Tool exposed surfaces of glazing materials to provide a slight wash away from the glass. Install exposed tapes and gaskets with a slight protrusion above stops in the final compressed condition.
18. Clean excess sealant or compound from glass and framing members immediately after application, using solvent or cleaners recommended by manufacturer.
19. Cure sealants in accordance with the manufacturer's instruction to attain maximum durability and adhesion in glass and framing as soon as possible.
20. Protect glass from breakage immediately upon installation. Use streamers of ribbons suitably attached to framing and held free of the glass. Warning markings shall not be applied directly to the glass.
21. Remove and replace glass which is broken, cracked, or scratched, by any source, including weather, vandalism and accidents prior to DPW acceptance.
22. Maintain glass in a reasonably clean condition during construction so that it will not become stained and will not contribute to the deterioration of the glazing materials.
23. Wash and polish on both faces, not more than four (4) days prior to date of final acceptance. Comply with instructions and recommendations of glass manufacturer of cleaning in each case.

DIVISION 9 FINISHES

09250 GYPSUM BOARD AND FRAMING ACCESSORIES

1. Provide gypsum drywall made in the USA and associated work as shown on the drawings for a complete installation.
2. General Standards: GA-216 by Gypsum Association, and manufacturer's instructions (whichever is general standard: GA.2 - More stringent).
3. Provide building standard steel studs 3-5/8" thick (or as shown on the drawings) in lengths required at 16" O.C. Provide metal runners for studs, metal furring channels, clips Z-furring channels, 1 1/2" cold rolled channels, wire hangers, ties and other incidentals necessary for the work. Studs shall be ASTM C-645, 25 Gage C-shaped studs with associate runner channels.
4. Provide ASTM C36 Type "X", 5/8" thick, 48" wide gypsum drywall, regular, Fire Code C, or other board as required or as shown on the plans.
5. Trim accessories: Provide manufacturer's standard metal trim accessories, of the beaded type with face flanges for concealment in joint compound except where semi-finishing or exposed type is indicated. Provide metal corner beads, L-shaped edge trim beads, U-type trimbeads, special L-Kerf type edge tri beads and one piece control beads.
6. Gypsum board fasteners: Type recommended by gypsum board manufacturer except as otherwise indicated.
7.Sound attenuation blankets: Semi rigid mineral fiber without membrane, FS HH-1-521, Type I, S Thickness as shown.
8. Frame door and other openings with studs, runners, gage, number and arrangement to comply with manufacturer's recommendations for size of opening, weight of doors and height of stud size, unless otherwise indicated.
9. Install gypsum boards in lengths and directions which will minimize number of end joints, and avoid end joints in central areas. Install walls and partitions with exposed gypsum boards vertical, with joints offset on opposite sides of partitions. Otherwise, install boards with edges perpendicular to supports with end joint staggered over supports, except where recommended in a different arrangement by manufacturer.

DIVISION 9 FINISHES

09250 GYPSUM BOARD AND FRAMING ACCESSORIES Continued.

10. Partial height partitions (as shown on the drawings): 3-5/8" steel studs spaced at 16" O.C. with 5/8" gypsum board on each side, taped, sanded and ready for paint. Provide steel tube bracing at partition ends and corners.
11. For any partition exceeding 15' in length without being braced by another partition, provide metal stud kickers above finished ceiling to the structure above at a maximum space of 15' on center.
12. Provide flush hinged metal access panels as required for appropriate mechanical, electrical and 14. Plumbing systems in gypsum board ceilings and soffits. Submit proposed locations and access panel product information for architect's review.
13. Provide water resistant "Purple board" gypsum board at all locations where plumbing fixtures (sinks, water heaters, toilets, etc) are located, including all walls of toilet rooms and full wall behind drinking fountains/eyewash wall.
14. All exposed gypsum board walls and ceilings to be finished to a Level 4 finish.
15. Paint colors not final till approved through submittals by DPW

09512 - ACOUSTICAL TILE CEILINGS

1. See reflected ceiling plan.
2. Ceiling suspension system shall comply with ASTM - C635 as applicable to type of suspension system required for type of ceiling units indicated on drawings. Coordinate all work supported by or penetrating through ceilings including light fixtures, HVAC equipment and partition systems. The structural class shall be intermediate duty system with attachment devices for five (5) times design load indicated in ASTM C-635
3. Provide ceiling trim reveal edge molding of manufactured standard channel molding for edges and penetrations of ceiling with baked enamel white finish. Provide all other necessary miscellaneous items required to complete this work.
4. Provide edge molding of manufactured standard channel molding for edges and penetrations of ceiling with baked enamel finish. Provide all other necessary miscellaneous items required to complete this work.
5. Install ceiling system in accordance with manufacturer's recommendations.
6. At wall or edge conditions for tile 6" or smaller, provide an oversized acoustical tile cut down with tegular edge at these conditions. Touch up cut tile edge to match tile face.

09660 - RESILIENT FLOOR TILE

1. Provide 12" x 12" x .125" vinyl composite tile flooring in gauge, size and color as specified. Install in accordance with manufacturer's recommendations.
2. Clean substrate, fill cracks, holes and depressions, and level within 1/8" in 10' before laying resilient floor.
3. Adhesive: as recommended by manufacturer.
4. Cleaning: not more than four (4) days before acceptance or occupancy, clean the resilient flooring. Wash thoroughly with a cleaner recommended by the flooring manufacturer, using a cleaning process per flooring manufacturer recommendations.
5. Submit two (2) sets of maintenance specifications to the architect.

09678 - RESILIENT BASE & ACCESSORIES

1. Provide base in gauge, height, and color as specified. Install in accordance with manufacturer's recommendations.
2. Base shall be straight at carpet and coved at tile floor unless otherwise noted.
3. Clean substrate, fill cracks holes and depressions, and level wall within 1/8" in 10' before installing resilient base.
4. Adhesive: As recommended by manufacturer.
5. Run base corner to corner without splices.

09795 - FLOORING UNDERLAYMENT

1. Floors shall be latexed to be true and level at all areas where it is required to maintain the proper relationship between the floor, adjacent floor finishes. These areas shall include, but not be limited to, door openings, glass partitions, millwork installation, cabinet and shelving units, files etc. The maximum slope allowed to be 1/4" in 10'-0". In addition, provide latex where shown on drawings.
2. Provide flooring underlayments to accommodate specified finish flooring. All materials to be in accordance with manufacturer's recommendations for each finished flooring material.

09910 - PAINT

1. Provide painting and related work as shown on the drawings.
2. Apply two (2) finish coats of architect's approved paint (over the prime coat), unless indicated otherwise. Finish coats to be as follows: A) Walls: Eggshell Finish, B) Ceilings: Flat Finish, C) Doors/Door frames: semi gloss finish.
3. Provide all materials, brushes, tools, ladder, any equipment of any kind necessary for proper execution of the work.
4. Protect all hardware, lighting fixtures, and all finish surfaces before and during painting.
5. Workmanship shall be of high quality in every respect with all materials evenly spread and smooth without runs or sags.
6. All prime coats to be tinted to approximate shade of final coat. First coat to be primer/sealer for all previously finished surfaces as indicated on finish specifications. For new construction scheduled to receive paint, apply first coat of primer/sealer as recommended by receiving material and finish paint manufacturer. All sections, spots or hot spots, after primary application shall be touched up before applying the second coat to produce a smooth and level surface. The final coat shall not be applied until after other trades whose operations would be detrimental to finish painting have finished their work in the areas to be painted.
7. Work not included: Unless otherwise indicated, shop priming of ferrous metal items and fabricated components are included under their respective trades. Pre-finished items, are not included. Finished metals such as anodized aluminum, stainless steel, bronze , and similar metals will not be painted. Do not paint any moving parts of operating units, or over any equipment identification, performance rating, names or nomenclature plates or code required labels.
8. Protect work of other trades. Correct painting related damages by cleaning, repairing or replacing, and refinishing, as approved by architect.
9. Apply painting and finishing materials in accordance with manufacturer's directions. Use applicators and techniques best suited for materials and surfaces to which applied.
10. Apply additional coats when undercoats, stain or other conditions shown through final paint coat, until paint film is of uniform finish, color and appearance.
11. Paint surfaces behind moveable equipment (refrigerators, etc.) and furniture same as similar exposed surfaces. Paint completely concealed surfaces behind new permanently fixed equipment, furniture or millwork, with a prime coat only before equipment is installed.
12. Paint back sides of access panels and removable or hinged covers to match surfaces.
13. Apply prime coat to material which is required to be painted or finished and which has not been prime coated by others.
14. Apply each material at not less than the manufacturer's recommended spreading rate, to provide a total dry film to thickness of not less than 2.5 mils. for entire coating system of prime and finish coats for coat work.
15. All exposed or visible ductwork, conduit, pipes, raceways, etc. shall be painted to match the ceiling (Benjamin Moore - Superspec Latex Pain, Color: black, flat finish)
16. Permanent building lighting shall be in operation during to painting.

FORT LEAVENWORTH DPW

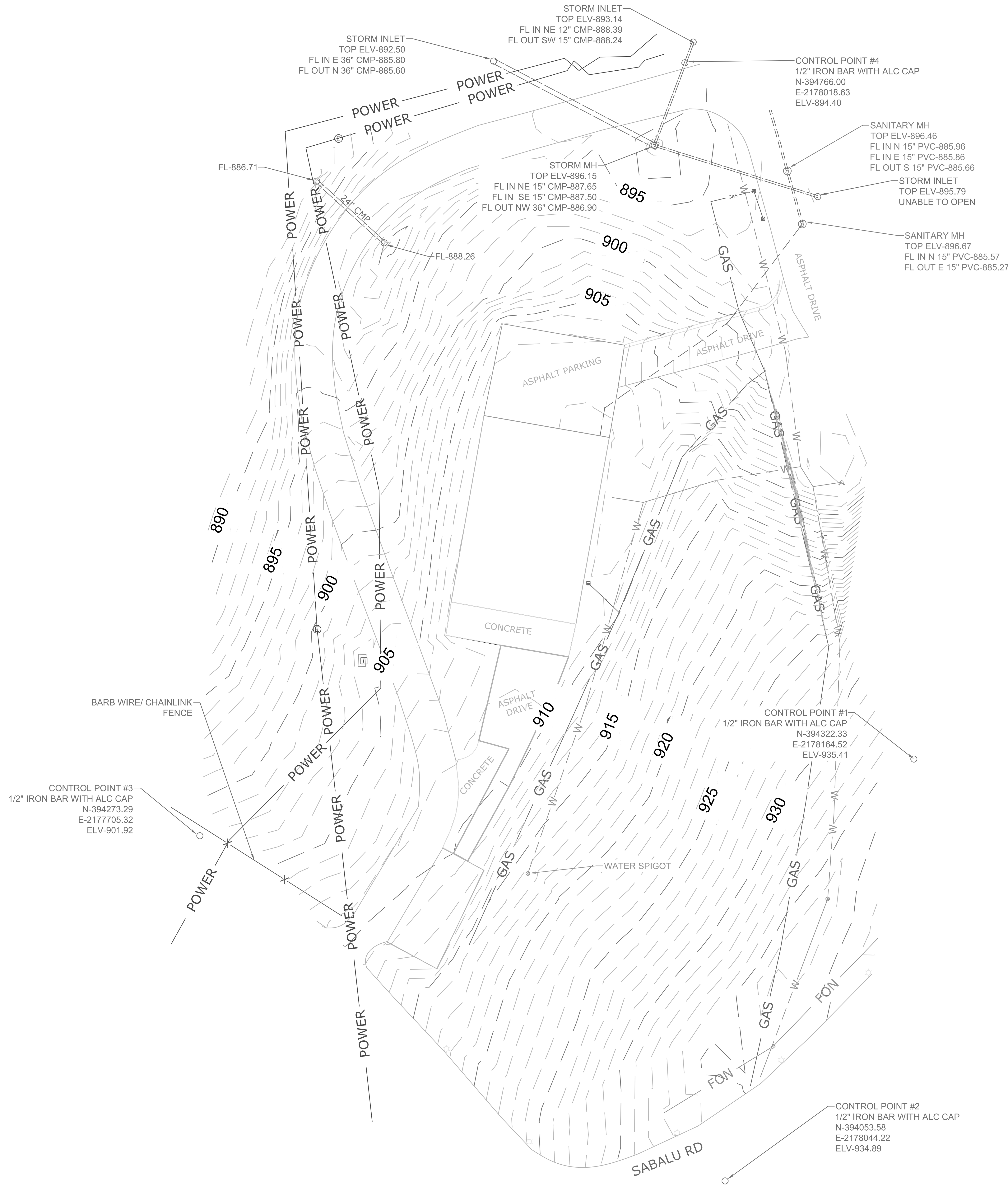
MARK	DESCRIPTION	DATE
2	Revision 2, Addendum 2	4 / 1 / 2026

DESIGNED BY: Designer
DRAWN BY: DB
CHECKED BY: DB
SIZE: ANSI 'D'
ISSUE DATE: 2/21/2025
PROJECT NO.: 30000031343
100% Drawings

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH
820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition
Specifications

SHEET ID
A6



LEGEND

- SANITARY MANHOLE
- ELECTRICAL MANHOLE
- TRANSFORMER
- GUY WIRE
- POWER POLE
- TELEPHONE PEDESTAL
- TREE
- WATER VALVE
- GAS VALVE
- LIGHT POLE
- POST INDICATOR VALVE
- OHP — EXISTING POWER LINE
- FON — EXISTING BURIED TELEPHONE LINE
- W — WATER LINE
- EXISTING SANITARY LINE
- EXISTING CHAIN LINK FENCE
- EXISTING STORM SEWER
- X — EXISTING FENCE
- POWER — EXISTING ELECTRICAL
- GAS — EXISTING NATURAL GAS
- 900 — CONTOUR

1 / C1 EXISTING CONDITIONS
SCALE: 1" = 40'

N

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE

ISSUE DATE: 02/21/2025
PROJECT NO.: 3000003313343

DESIGNED BY: DPL
DRAWN BY: DPL
CHECKED BY: DPL
SIZE:

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

820 McCLELLAN AVE,
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

Existing Conditions

SHEET ID
C1

SHEET KEYNOTES

1. REMOVE 2,850 SQ FT CONCRETE
2. REMOVE 3,254 SQ FT ASPHALT
3. REMOVE WATER SERVICE LINE. APPROX. 25 LF
   RE: PLUMBING

1 / C2 DEMOLITION PLAN
SCALE: 1" = 30'

N

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE

ISSUE DATE: 02/21/2025
PROJECT NO.: 30000331343

DESIGNED BY: DPL
DRAWN BY: DPL
CHECKED BY: DPL
SIZE:

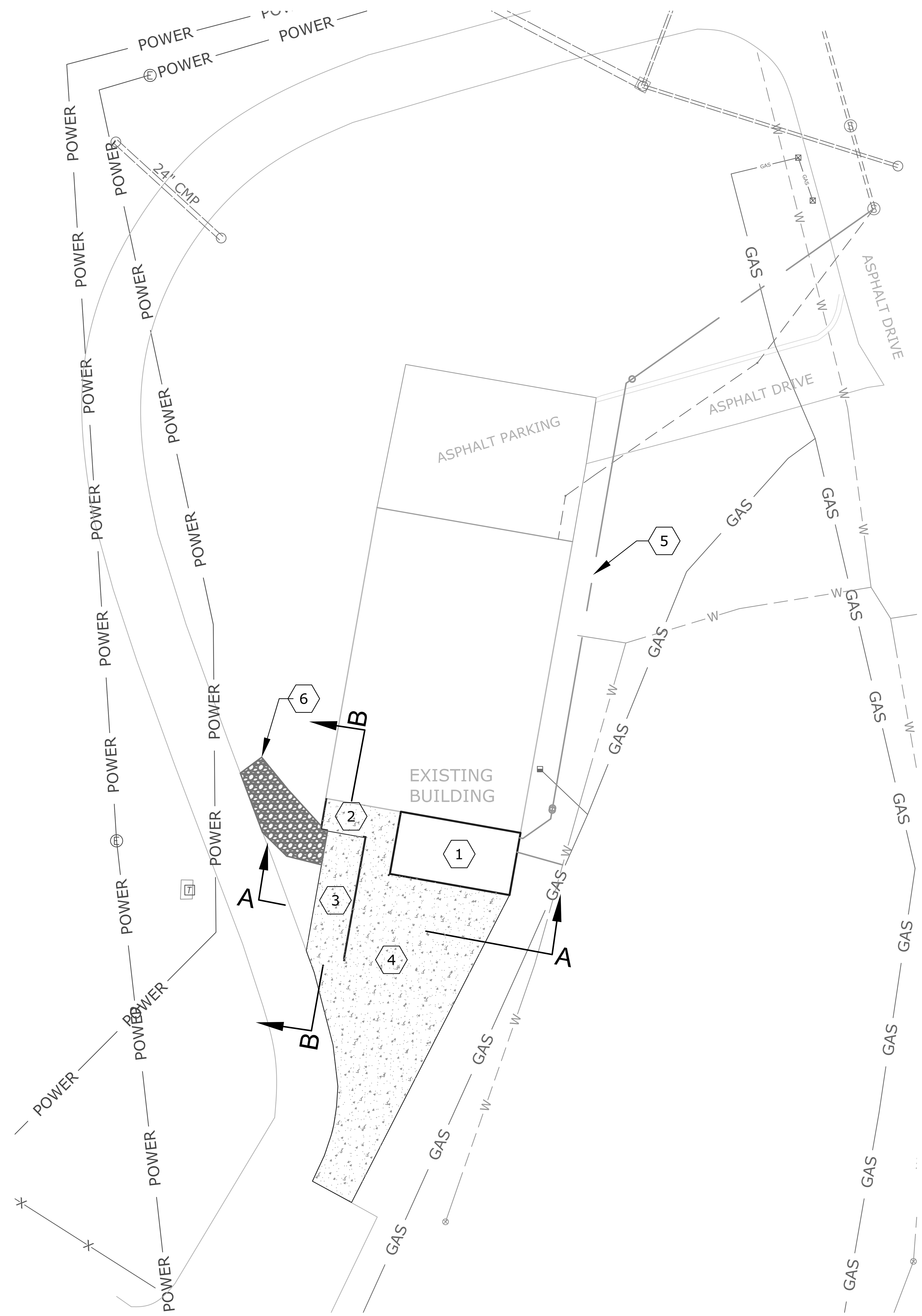
DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

820 McCLELLAN AVE,
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

Demolition Plan

SHEET ID
C2

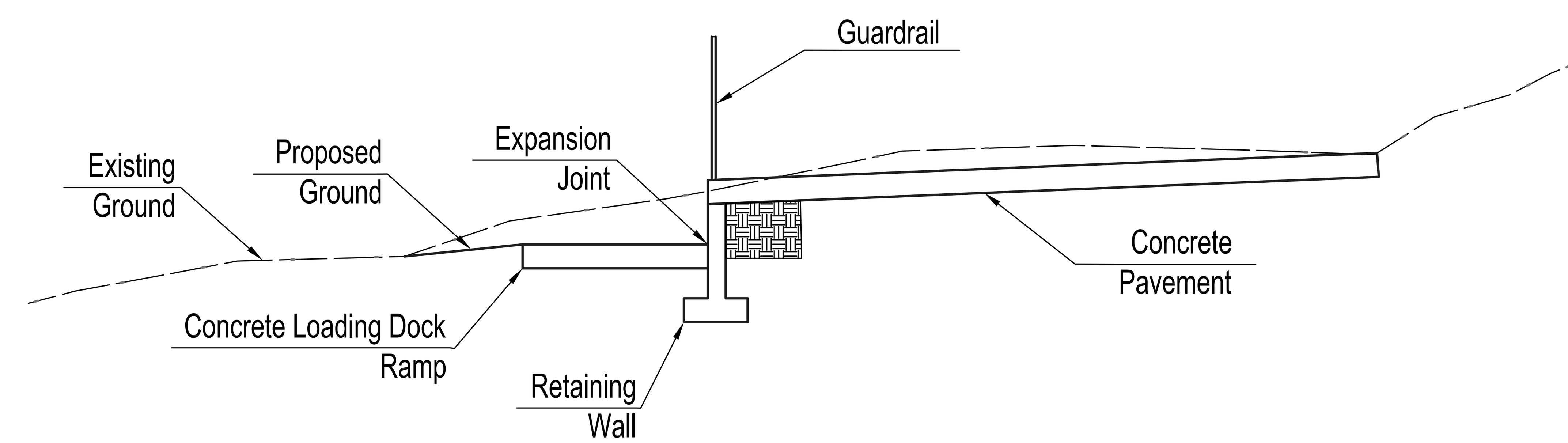


1
C3
SITE PLAN
SCALE: 1" = 30'

N

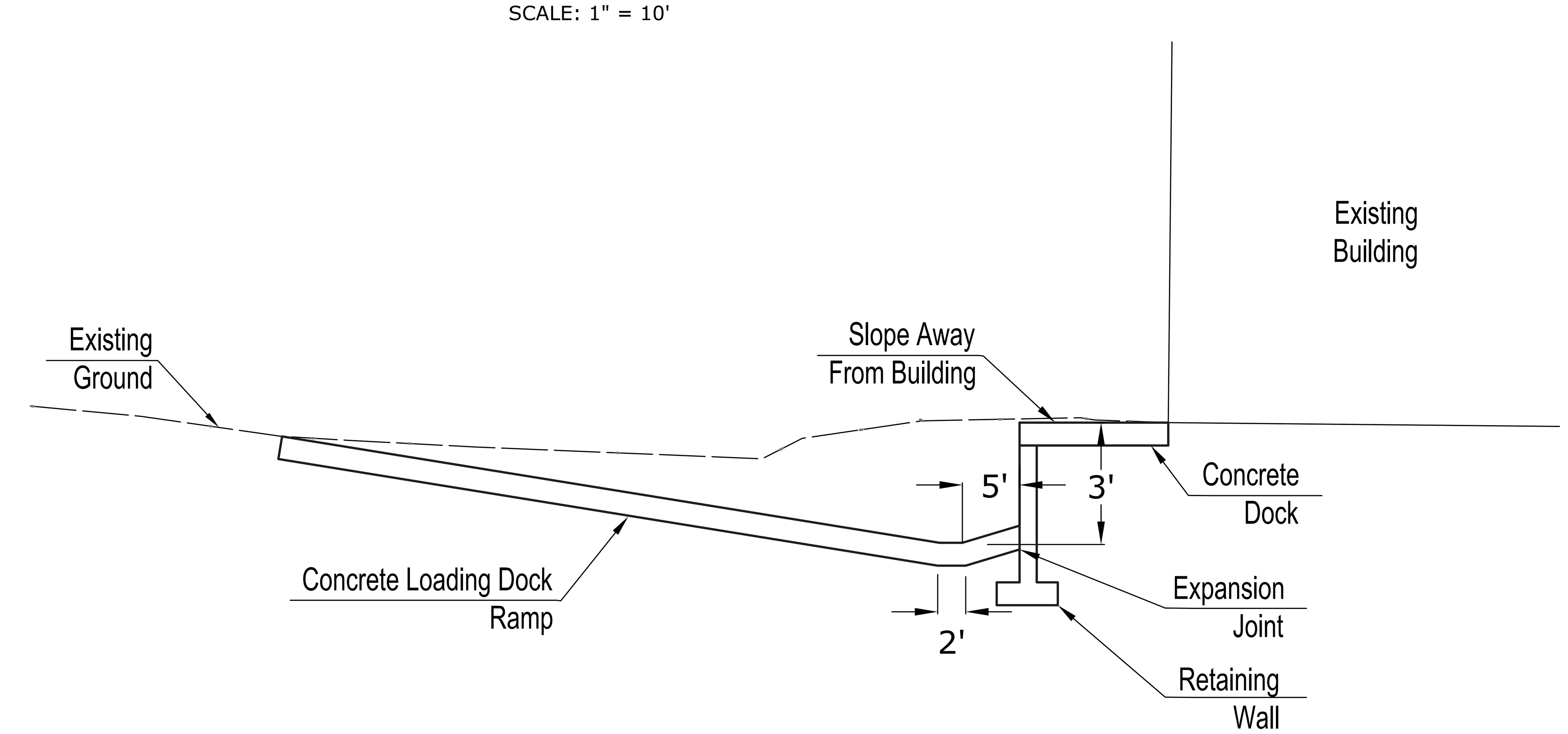
SHEET KEYNOTES

1. NEW BUILDING ADDITION, RE:ARCHITECTURAL
2. NEW CONCRETE LOADING DOCK W/DECK LEVELER AND DOCK BUMPERS, RE:ARCHITECTURAL
3. NEW CONCRETE LOADING DOCK RAMP
4. NEW CONCRETE PAVEMENT
5. INSTALL NEW 4" SANITARY SEWER SERVICE LINE W/TWO-WAY CLEANOUT, RE:PLUMBING
6. 30 CY 6" DITCH LINING



SECTION A-A

SCALE: 1" = 10'



SECTION B-B

SCALE: 1" = 10'

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE
	Revised Per Comments	1/14/2025

ISSUE DATE: 02/21/2025
PROJECT NO.: 300000331343

DESIGNED BY: DPL
DRAWN BY: DPL
CHECKED BY: DPL
SIZE:

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

820 McCLELLAN AVE,
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

Site Plan

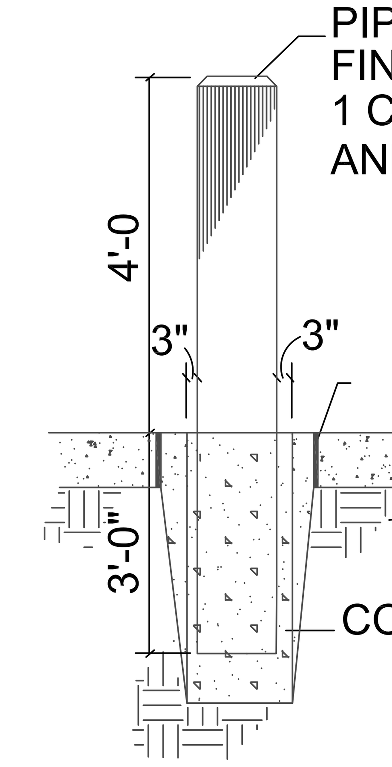
SHEET ID

C3

1 / C4 GRADING PLAN
SCALE: 1" = 10'

SHEET KEYNOTES

1. NEW BUILDING ADDITION, RE:ARCHITECTURAL
2. NEW CONCRETE LOADING DOCK W/DECK LEVELER AND DOCK BUMPERS, RE:ARCHITECTURAL
3. NEW CONCRETE LOADING DOCK RAMP. MAXIMUM SLOPE 6.0%
4. NEW CONCRETE PAVEMENT, RE:DETAIL D1 SHEET C5
5. NEW RETAINING WALL. CONTRACTOR SHALL SUBMIT RETAINING WALL DESIGN PREPARED BY A KANSAS LICENSED PE.
6. 42" GUARDRAIL, RE: DETAILS SHEET C5
7. INSTALL NEW WATER METER AND 1" WATER LINE, RE:PLUMBING
8. STEEL BOLLARD



STEEL BOLLARD

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE
	Revised Per Comments	1/14/2025

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DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH
820 McCLELLAN AVE,
FORT LEAVENWORTH, KS
66027-1292

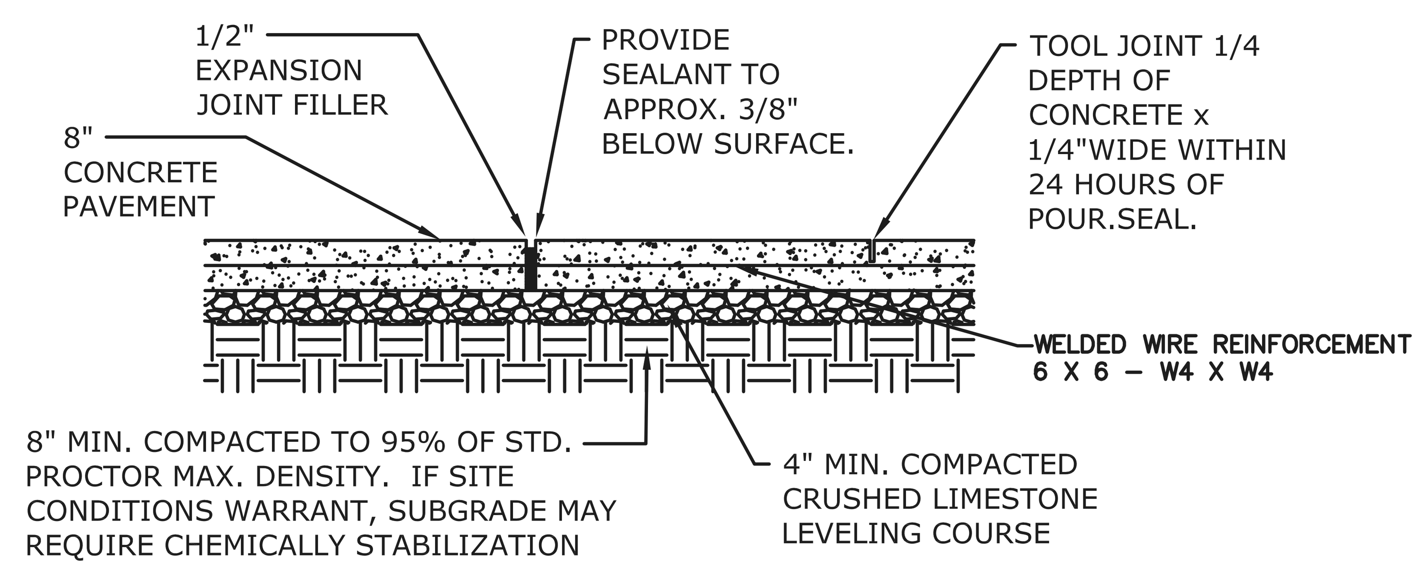
B1146 - Textiles Warehouse - Office Addition

Grading Plan

SHEET ID
C4

GENERAL PROJECT NOTES

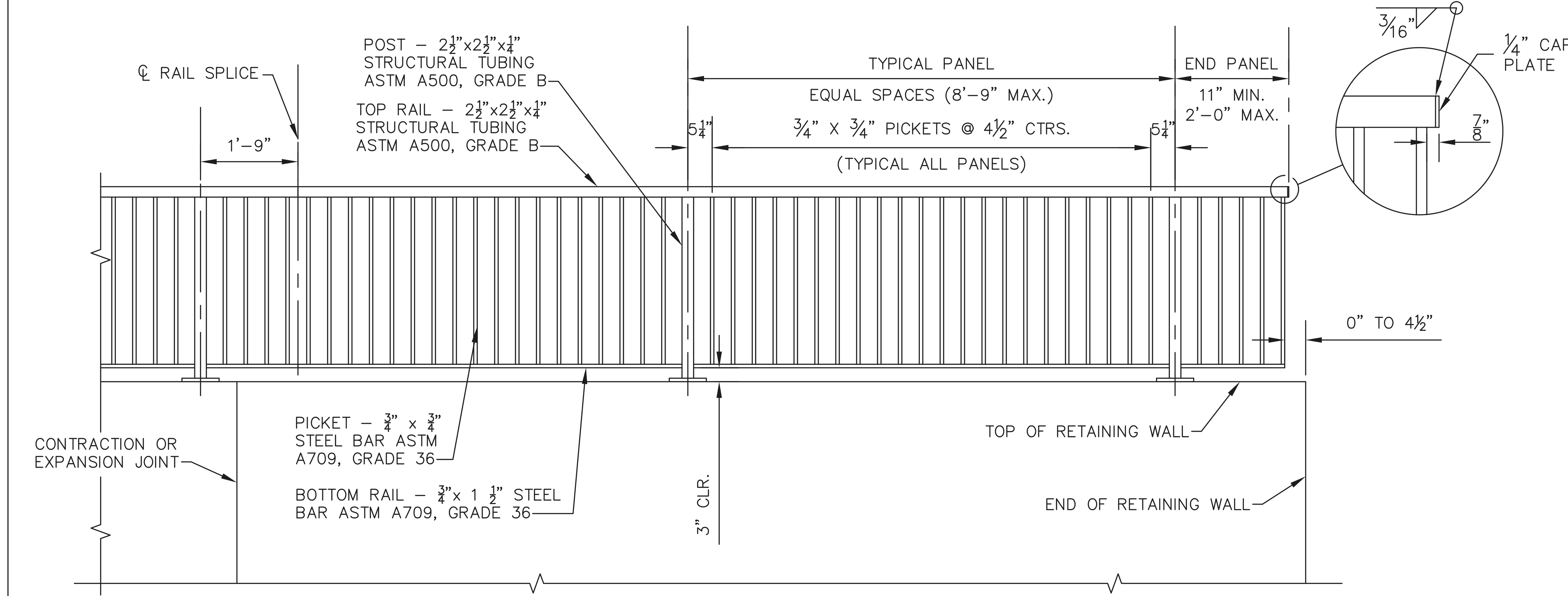
1. ALL WORK SHALL BE COORDINATED WITH THE DIRECTORATE OF PUBLIC WORKS (DPW) AND THE CONTRACTING OFFICER. CONTRACTOR TO VERIFY LOCATION OF ALL EXISTING UTILITIES. CONTRACTOR TO HAVE LOCATES PERFORMED PRIOR TO ANY EXCAVATION. IN THE EVENT THAT A UTILITY IS SITUATED SUCH THAT CONSTRUCTION CANNOT PROCEED AS SHOWN ON THE PLANS, NOTIFY THE CONTRACTING OFFICER REPRESENTATIVE(COR) IMMEDIATELY. ANY DISCREPANCIES BETWEEN THE DRAWINGS AND ON SITE CONDITIONS ARE TO BE CAPTURED VIA REDLINES AND UPDATED ON THE AS BUILT SUBMITTAL.
2. CONTRACTOR SHALL INSTALL TRAFFIC CONTROL SIGNAGE AND CONSTRUCTION BARRICADES IN ACCORDANCE WITH MANUAL OF UNIFORM TRAFFIC CONTROL DEVICES (MUTCD). CONSTRUCTION LIMITS SHALL BE INDICATED VIA BLACK MESH TEMPORARY CONSTRUCTION FENCING.
3. CONTRACTOR TO FIELD ADJUST ANY EXISTING UTILITY MANHOLES OR VALVES TO GRADE AS REQUIRED.
4. CONSTRUCTION STAGING AND STORAGE SHALL BE LOCATED ON SITE. SMALLER MATERIALS WILL BE STAGED AT A CONTRACTOR YARD OR AS ASSIGNED BY THE INSPECTOR. MATERIALS FOR THIS PROJECT SHALL BE STORED AS DETERMINED IN THE PRE-CONSTRUCTION MEETING.
5. CONTRACTOR PARKING SHALL BE ONLY IN THE PARKING LOT OF B 1146 OR AS OTHERWISE DICTATED BY THE COR.
6. CONTRACTOR SHALL ENSURE THAT ALL ADJACENT PROPERTY IS PROTECTED FROM DUST, DIRT, RUNOFF, DEBRIS, AND ANY OTHER DAMAGES RESULTING FROM PROJECT ACTIVITIES.
7. CONTRACTOR TO REPLACE OR REPAIR ALL EXISTING ADJACENT FINISHED SURFACE, UTILITIES, EQUIPMENT, PRODUCTS, VEGETATION, AND STRUCTURES, OR PARTS THEREOF, THAT HAVE BEEN DAMAGED OR REMOVED TO PERFORM THE WORK.
8. CONTRACTOR SHALL ENSURE THAT ALL NON-RECYCLABLE TRASH, DEBRIS, REFUSE, GARBAGE, ETC. IS TAKEN OFF BASE AND DISPOSED OF IN A SAFE AND LEGAL MANNER.
9. CONTRACTOR SHALL ENSURE THAT THE PROJECT SITE IS COMPLETELY CLEAN BEFORE THE PROJECT FINAL WALK-THROUGH. ALL MATERIALS SHALL BE NEW, PROPERLY CLEANED, AND SHALL BE FREE OF SCUFFS, DIRT, AND DUST. AREAS SURROUNDING THE CONSTRUCTION SITE SHALL ALSO BE CLEANED OF ANY DIRT, DUST, OR DEBRIS.



JOINT SPACING:
EXPANSION – 40' MAX.
CONTROL – 10' MAX.

NOTE:
1. CROSS SLOPE SHALL NOT EXCEED 2% ALONG ALL ADA ROUTES.
2. MACRO FIBER REINFORCEMENT MAY BE SUBSTITUTED FOR WELDED WIRE

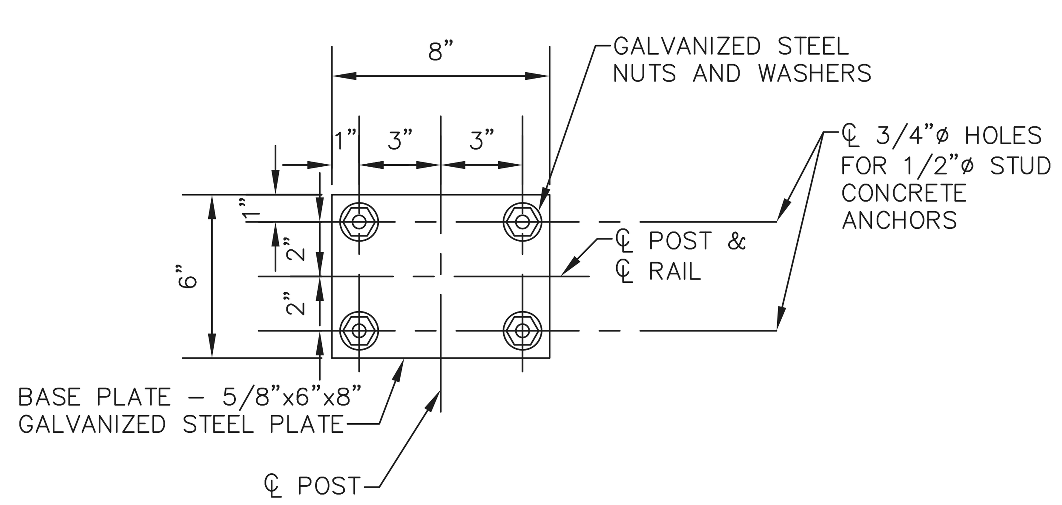
D1 REINFORCED CONCRETE PAVEMENT
NOT TO SCALE



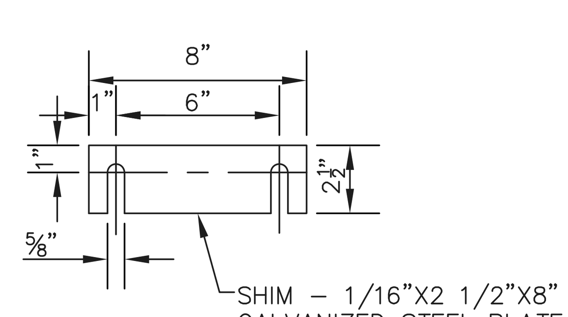
TYPICAL METAL HANDRAIL ELEVATION
ON CONCRETE RETAINING WALL

HANDRAIL GENERAL NOTES:
1. GALVANIZE ALL RAIL, POSTS, PICKETS, AND BASE PLATES AFTER FABRICATION IN ACCORDANCE WITH THE REQUIREMENTS OF ASTM A123. HANDRAILS SHALL BE PROTECTED FROM ANY DAMAGE TO THE FINISHED SURFACE. NO FIELD GALVANIZING WILL BE ALLOWED.
2. SET RAILS PARALLEL TO THE TOP OF THE WALL. SET POSTS AND PICKETS VERTICAL IN BOTH PLANES OF THE HANDRAIL. SHIMS MAY BE USED BETWEEN CONCRETE AND BASE PLATE OF STEEL POST.
3. GRIND SMOOTH ALL TOP AND BOTTOM RAIL–TO–POST WELDED CONNECTIONS. FIELD WELDING IS NOT PERMITTED. THE CONTRACTOR SHALL SUBMIT SHOP DRAWINGS TO THE CONTRACTING OFFICER FOR APPROVAL PRIOR TO FABRICATION OF THE HANDRAIL. FIELD VERIFY AS–BUILT DIMENSIONS AND ELEVATIONS OF RETAINING WALLS, HUBGUARDS OR WINGWALLS PRIOR TO FABRICATION OF HANDRAIL. POST SPACING WILL BE REVIEWED DURING THE DESIGN REVIEW PERIOD TO COORDINATE WITH EXPANSION AND CONTRACTION JOINT SPACING IN THE WALL.
4. FABRICATE HANDRAIL IN LENGTHS TO INCLUDE A MINIMUM OF ONE AND A MAXIMUM OF THREE PANELS.
5. ATTACH TOP RAIL MEMBERS TO AT LEAST TWO POSTS. LOCATE CENTER OF RAIL SPLICES 1'–9" FROM CENTERLINE OF POST UNLESS SHOWN OTHERWISE.
6. THE THREADED ROD CONCRETE ANCHORS SHALL BE GALVANIZED AND PROVIDE A MINIMUM PULLOUT STRENGTH OF 8000 POUNDS ON 4000 PSI CONCRETE. LENGTH OF EMBEDMENT INTO CONCRETE SHALL CONFORM TO MANUFACTURER'S RECOMMENDATIONS. AFTER THE HOLES ARE DRILLED, REMOVE ALL LOOSE MATERIAL BY USING A WIRE BRUSH TO FREE THE DUST FROM THE SIDE OF THE HOLE AND THEN VACUUMING TO REMOVE MATERIAL AND DUST. PLACE EPOXY CAPSULE SYSTEM IN THE HOLE AND INSERT THE THREADED ROD IN ACCORDANCE WITH THE MANUFACTURER'S DIRECTIONS.

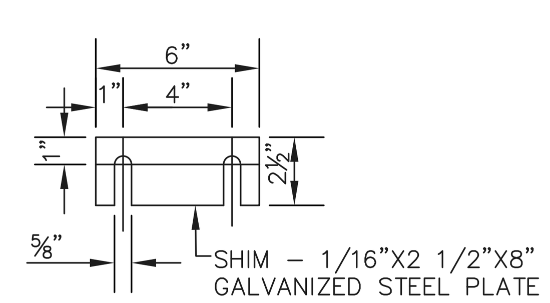
D6 HANDRAIL ON RETAINING WALL
NOT TO SCALE



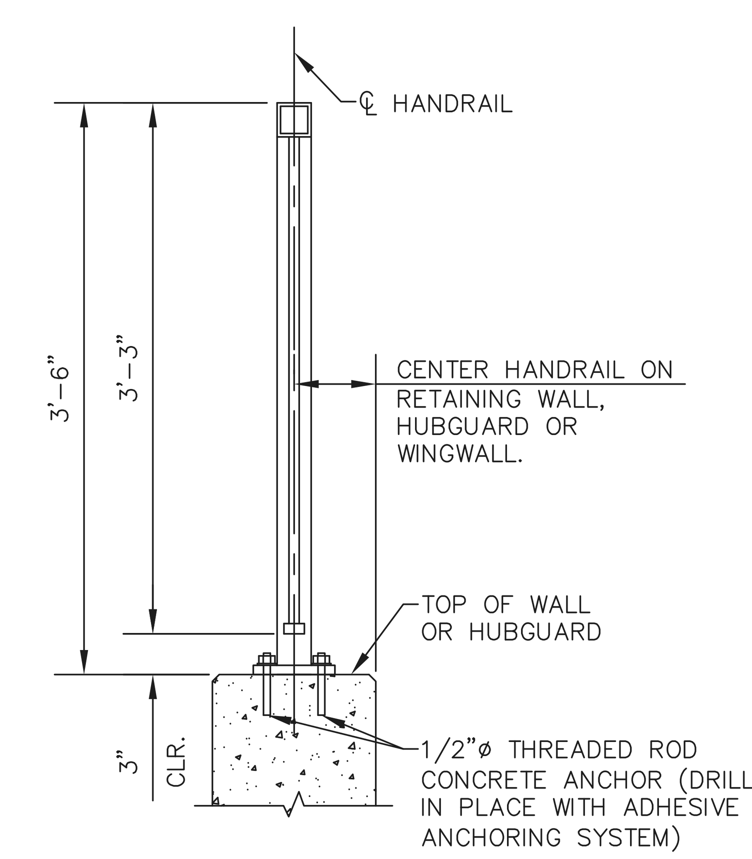
BASE PLATE DETAIL



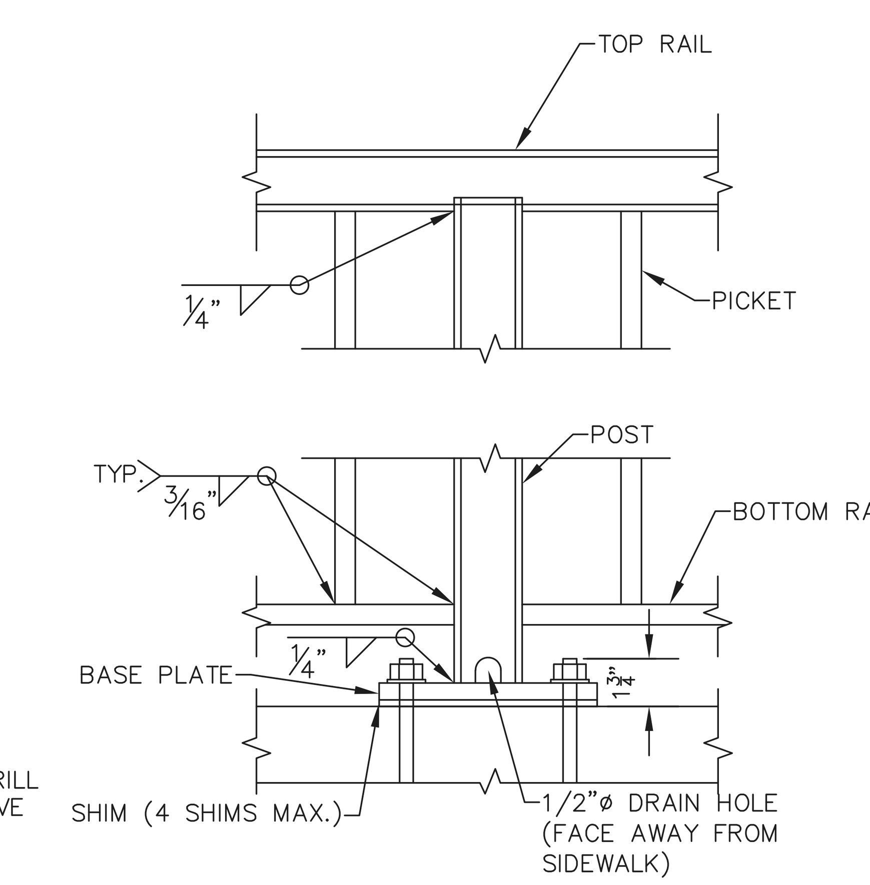
8" SHIM DETAIL



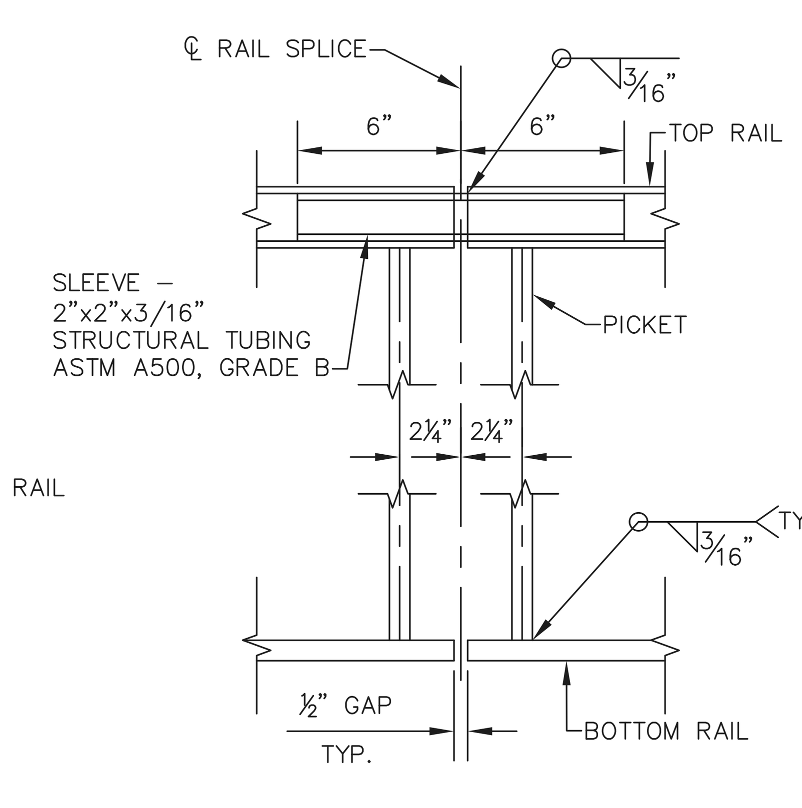
6" SHIM DETAIL



SECTION THRU RAIL



POST DETAIL



RAIL SPLICE DETAIL

A1 HANDRAIL ON RETAINING WALL DETAILS
NOT TO SCALE

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE
	Revised Per Comments	1/14/2025

ISSUE DATE: 02/21/2025
PROJECT NO.: 300000331343
DESIGNED BY: DPL
DRAWN BY: DPL
CHECKED BY: DPL
SIZE:

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH
820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

Details

SHEET ID
C5

General Notes

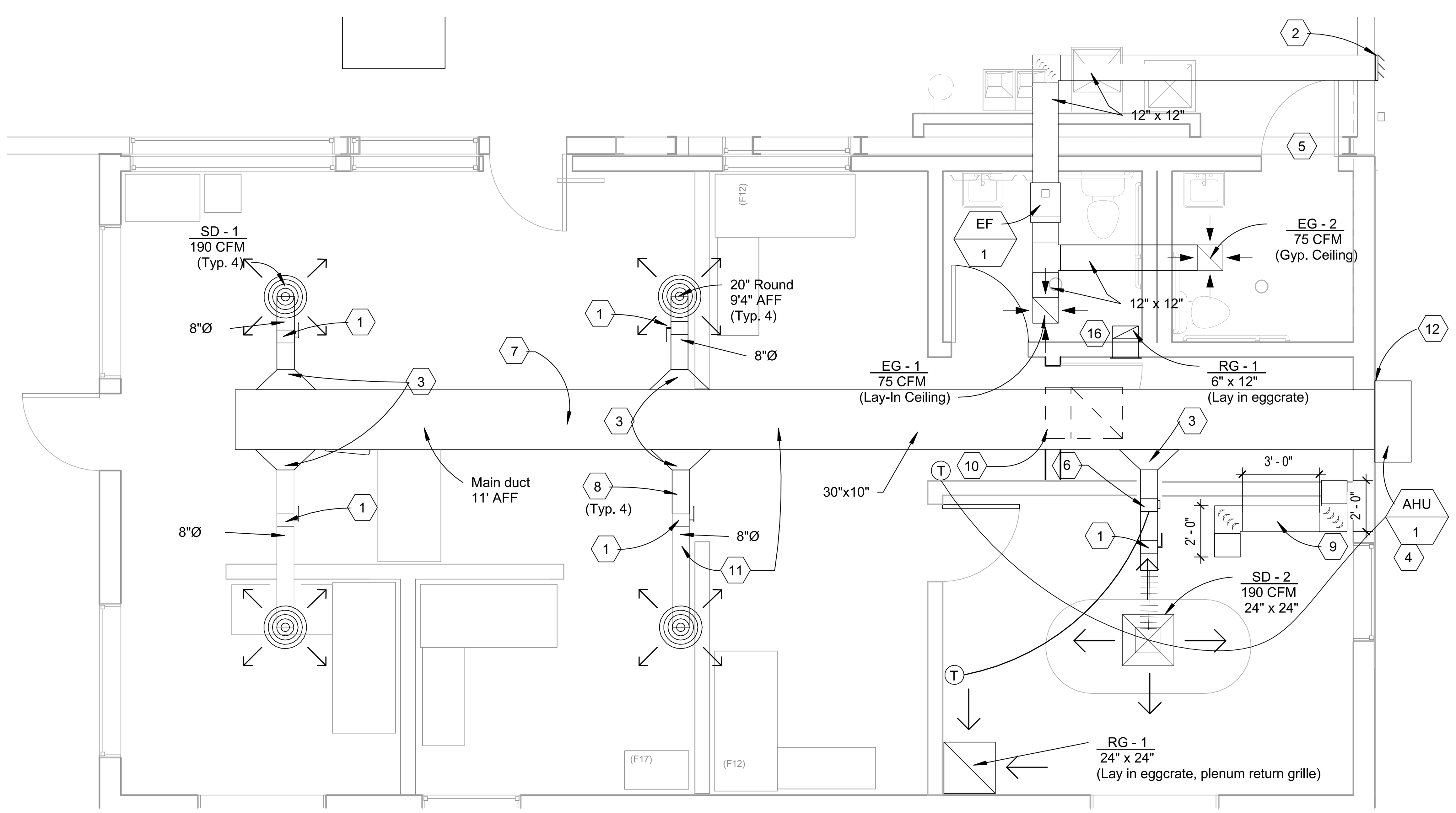
1. Confirm duct path and final diffuser locations with Government prior to installation.
2. Provide test and balance for all supply air diffusers as indicated. Airflows shall be within 10%.
3. The 24"x24" diffuser is a 4 way diffuser, the round diffusers are 360° diffusers.
4. Seal ductwork as called out below using hardcast DT tape and FTA-20 adhesive or hardcast AFG-1402 "FOIL GRIP" or similar per manufacturers instructions. Seal to SMACNA class A.

Type of Duct	Apply to Joints
Exhaust Duct (Rectangular)	Transverse and Longitudinal
Supply (Rectangular)	Transverse and Longitudinal
Supply (Round)	Transverse and Longitudinal

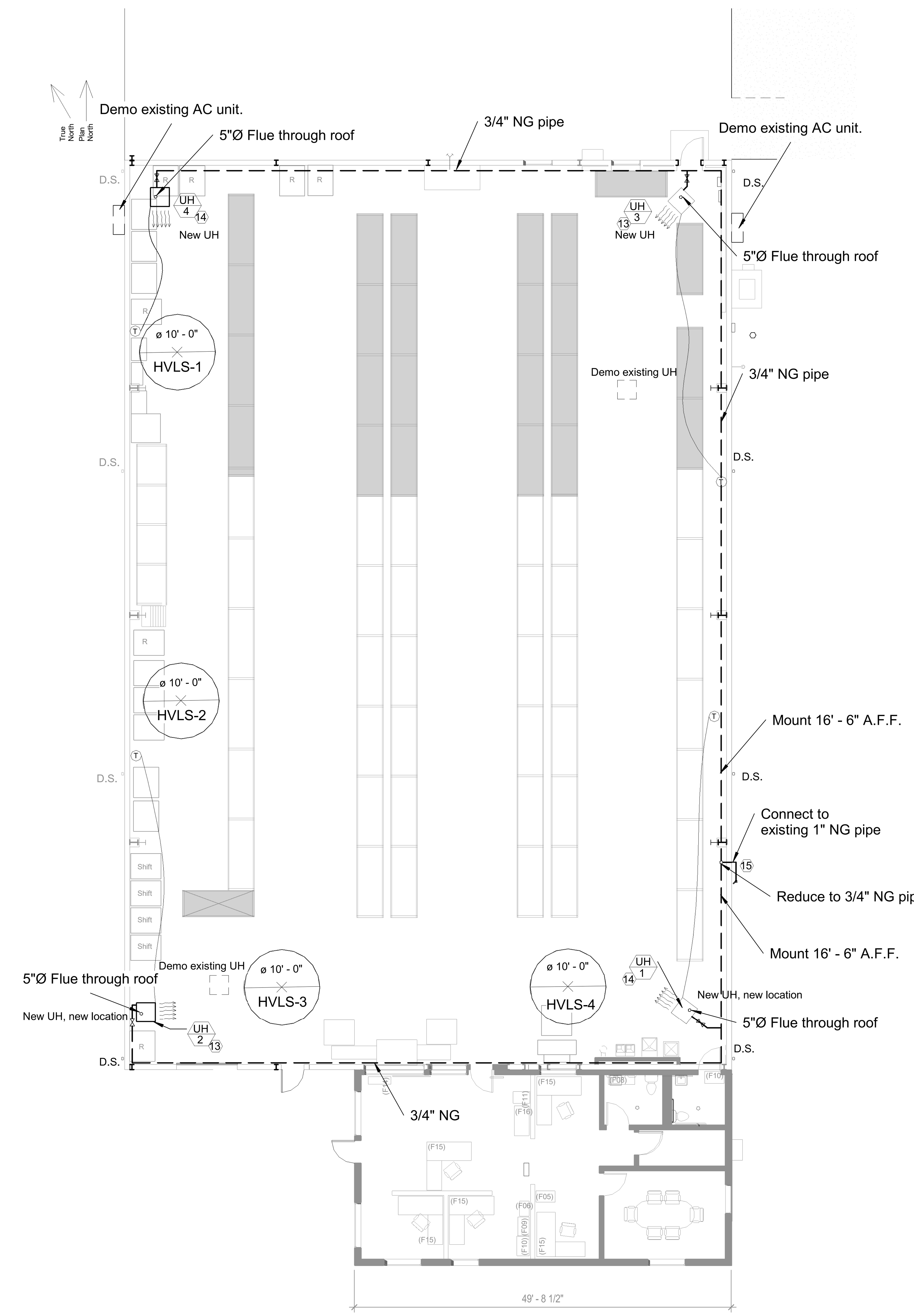
5. All work is to conform to applicable UFC, Building codes and standards.
6. Flexible ductwork is allowed on runouts to supply diffusers only. Utilize only above lay-in accessible ceilings. Do not install flex duct above hard ceilings or where exposed. A maximum length of 5'-0" may be used at each connection.
7. Coordinate all work with other trades prior to installation.
8. Coordinate routing of plumbing, and HVAC piping with ductwork, lights, architectural ceiling and structural elements. Piping shall rise and drop, jog or offset as required to avoid conflicts. Ductwork shall take precedence over all piping, except where grade must be maintained for drainage. Rework of installed work to resolve conflicts rising from lack of coordination shall not justify an increase in the contract amount.
9. Temperature controls contractor shall furnish and install all low voltage wiring and associated conduit required for mechanical controls system. Wiring shall be in conduit inside walls, in rooms with exposed ceilings, and above hard ceilings. Line voltage wiring and associated conduit shall be provided and installed by E.C. Control system shall be installed in accordance with specifications.
10. Contractor to install temporary filters over all return and exhaust grilles in work area during construction. Contractor shall provide new filters at time of turnover.

Keyed Notes

1. Provide round air balance dampers as indicated.
2. Provide weatherproof backdraft damper for bathroom exhaust fan, as indicated.
3. Transition from 30" x 10" rectangular duct to 8" round duct.
4. VPAC shall include fresh air and dehumidification operations. Maintain 18" clearance from exterior door.
5. Undercut restroom door 1.5" for exhaust airflow. Ref. Architecture Floorplan.
6. Conference Room climate control to include zone damper control in addition to a manual air balance damper. Two position control damper shall modulate open when thermostat calls for heating or cooling. When zone is within +/- 2°F of the setpoint damper shall remain closed. Initial thermostat setpoints shall be 76°F/68°F (adj.) with 2°F adjustability.
7. Rectangular hard ducting, with internal duct insulation, typical. Paint to match exposed gypsum board ceiling. Match main ducting trunk slope with roof slope.
8. Round hard ducting connected to round supply diffusers. No flex ducting to be used in exposed ceiling areas. Internal duct insulation shall be used. Paint all exposed mechanical ducting and devices to match ceiling. Do not inhibit the operation of mechanical devices.
9. Provide Z-bend transfer duct above drop ceiling to provide return airflow from conference room to storage room. Provide ducting with acoustic media lining the interior of duct. Two 12"x12" transfer grilles and 3' rect. ducting.
10. Provide L-bend transfer duct above storage room door. Utilizing drop in eggcrate return grille, 24" x 24" in storage and a wall return grille above door 18" x 24".
11. All exposed ducting and diffusers shall be painted to match gypsum board on underside of the exposed roof.
12. Provide insulated weather-tight seal between unit and well metal siding.
13. Remove and dispose of existing unit heater and all associated piping, regulators, valves etc. Replace and relocate new unit heater as shown. (Re: M4 schedule). Provide and install natural gas piping as shown. Remove and dispose of existing unit heater flue ducts. Patch roof penetrations weather tight, match roof type. Install new unit heater concentric flue vent directly above units through roof. Seal all new penetrations weather tight.
14. Install additional new unit heater. Install new unit heater concentric flue vent directly above units through roof. Seal all new penetrations weather tight.
15. Provide new gas regulator and meter. Coordinate to ensure sufficient outlet pressure from regulator to UHs. Main natural gas supply pressure is ~40 psi. Re: Sheet M4, NG Detail.
16. Provide a "L" type transfer air duct assembly, to include acoustic dampening media lining. Toilet Room grille shall be 6"x12" drop ceiling lay-in eggcrate type and Storage Room grille shall be a 6"x12" wall mounted transfer grille



1	Floor Plan - Construction Mechanical
A3	1/4" = 1'-0"



2	Floor Plan - Construction
A3	3/32" = 1'-0"

FORT LEAVENWORTH
DPW

DATE
MARK

ISSUE DATE:
2/21/2025
PROJECT NO.:

100% Drawings

DESIGNED BY: TS
DRAWN BY: TS
CHECKED BY: JM
SIZE: ANSI 'D'

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH
820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

Mechanical Floorplan

SHEET ID

M1

VERTICAL PACKAGED AIR CONDITIONING UNIT (DESIGN BASED ON BARD W30AB-A10XNXXXE)

MARK	AIRFLOW (CFM)	Static Pressure (E.S.P.)	MIN COOLING CAPACITY (BTU/H)	MIN HEATING CAPACITY (kW)	Compressor Type	Indoor Motor HP	Motor Operating AMPS	Blower Speeds	ELECTRICAL		
									Volt	Ø	HZ
AHU-1	950	1"	30,000	10	Scroll	0.5	2.3	5	208/230	3	60

NOTES:
1. Unit to be mounted on the exterior of the building as indicated on drawings.
2. Provide local thermostat to control unit and have 7 day schedule capability.
3. Provide disconnect switch at each unit.
4. Unit to provide electric heating strip.
5. Provide supply duct path as annotated.
6. Provide wall supply/return diffuser/grilles as annotated.
7. Provide wall return grille.
8. Provide new MERV 13 filter media.
9. Provide low ambient control.
10. Provide dehumidification controls.
11. Provide fresh air intake.
12. Mount unit at such height that distribution ductwork extends straight through wall from HVAC unit outlet to the end of the run with NO 90° bends to minimize static pressure drop out of unit to each diffuser.

Exhaust Fan Schedule

Mark	Type	Min. Airflow (CFM)	E.S.P.	Fan Type	Max RPM	Drive	Electrical (V/Ø/Hz)			HP	Unit Location
EF-1	INLINE	150	0.15" W.G.	CENTRIFUGAL	1550	Direct	120	3	60	1/12	Restroom

Note:
1. Exhaust fan shall be integrated into the restroom lighting controls. Fan shall be enabled when either light switch is on. Coordinate operation with electrical sub.
2. Provide with separate disconnect switch.
3. Exhaust fan unit shall be maintenance accessible from a ladder in the restroom.

Unit Heater Schedule

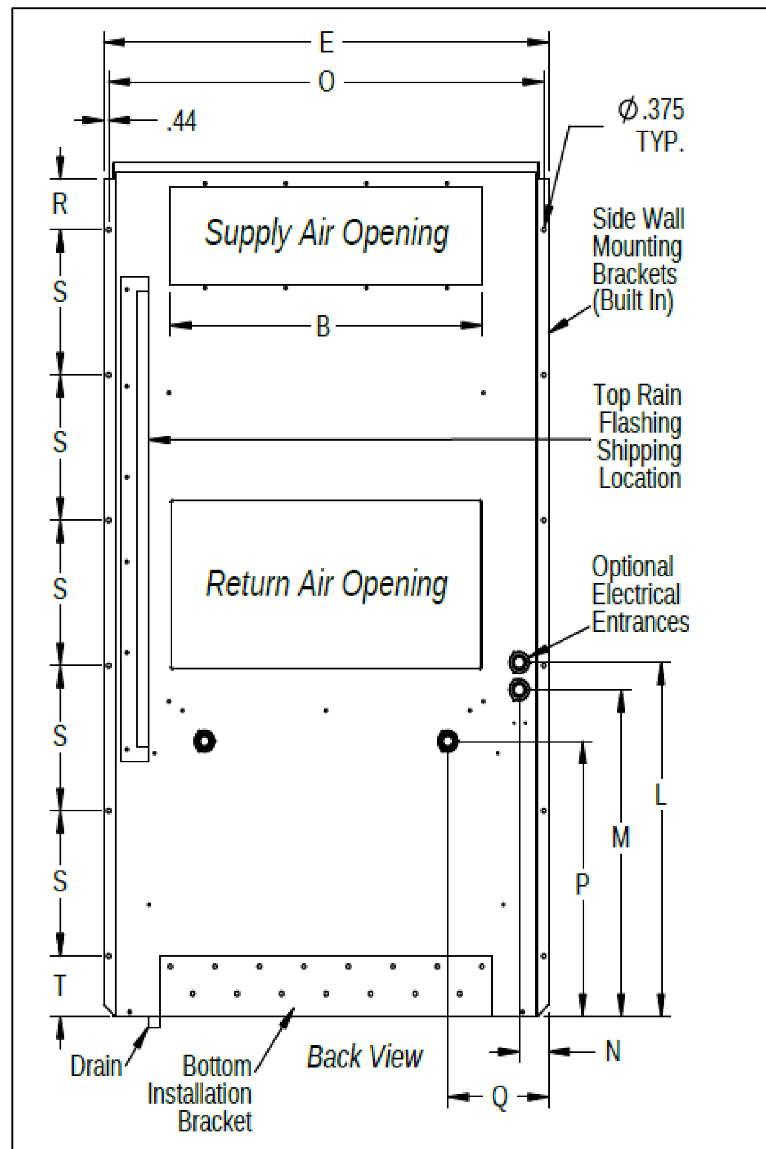
Mark	Type	Capacity	Supply Fan		Electrical			Unit Location
		Input/Output MBH	Airflow (CFM)	Motor (HP)	V	Ø	Hz	
UH-1	Horizontal Discharge	125.0/101.0	2200	1/4	115	1	60	Southeast
UH-2	Horizontal Discharge	125.0/101.0	2200	1/4	115	1	60	Southwest
UH-3	Horizontal Discharge	125.0/101.0	2200	1/4	115	1	60	Northeast
UH-4	Horizontal Discharge	125.0/101.0	2200	1/4	115	1	60	Northwest

Note:
1. Mount 120" A.F.F
2. Provide single stage control valve and integral disconnect.
3. Provide wall mounted adjustable thermostat for each unit.

High Volume Low Speed Fans

Mark	Type	Speed Settings	Diameter	Min/Max CFM	Motor Type	Electrical				Unit Location
						V	Ø	Hz	Amps	
HVLS-1	Ceiling	5	10'	1788/8944	Direct	120	1	60	0.6	Northwest
HVLS-2	Ceiling	5	10'	1788/8944	Direct	120	1	60	0.6	Central West
HVLS-3	Ceiling	5	10'	1788/8944	Direct	120	1	60	0.6	Southwest
HVLS-4	Ceiling	5	10'	1788/8944	Direct	120	1	60	0.6	Southeast

Note:
1. HVLS fans to have directional reversable ability.
2. Provide wall mounted control module for speed and direction adjustments.
3. Ensure fan downrods provide clearance of roof structure.
4. Coordinate all electrical requirements with electrical sub. Re: Electrical Sheets
5. All electrical wires shall be routed through EMT conduit. Re: Electrical Sheets

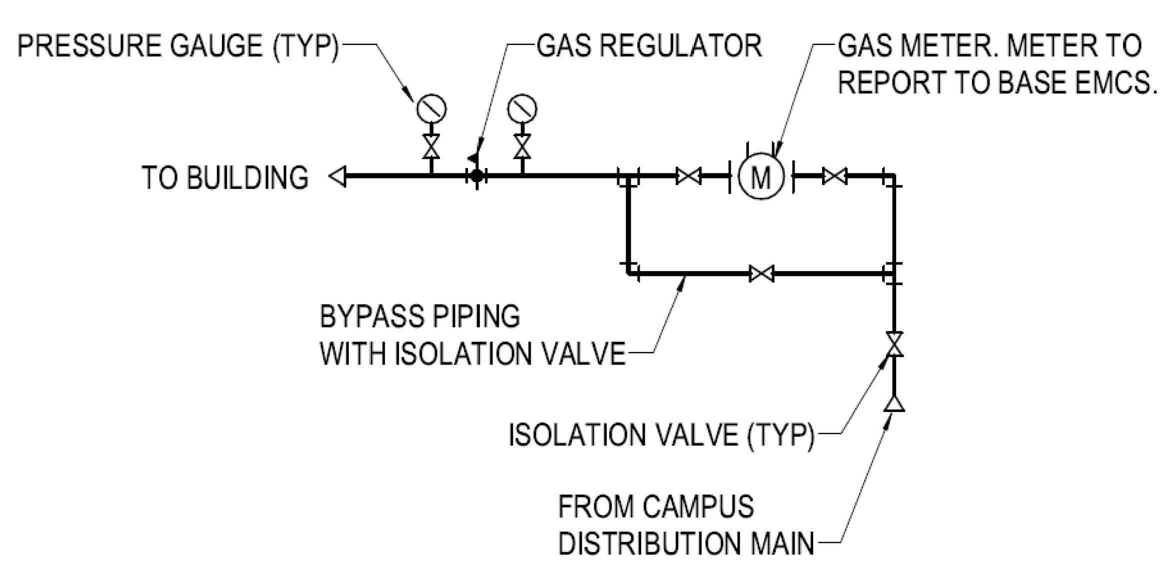


C4 Front View Detail AHU-1

N.T.S.

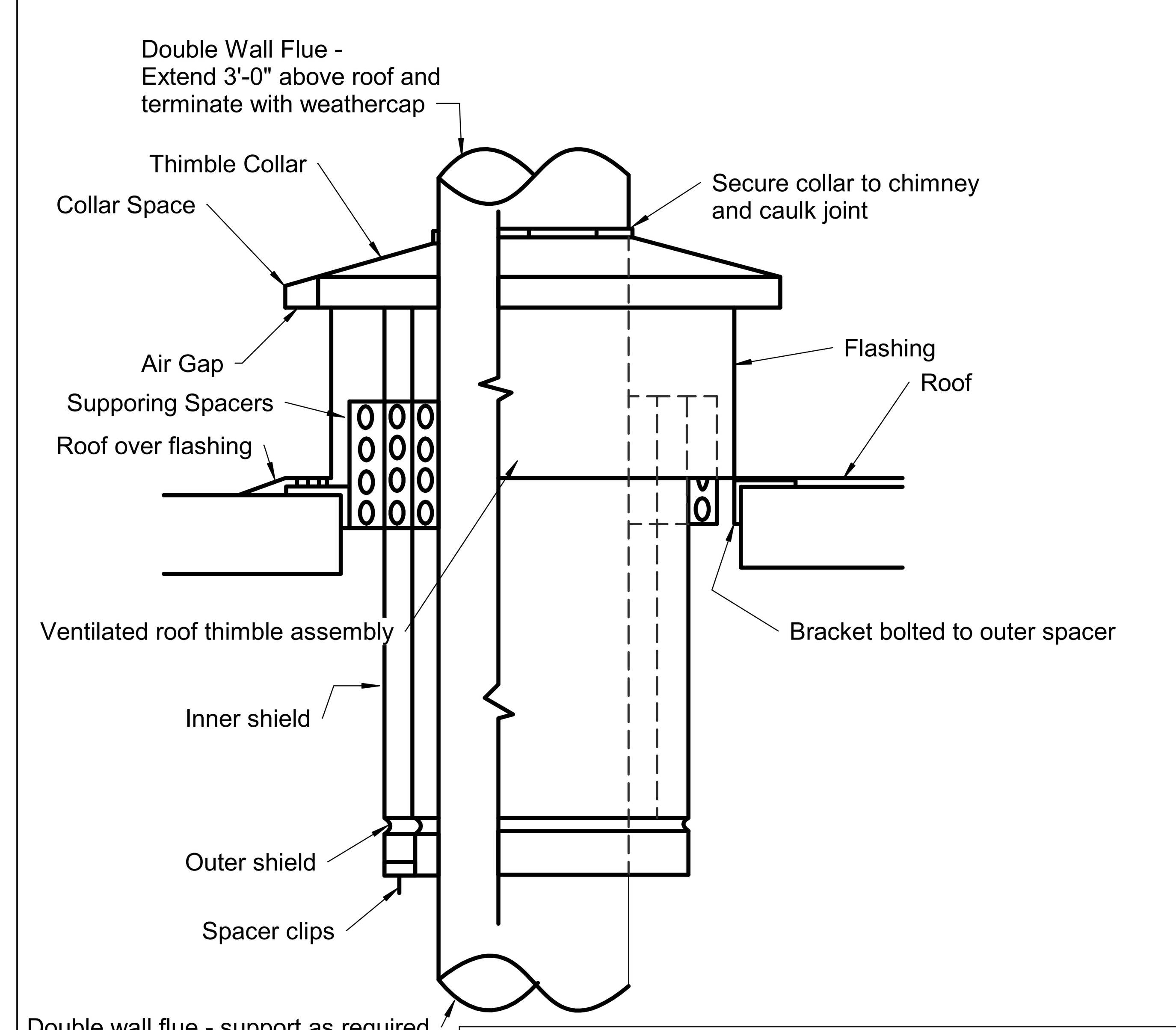
DIMENSIONS OF W18-36L BASIC UNIT FOR ARCHITECTURAL & INSTALLATION REQUIREMENTS (NOMINAL) (Basis of Design Bard)

MODEL	WIDTH (W)	DEPTH (D)	HEIGHT (H)	SUPPLY		RETURN																
				A	B	C	B	E	F	G	I	J	K	L	M	N	O	P	Q	R	S	T
W18LB W24LB	33.300	17.125	74.563	7.88	19.88	11.88	19.88	35.00	10.88	29.75	20.56	30.75	32.06	33.25	31.00	2.63	34.13	26.06	10.55	3.94	12.00	9.00
W30LB W36LB	38.200	17.125	74.563	7.88	27.88	13.88	27.88	40.00	10.88	29.75	17.93	30.75	32.75	33.25	31.00	2.75	39.13	26.75	9.14	3.94	12.00	9.00



A1 Gas Entrance Detail

N.T.S.



A3 UH Flue Penetration Detail

N.T.S.

FORT LEAVENWORTH DPW

DATE

MARK

ISSUE DATE: 2/21/2025

PROJECT NO.:

DESIGNED BY: TS

DRAWN BY: TS

CHECKED BY: JM

SIZE: ANSI 'D'

100% Drawings

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

820 McCLELLAN AVE.
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66027-1292

B1146 - Textiles Warehouse - Office Addition

Mechanical Details

SHEET ID

M2

Symbol	Description
	Supply Diffuser
	Airflow Control Damper
	Flex Ducting
	Exhaust Grille
	Exhaust Fan
T	Thermostat
	Balance Damper
	Round Diffuser
	Return Grille

GENERAL MECHANICAL SPECIFICATIONS

1. APPLICABILITY
   a. These general requirements apply to all divisions (22, 23, 26). Refer to individual divisions as included for specific information regarding each trade or scope of work.
2. GENERAL REQUIREMENTS
   a. Furnish & install all labor & materials required for complete, functioning, mechanical & plumbing systems w/ all associated equipment & apparatus as shown on plans.
   b. All materials shall be new & shall bare UL label where applicable.
   c. Visit site & observe conditions under which work will be done. Any discrepancies shall be called to government's attention. No subsequent allowance will be made in contract for any error or negligence on contractor's part.
   d. Final acceptance of work shall be subject to condition that all systems, equipment, apparatus & appliances operate satisfactorily as designed & intended. Work shall include required adjustment of systems & control equipment installed under these specifications.
   e. Warrant to owner quality of materials, equipment, workmanship & operation of equipment provided under these specifications for one year from & after completion of building & acceptance of mechanical systems by owner.
   f. The specifications & drawings for project are complementary, & portions of work described in one, shall be provided as if described in both. In event of discrepancies, notify engineer & request clarification prior to proceeding w/ work involved.
3. EXTENT OF CONTRACT WORK
   a. Provide MEP systems indicated on drawings, specified or reasonably implied. In addition to specific equipment called out in plans and specifications, provide every device, component, programming, interlocking and accessory necessary for proper operation and completion of totally functional MEP systems.
   b. In no case will claims for "Extra Work" be allowed for work about which Contractor could have been informed before bids were taken.
   c. Contractor shall become familiar with equipment provided by other contractors that require plumbing connections and controls.
   d. Electrical work required to install and control plumbing equipment, which is not shown on plans or specified under Division 26, shall be included in Contractor's base bid proposal.
   e. All automatic temperature control devices shall be mounted as indicated on the mechanical floorplan sheets.
   f. The cost of larger wiring, conduit, control and protective devices resulting from installation of equipment which was not used for basis of design as outlined in specifications shall be paid for by the supplying Contractor at no cost to the government.
   g. Contractor shall be responsible for providing supervision to other trade Contractors to ensure that required connections, interlocking and interconnection of MEP equipment is made to attain intended control sequences and system operation.
   h. Contractor shall obtain complete MEP data on shop drawings and shall list this data on an approved form that shall be presented on request, to other trade Contractors. Data shall be complete with wiring diagrams received to date and shall contain necessary data on electrical components of plumbing equipment such as HP, voltage, amperes, watts, locked rotor current to allow other trade Contractors to order support or other equipment coordinated as required in his contract.
4. MATERIAL & WORKMANSHIP
   a. Provide new material, equipment. & apparatus under this contract unless otherwise stated herein. Of best quality normally used for purpose in good commercial practice & free from defects. Model numbers listed in specifications or shown on drawings are not necessarily intended to designate required trim, written descriptions of trim govern model numbers.
5. ARCHITECTURAL VERIFICATION AND RELATED DOCUMENTS
   a. Contractor shall consult all Architectural Drawings and specifications in their entirety incorporating and certifying all millwork, furniture, and equipment rough-in including utility characteristics such as voltage, phase, amperage, pipe sizes, duct sizes, including height, location and orientation. Shop drawings incorporating these requirements should be submitted to the Architect for approval prior to installation or rough in.
6. ORDINANCES & CODES
   a. Work performed under this contract shall. At minimum, be in conformance with applicable Unified Facilities Criteria's (UFC), national, state & local codes having jurisdiction.
   b. Installation work performed under this contract shall be in strict compliance w/current applicable codes adopted by local AHJ including any amendments & standards as set forth by National Fire Protection Association (NFPA). Underwriters Laboratories (UL), Occupational Safety & Health Administration (OSHA). American Society of Mechanical Engineers (ASME), American Society of Heating, Refrigeration, & Air Conditioning Engineers (ASHRAE). American national standards institute (ANSI), American Society of Testing Materials (ASTM) & other national standards & codes where applicable.
   c. Where contract documents exceed requirements of referenced codes. Standards, etc., contract documents shall take precedence.
7. STANDARDS
   a. Drawings and specifications indicate minimum construction standard. Should any work indicated be substandard to any ordinances, laws, codes, rules or regulations bearing on work, Contractor shall promptly notify government in writing before proceeding with work so that necessary changes can be made. However, if the Contractor proceeds with work knowing it to be contrary to any ordinances, laws, rules, and regulations, Contractor shall thereby have assumed full responsibility for and shall bear all costs required to correct non complying work.
8. SHOP DRAWINGS
   a. Equipment to be furnished under this contract, items requiring coordination between contractors & sheet metal ductwork fabrication drawings. Before submitting shop drawings verify equipment submitted is mutually compatible & suitable for intended use & will fit available space & allow ample room for maintenance. Government checking & subsequent approval of such shop drawings will not relieve contractor from responsibility for errors in dimensions, details, size of members, quantities, omissions of components or fittings; coordination of electrical requirements; or for coordinating items with actual building conditions. Proceed with procurement & installation of equipment only after receiving approved shop drawings relative to each item.
   b. Submittal data shall be neatly organized, identified & indexed. Each item or model number shall be clearly marked & accessories indicated. Label catalog data with equipment identification acronym or number as used on drawings & include performance curves, capacities, sizes, materials, finishes, wiring diagrams & deviations from specified equipment or materials. Mark out inapplicable items. Shop drawings will be returned without review if above mentioned requirements are not met.
   c. Requirements shall be met electronically & submitted as pdf in files less than 10mb.
   d. Contractor's stamp, which shall certify that drawings have been checked by Contractor, comply with drawings & specifications, & have been coordinated with other trades, shall be affixed to each shop drawing prior to submitting.
   e. Transmit submittals as early as required to support project schedule. Allow for two weeks government review time, plus duplication of this time for resubmittals, if required. Transmit submittals as soon as possible after notice to proceed & before construction starts. Government's submittal reviews will not relieve contractor from responsibility for errors in dimensions, details, size of members, or quantities; or for omitting components or fittings; or for not coordinating items w/actual building conditions.
   f. Final copies shall be furnished to owner as part of O&M documents in hard & electronic formats.
9. OPERATION & MAINTENANCE INSTRUCTIONS
   a. Collect & compile complete brochure of equipment furnished & installed on this project. Include operational & maintenance instructions, manufacturer's catalog sheets, wiring diagrams, parts lists, approved shop drawings, test & balance reports, & descriptive literature as furnished by equipment manufacturer. Include an inside cover sheet that lists project name, date, Owner, Architect, consulting Engineer, general Contractor, sub-contractor, & an index of contents. Submit three copies of literature bound in 3-ring binders with index & tabs separating equipment types to architect at termination of work. Final approval of plumbing systems will be withheld until manual is received & deemed complete by the government. Provide "as-built" drawings at the completion of the work.
   b. These requirements shall also be provided to the owner in a well-organized pdf electronic submission & delivered on a DVD.
10. TRAINING
   a. Provide factory trained & authorized representative to train owner's designated personnel on operation & maintenance of equipment provided for this project. Provide training to include but not be limited to an overview of system &/or equipment as it relates to facility as whole; operation & maintenance procedures & schedules related to startup & shutdown, troubleshooting, servicing, preventive maintenance & appropriate operator intervention; & review of data included in operation & maintenance manuals. Submit certification letter to Government stating that owner's designated representative has been trained as specified herein. Letter shall include date, time, attendees & subject of training. Contractor & Owner's representative shall sign certification letter indicating agreement that training has been provided. Schedule Owner training with at least 7 days' advance notice.
11. SPARE PARTS
   a. Furnish to owner, with receipt one set of spare filters of each type required for each unit. In addition to spare set of filters, install new filters prior to testing, adjusting, & balancing work & before turning system over to owner.
12. EQUIPMENT LABELS:
   a. Material and thickness: multilayer, multicolor, plastic labels for mechanical engraving, 1/16 inch thick, and having predrilled holes for attachment hardware. Black letters on white background.
   b. Minimum label size: length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
   c. Minimum letter size: 1/4" for name of units if viewing distance is less than 24 inches, 1/2" for viewing distances up to 72" & proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
13. WARRANTIES
   a. Warrant each system & each element thereof against all defects due to faulty workmanship design or material for period of 12 months from date of substantial completion unless specific items are noted to carry longer warranty in construction documents or manufacturer's standard warranty exceeds 12 months. Remedy all defects, occurring within warranty period(s) stated in general conditions & Division 1. Warranties shall include labor & material. Make repairs or replacements without any additional costs to the Government. Perform remedial work promptly, upon written notice from Government.
   b. At time of substantial completion, deliver to Owner all warranties in writing & properly executed including term limits for warranties extending beyond one year period. Each warranty instrument being addressed to Government & stating commencement date & term.
14. CUTTING & PATCHING
   a. Perform cutting of walls, floors, ceilings, etc. As required to install work under this section. Obtain permission from architect prior to cutting. Do not cut or disturb structural members without prior approval from Architect. Cut holes as small as possible. General contractor shall patch walls, floors, etc. As required by work under this section. Patching shall match original material & construction. Repair & refinish areas disturbed by work to condition of adjoining surfaces in manner satisfactory to government.
   b. Pipe, fittings, specialties & valves shall be manufactured in USA. Work performed under this contract shall provide neat & "workmanlike" appearance when completed to satisfaction of the government. shall comply w/ applicable UFCs, building codes & laws. Complete installation shall function as designed & intended w/respect to efficiency, capacity, noise level. etc. Abnormal noise caused by rattling equipment, piping, ducts, air devices & squeaks in rotating components will not be acceptable.
   c. Remove from premises waste material present from work, including cartons, crating, paper, stickers, &/or excavation material not used.
   d. Clean equipment installed under this contract to present neat & clean i nstallation at completion.
   e. Repair or replace public & private property damaged as result of work performed under this contract to satisfaction of authorities & regulations having jurisdiction.
15. ROUGH-IN
   a. Coordinate rough-in with general construction & other trades. Conceal piping & conduit rough-in except in unfinished areas & where otherwise shown.
16. PENETRATIONS
   a. Seal mechanical floor, exterior wall & roof penetrations watertight & weathertight. Seal around mechanical penetrations w/ 3M CP-25 fire barrier caulk (thickness as required & recommended by manufacturer) to maintain resistance rating of fire-rated assemblies. Provide prefabricated roof curbs manufactured by Custom Curb, Pate, Thycurb or approved equal. Provide roof curb with factory installed wood nailer; welded, 18-gauge galvanized steel shell, base plate & flashing; 1-1/2" thick, 3 pound rigid insulation; fully mitered 3-inch raised cant; cover of weather-resistant, weather-proof material & pipe collar of weather-resistant material with stainless steel pipe clamps. Make roof penetrations by authorized roofing contractor when required.
17. MOTORS & STARTERS
   a. Provide motors & starting equipment where not furnished w/ equipment package. Motors shall have copper windings, class b insulation, & standard squirrel cage with starting torque characteristics suitable for equipment served. Motors for air handling equipment shall be selected for quiet operation. Each motor shall be checked for proper rotation after electrical connection has been completed. Provide drip proof enclosure for locations protected from weather & not in air stream of fan; & totally enclosed fan cooled enclosure for motors exposed to weather. Provide every motor, except fractional horsepower single phase motors w/ an approved type of "built-in" thermal overload protection, w/ motor starter. Each starter shall be provided w/ overload heaters sized to motor rating, & every three-phase motor starter shall have overload heaters in each phase. Ambient compensated heaters shall be installed wherever necessary. Unless noted otherwise, motor starters shall be furnished by Division 22/23 contractor for installation & connection by Division 26 contractor.
18. ELECTRICAL WIRING
   a. Line voltage wiring shall be provided by Division 26. Line voltage control & interlock wiring for mechanical systems shall also be provided by Division 26 contractor. Low voltage control wiring shall be provided by Division 22/23 contractor. Furnish wiring diagrams to Division 26 contractor as required for proper equipment hookup. Coordinate w/ Division 26 contractor actual wire sizing amps for submitted mechanical equipment to ensure proper installation
19. DISCONNECT SWITCHES
   a. Provide heavy-duty horsepower rated safety switches rated in accordance with NEMA enclosed switch standard KS 1_1969 and I98 standard.
   b. Each piece of electrical equipment shall be provided with a disconnecting means.
20. EQUIPMENT FURNISHED BY OTHERS
   a. Provide necessary equipment & accessories that are not provided by equipment supplier or Government to complete installation. Equipment & accessories not provided by equipment supplier may include flues, vents, intakes, associated roof jacks & caps to outdoors, dampers. In-line fans, roof fans, control interlocks, etc. As required for proper operation of complete system in accordance w/manufacturer's instructions. Contractor shall be responsible for correct rough-in dimensions and shall verify same with Government &/or equipment supplier prior to service installations.
21. SETTING, ADJUSTMENT AND EQUIPMENT SUPPORTS
   a. Work shall include mounting, alignment and adjustment of systems and equipment. Set equipment level on adequate foundation and provide proper anchor bolts and isolation as shown, specified or required by manufacturers in installation instructions. Level, shim and grout equipment bases as recommended by manufacturer. Mount motors, align and adjust drive shafts and belts according to manufacturer's instructions.
   b. Equipment failures resulting from improper installation or field alignment shall be repaired or replaced by Contractor at no cost to government.
   c. Floor or pad mounted equipment shall not be held in place solely by its own dead weight. Include anchor fastening in all cases.
   d. Provide floor or slab mounted equipment with 3-1/2" high concrete bases unless specified otherwise. Individual concrete pad shall be no less than 4" wider and 4" longer than equipment and shall extend no less than 2" from each side of equipment.
   e. Provide each piece of equipment or apparatus suspended from ceiling or mounted above floor level with suitable structural support, platform, or carrier in accordance with best-recognized practice. Verify that structural members of buildings are adequate to support equipment and unless otherwise indicated on plans or specified, arrange for their inclusion and attachment to building structure. Provide hangers with vibration isolators.
   f. Submit details of hangers, platforms and supports together with total weights of mounted equipment to government for review before proceeding with fabrication or installation.
22. MISCELLANEOUS REMODELING WORK
   a. Remove all unused equipment, ductwork, piping & associated supports. Cap ductwork & piping at mains & seal air and watertight. Provide items of HVAC systems modification required because of building remodeling, as noted on drawings or necessary for proper operation. Match existing materials & construction techniques when modifying existing systems unless specified otherwise. Coordinate additional requirements with general contractor & government. Seal airtight existing ductwork required to be abandoned in place or not in use at termination of work. Cap & seal weathertight existing roof curbs & roof openings to be abandoned in place as result of equipment removal. Clean & rebalance existing ductwork, diffusers, registers, & grilles intended for reuse as required or as indicated on drawings.
23. BUILDING OPERATION
   a. Comply with schedule of operations as outlined in Architectural portions of this specification. Building shall be in continuous operation. Accomplish work requiring interruption of building operation at time when building is not in operation, & only with written approval of Government. Coordinate interruption of building operation with government minimum of 14 days in advance of work.
   b. The following Work shall be performed at night or weekend other than holiday weekends as directed and coordinated with the government: All tie-in, cut-over and modifications to the existing electrical system and other existing system requiring tie-ins or modifications shall be arranged and scheduled with the Government to be done at a time as to maintain continuity of the service and not interfere with normal building operations.
24. VIBRATION ISOLATION
   a. Provide vibration isolation equipment & materials by single manufacturer. Amber booth, kinetics noise control, mason industries, inc., vibration eliminator co., inc., & vibration mounting & controls. General requirements: select vibration isolators by weight distribution to produce uniform deflection. Isolators shall operate in linear portion of their load versus deflection curves. Spring isolators shall have 50 percent excess capacity without becoming coil bound. Coat vibration isolators with factory-applied paint. Coat vibration isolators exposed to weather & corrosion with factory-applied protection. Install & adjust isolators in accordance w/manufacturers instructions.

FORT LEAVENWORTH DPW

MARK	DATE

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

DESIGNED BY: TS	ISSUE DATE: 2/21/2025	
DRAWN BY: TS	PROJECT NO.:	
CHECKED BY: JM		
SIZE: ANSI 'D'	100% Drawings	

B1146 - Textiles Warehouse - Office Addition

Mechanical Specifications

SHEET ID

M3

DIVISION 230000 – MECHANICAL

1. MECHANICAL GENERAL REQUIREMENTS
    a. Refer to GENERAL MECHANICAL, ELECTRICAL & PLUMBING requirements.
2. SHEET METAL WORK
    a. HVAC ductwork shall be galv sheet metal of gauges & joint types specified in SMACNA manual. Provide turning vanes in elbows.
    b. Coordinate routing of ductwork w/ other contractors such that piping, electrical conduit, & associated supports are not routed through ductwork. Construct supply ducts to meet SMACNA positive pressure of 3" WG. Construct return, outdoor & exhaust ductwork upstream of fans to meet SMACNA negative pressure of 1" WG. construct exhaust ductwork downstream of fans to meet SMACNA positive pressure of 1" WG.
    c. Exposed ductwork to be field painted shall have galvanized metal primer applied in shop after fabrication & prior to shipping.
    d. Seal ductwork w/ heavy liquid sealant, applied according to sealant manufacturer's instructions.
    e. Exposed spiral duct shall be gasketed style.
    f. Ducts shall be connected to fans, fan casings & fan plenums by means of flexible connectors. Flexible connectors shall be neoprene coated glass cloth canvas connections. Flexible connectors shall have flame spread of 25 or less & smoke developed rating not higher than 50. Make airtight joints & install w/ minimum 1-1/2" slack.
    g. All ductwork must be supported properly from structure.
3. DUCTWORK SPECIALTIES
    a. Flexible ducts – shall be sound rated type G-KM insulated. (duct w/o published acoustical attenuation ratings not acceptable). Take off fitting shall be hi-eff style w/ locking damper. Maximum length of flexible ductwork shall be 5'-0".
    b. Diffusers & grilles - see schedule. Coordinate color, mounting w/ duct, ceilings, with government. Select air devices to limit room noise level to no higher than NC-30 unless otherwise shown. Provide devices w/ soft plastic gasket to make an airtight seal against mounting surface. Coordinate final location, frame, & mounting type of air devices w/ architectural reflected ceiling plans. Submit complete shop drawings including information on noise level, pressure drop, throw, cfm for each air device, styles, borders, etc. Clearly marked with specified equipment number. Provide ceiling supply air diffusers & return air grilles of lay-in or surface mounted type as required to be compatible w/ ceiling construction. Provide ceiling diffusers & grilles w/ white enamel finish unless noted otherwise. Provide slot plenums by diffuser manufacturer. Plenums shall be internally insulated by manufacturer.
    c. Provide balancing dampers where shown on drawings & wherever necessary for complete control of air flow. Splitter dampers shall be controlled by locking quadrants; provide end bearings for damper rod. Rectangular volume dampers shall be opposed blade interlocking type. Round volume dampers shall be butterfly type consisting of circular blade mounted to shaft.
    d. Damper leakage for outside air dampers shall not exceed 6.5 cfm/square foot in full closed position at 4" wg pressure differential across damper. Reference manufacturer & model number for outside air dampers.
4. DUCT INSULATION WORK
    a. Duct insulation & wraps shall meet flame/smoke rating of 25/50 per ASTM E 84.
    b. Line all low-pressure supply, return and transfer air ductwork with 1/2" liner. Line all medium pressure supply with 1" liner.
    c. Line all transfer boots w/ ½" liner.
    d. Do no wrap exposed spiral ducts. Provide pre-manufactured 1/2" or 1" round liner for all exposed round ducts. Contractor has the option to use double wall perforated lined round spiral ducts for exposed ducts. Wrap all concealed round supply HVAC ductwork with 1-1/2" thick insulation with vapor barrier in concealed locations or in unfinished shell spaces.
    e. Wrap all outside air HVAC ductwork w/ 1-1/2" thick insulation w/vapor barrier in concealed locations. Exposed installations shall use 1-1/2" thick rigid board insulation or lined with 1" liner.
5. PIPING
    a. COPPER TUBING
        1. Provide copper tubing and fittings with an ANSI/ASME Class 125 service rating, which for 66 degrees C, the pressure rating is 1207 kPa 150 degrees F., the pressure rating is 175 psig.
        2. Tube
            1. Use copper tube conforming to ASTM B88M, ASTM B88, Type L or M for aboveground tubing, and Type K for buried tubing.
        3. Fittings and End Connections (Solder and Flared Joints).
            1. Wrought copper and bronze solder joint pressure fittings, including unions and flanges, shall conform to ASME B16.22 and ASTM B75/B75M. Provide adapters as required. Cast copper alloy solder-joint pressure fittings , including unions and flanges, shall conform to ASME B16.18. Cast copper alloy fittings for flared copper tube shall conform to ASME B16.26 and ASTM B62. ASTM B42 copper pipe nipples with threaded end connections shall conform to <RID>ASTM B42. Copper tubing of sizes larger than 100 mm 4 inches shall have brazed joints. Brass or bronze adapters for brazed tubing may be used for connecting tubing to flanges and to threaded ends of valves and equipment. Extracted brazed tee joints may be used if produced with an acceptable tool and installed in accordance with tool manufacturer's written procedures.
        4. Grooved Mechanical Connections For Copper
            1. Rigid grooved mechanical connections may only be used in serviceable aboveground locations where the temperature of the circulating medium does not exceed 110 degrees C 230 degrees F. Flexible grooved connections shall be used only as a flexible connector with grooved pipe system. Unless otherwise specified, grooved piping components shall meet the corresponding criteria specified for the similar welded, flanged, or threaded component specified herein. Each grooved mechanical joint shall be a system, including coupling housing, gasket, fasteners, all furnished by the same manufacturer. Joint installation shall be in compliance with joint manufacturer's written instructions. Grooved fitting and mechanical coupling housing shall be ductile iron conforming to. Provide gaskets for use in grooved joints shall constructed of molded synthetic polymer of pressure responsive design and shall conform to ASTM for circulating medium up to 110 degrees C 230 degrees F. Provide grooved joints in conformance with AWWA C606.
        5. Solder
            1. Provide solder in conformance with ASTM B32, grade Sb5, tin-antimony alloy. Solder flux shall be liquid or paste form, non-corrosive and conform to ASTM B813.
        6. Brazing Filler Metal
            1. Filler metal shall conform to AWS A5.8/A5.8M, Type BAg-5 with AWS Type 3 flux, except Type BCuP-5 or BCuP-6 may be used for brazing copper-to-copper joints.
6. VALVES
    a. Provide valves with a ANSI/ASME Class 125 service rating, which for 66 degrees C, the pressure rating is 1207 kPa 150 degrees F, the pressure rating is 175 psig. Valves in sizes larger than 25 mm 1 inch and used on steel pipe systems, may be provided with rigid grooved mechanical joint ends. Such grooved end valves shall be subject to the same requirements as rigid grooved mechanical joints and fittings and, shall be furnished by the same manufacturer as the grooved pipe joint and fitting system.
    b. Gate valves
        1. Gate valves 65 mm 2-1/2 inches and smaller shall conform to MSS SP-80 Class 125 and shall be bronze with wedge disc, rising stem and threaded, soldered, or flanged ends. Gate valves 80 mm 3 inches and larger shall conform to MSS SP-70, Class 125, cast iron with bronze trim, outside screw and yoke, and flanged or threaded ends.
    c. Check Valve
        1. Check valves 65 mm 2-1/2 inches and smaller shall conform to MSS SP-80. Check valves 80 mm 3 inches and larger shall conform to MSS SP-71, Class 125.
    d. Butterfly Valve
        1. Butterfly valves shall conform to MSS SP-67, Type 1 and shall be either the wafer or lug type. Valves smaller than 200 mm 8 inches shall have throttling handles with a minimum of two locking positions. Valves 200 mm 8 inches and larger shall have totally enclosed manual gear operators with adjustable balance return stops and position indicators.
    e. Ball Valve
        1. Ball valves 15 mm 1/2 inch and larger shall conform to MSS SP-72 or MSS SP-110 and shall be cast iron or bronze with threaded, soldered, or flanged ends. Valves 200 mm 8 inches or larger shall be provided with manual gear operators with position indicators. Ball valves may be provided in lieu of gate valves.
7. EXHAUST FANS
    a. Bearings shall be designed for 200,000 hours operation. Variable pitch motor sheaves shall be standard.
    b. Fans shall be furnished with acceptable electrical disconnect. Provide single phase motor equipped fans with motor rated start relay. Provide single phase motor equipped fans with magnetic motor starter. Provide local disconnect means for all fans. Coordinate location of starter & disconnects with other trades.
8. PROGRAMMABLE THERMOSTATS
    a. Stages of cooling & heating as required by stages on specified equipment. 7-day programming capability w/ 2 occ/unocc periods/day. Auto heat/cool change over. Locking setpoints to prevent tampering. Provide w/ all interfaces to other equipment as required.
9. MECHANICAL EXECUTION
    a. Coordinate w/ e/c to provide all wiring between equipment, dampers, thermostats & all other required controls & devices. mechnical contractor is responsible for all control & interlock wiring unless specifically shown on electrical drawings. All electrical work shall comply w/ electrical specifications.
    b. All piping shall be properly supported with hangers & supports specifically intended for that purpose. Provide clevis hangers, unistrut brackets & pipe clamps & similar systems. Protect integrity of insulation & provide rigid insulation inserts or pipe saddles as necessary.
    c. All exterior control wiring shall be in conduit.
    d. Provide any required interfaces to fire alarm or similar systems.
    e. Provide factory-authorized service start up on equipment. Train owner's maintenance personnel on startup, shutdown, troubleshooting, servicing, preventive maintenance.
    f. Provide clean filters at time of project turnover.
10. FINAL TESTING & ADJUSTMENTS
    a. Final system Testing, Balancing & adjustments shall be performed by contractor certified by NEBB, AABC or other approved agency. Perform test readings on fans, units, coils, etc. & adjust equipment to deliver specified amounts of air or water. Prepare testing & balancing report log showing air supply quantities, air entering & leaving temperatures & pressures, fan & unit test readings, motor voltage & amp draws. etc., & submit PDF of final compilation of data to Architect for evaluation & approval before final inspection of project. Balance air systems to within plus or minus 10 percent for terminal devices & branch lines & plus or minus 5 percent for main ducts & air handling equipment of amount of air shown on drawings. Further adjustments shall be made to obtain uniform temperature in spaces. Adjust equipment to operate as intended by specification. Align bearings & replace bearings that have dirt or foreign material in them w/ new bearings without additional cost to owner. Balance contractor shall include in report any improperly installed or missing balancing devices that would negatively impact system operation. Adjust thermostats & control devices to operate as intended. Adjust burners, pumps, fans, etc. For proper & efficient operation. Certify to government that adjustments have been made & that system is operating satisfactorily. Further adjustments shall be made to obtain uniform temperature in spaces. Calibrate, set, & adjust automatic temperature controls. Check proper sequencing of interlock systems, & operation of safety controls. Verify clean filters are installed.
    b. TAB Reports
        1. After completion of the TAB field work, prepare the TAB field data for TAB supervisor's review and certification, using the reporting forms approved in the pre-field engineering report. Data required by those approved data report forms is to be furnished by the TAB team. Except as approved otherwise in writing by the Contracting Officer, the TAB work and thereby the TAB report is considered incomplete until the TAB work is accomplished to within the accuracy range specified in the Final Systems Testing paragraph.
        2. Prepare the report neatly and legibly; the pre-final TAB report is the final TAB report minus the TAB supervisor's review and certification. Obtain, at the contract site, the TAB supervisor's review and certification of the TAB report.
        3. Verbally notify the COR that the field check of the TAB report data can commence; give this verbal notice 48 hours in advance of field check commencement with the approval of the DPW COR. Do not schedule field check of the TAB report until the specified workmanship requirements have been met or written approval of the deviations from the requirements have been received from the Contracting Officer.
13. STARTUP SERVICE
    a. Engage a factory-authorized service representative to perform startup service for all equipment & systems.
    b. Complete installation & startup checks according to manufacturer's written instructions. Maintain written records of all startup activities & also do the following:
        1. Inspect for visible damage to any part, casing or component.
        2. Verify that labels are clearly visible.
        3. Verify service clearances are provided.
        4. Verify that controls are connected & operable.
        5. Verify that filters are installed.
        6. Clean all interior and exterior components of construction debris.
        7. Release and adjust vibration isolators.
        8. Inspect all rotating components for direction and correct.
        9. Start unit according to manufacturer's written instructions.
        10. Inspect & record performance of interlocks & protective devices; verify sequences.
        11. Calibrate thermostats, sensors and similar equipment.
        12. Inspect controls for correct sequencing of heating, mixing dampers, refrigeration, & normal & emergency shutdown.
        13. After startup & performance testing, change filters, vacuum heat exchanger & cooling & outside coils, lubricate bearings, adjust belt tension, & inspect operation of power vents.
        14. Provide one spare set of clean filters & deliver to owner.
    c. Adjusting
        1. Adjust initial temperature & humidity set points.
        2. Set field-adjustable switches & circuit-breaker trip ranges as indicated.
        3. Occupancy adjustments: when requested within 12 months of date of substantial completion, provide on-site assistance in adjusting system to suit actual occupied conditions. Provide up to two visits to site outside normal occupancy hours for this purpose, without additional cost.
    d. Demonstration
        1. Engage a factory-authorized service representative to train government's maintenance personnel to adjust, operate, & maintain all HVAC equipment & systems.

END OF DIVISION 230000

FORT LEAVENWORTH DPW

DATE	MARK

ISSUE DATE: 2/21/2025
PROJECT NO.:
100% Drawings
DESIGNED BY: TS
DRAWN BY: TS
CHECKED BY: JM
SIZE: ANSI 'D'

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

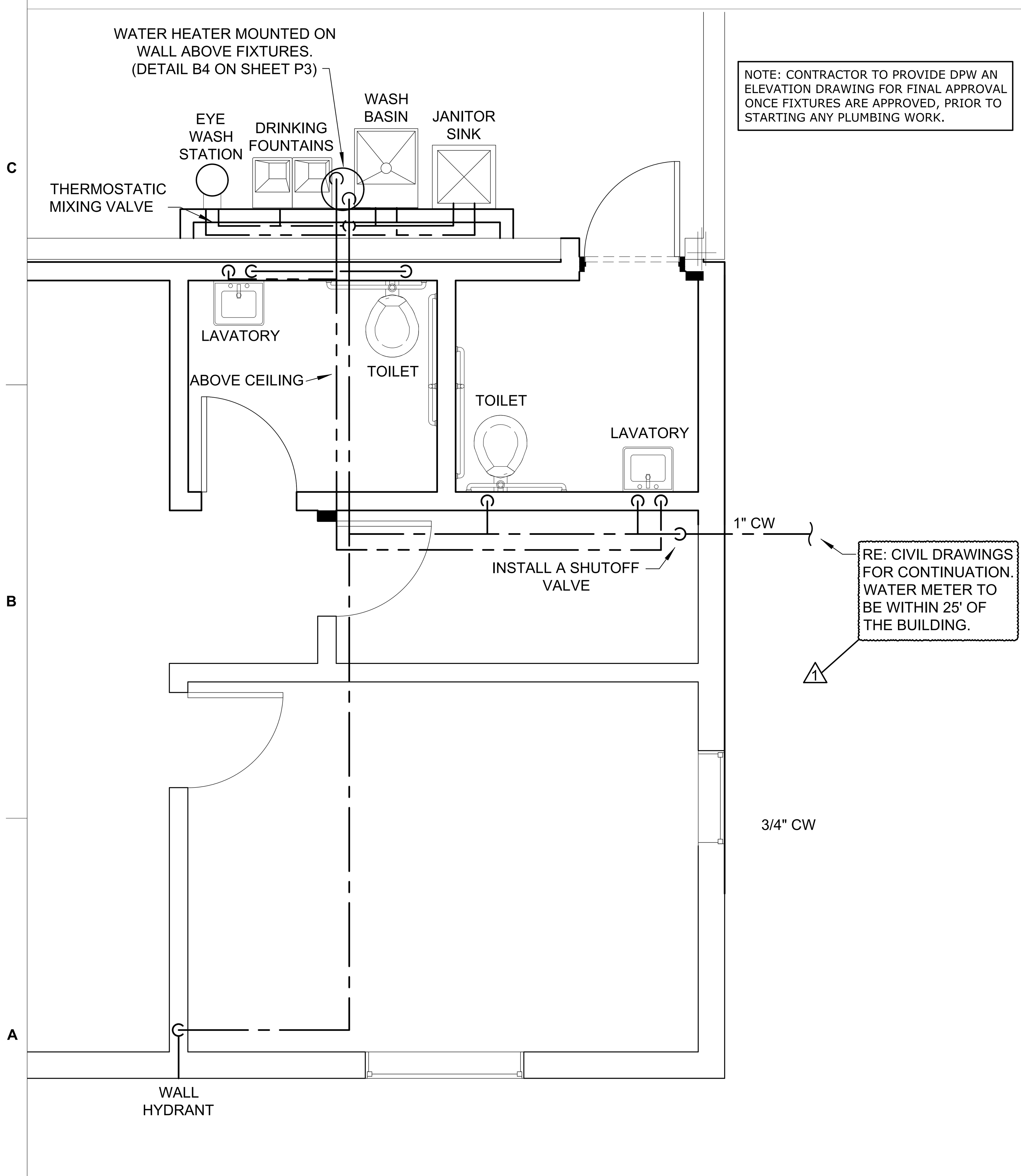
Mechanical Specifications Cont.

SHEET ID

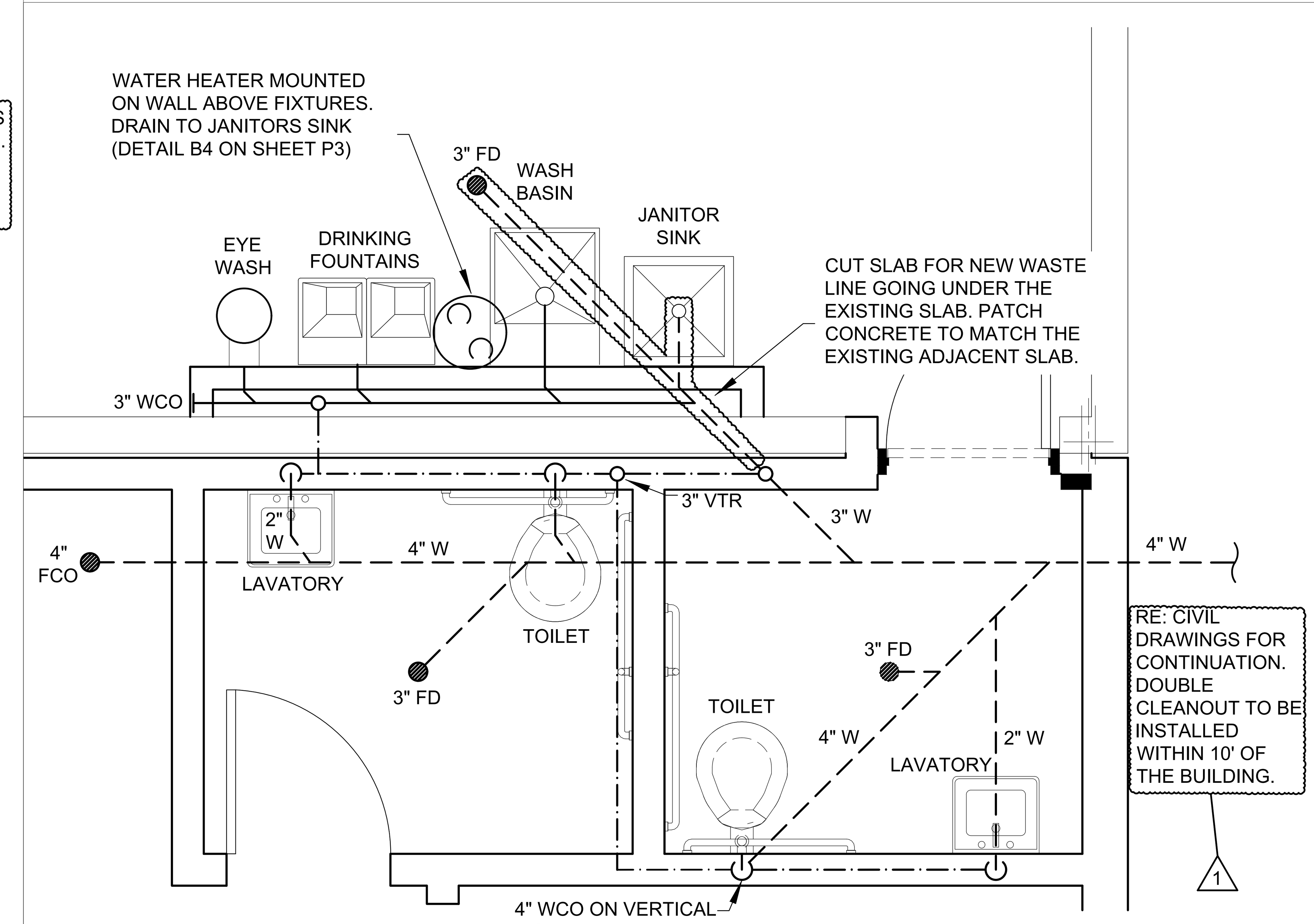
M4

PLUMBING FIXTURE SCHEDULE (BASIS OF DESIGN)

ITEM	MANUFACTURER & TYPE	FINISH/COLOR	CONNECTIONS			
			CW	HW	WASTE	VENT
TOILET	AMERICAN STANDARD "AFWALL MILLENIUM", FLUSHOMETER, WALL MOUNT, 1.1-1.6 GPF, TOP SPUD, RE: ARCHITECTURAL	WHITE	1"	-	4"	2"
LAVATORY	AMERICAN STANDARD 17" DEEP WALL-MOUNT RECTANGULAR TRADITIONAL BATHROOM SINK, FAUCET PROVIDED AND INSTALLED BY CONTRACTOR, RE: ARCHITECTURAL	WHITE	1/2"	1/2"	1 1/2"	1 1/2"
DRINKING FOUNTAINS	ELKAY - EMABFTL8WSLK - EZH20 - BOTTLE FILL, WITH CANE DETECTABLE APRON ON HIGH FOUNTAIN. CONTRACTOR TO PROVIDE P-TRAP, RE: ARCHITECTURAL	GRAY	1/2"	-	1 1/2"	1 1/2"
EYE WASH STATION	SPEAKMAN SE-582, TRADITIONAL SERIES WALL-MOUNTED EYEWASH STATION IN STAINLESS STEEL	STAINLESS	1/2"	1/2"	1 1/2"	1 1/2"
WASH BASIN	MUSTEE UTILATUB 23" W X 23-1/2" D X 33" H, WHITE THERMOPLASTIC FLOOR-MOUNT LAUNDRY/UTILITY SINK WITH FAUCET 93.600	WHITE	1/2"	1/2"	1 1/2"	1 1/2"
JANITORS SINK	FIAT NO. TSB-100 FLOOR SERVICE BASIN. 24" X 24" X 12" PRECAST TERRAZZO FLOOR SERVICE SINK WITH STAINLESS STEEL CAP, SPLASH PLATES ON THE WALL AROUND THE FLOOR SINK, 3" VANDAL PROOF CAST BRASS DRAIN AND STAINLESS STEEL STRAINER PLATE. FAUCET PROVIDED AND INSTALLED BY CONTRACTOR.	WHITE	3/4"	3/4"	3"	2"
WATER HEATER	RHEEM ELECTRIC WATER HEATER "EGSP15", 120V, 15 GALLON, 3000 W, RE: DETAIL B4 SHEET P3 1	-	3/4"	3/4"	-	-
WALL HYDRANT	ZURN Z1321-C, EXPOSED ANTI-SIPHON AUTOMATIC DRAINING NON-FREEZE WALL HYDRANT, 18" MOUNTING HEIGHT	-	3/4"	-	-	-



A2 P1 Enlarged Bathroom Domestic Water Plan
3/8" = 1'-0"



A3 P1 Enlarged Bathroom Waste and Vent Plan
1/2" = 1'-0"

GENERAL NOTES

1. ALL WORK IS TO CONFORM WITH APPLICABLE UFC, BUILDING CODES AND STANDARDS.
2. FIELD VERIFY ALL THE EXISTING CONDITIONS AND DIMENSIONS PRIOR TO STARTING WORK.
3. REFER TO ARCHITECTURAL AND CIVIL DRAWINGS FOR RELATED INFORMATION. REFER TO CIVIL FOR DOUBLE CLEANOUT LOCATION.
4. ALL EQUIPMENT AND PLUMBING COMPONENTS SHALL BE INSTALLED WITH ADEQUATE CLEARANCE TO PERMIT ACCESS FOR OPERATION AND MAINTENANCE.
5. ACCESS PANELS SHALL BE PROVIDED FOR ALL CONCEALED EQUIPMENT, CLEANOUTS, VALVES, AND TRAP PRIMERS INSTALLED IN NON-ACCESSIBLE CEILINGS OR WALLS.
6. GRAVITY DRAIN PIPING SHALL BE SLOPED AT A MINIMUM OF ⅛" PER FOOT FOR PIPES 3" OR GREATER AND ¼" PER FOOT FOR 3" AND SMALLER.
7. ALL DOMESTIC WATER PIPING SHALL BE SLOPED DOWN TOWARDS POINTS OF DRAINAGE.
8. FURNISH AND INSTALL PROPERLY SIZED WATER HAMMER ARRESTORS ON DOMESTIC HOT AND COLD WATER PIPING AT EACH FIXTURE BATTERY AND AT THE END OF ALL MAINS AND MAJOR BRANCHES. SHOCK ABSORBERS SHALL BE SIZED, SELECTED, MANUFACTURED AND LOCATED IN ACCORDANCE WITH PLUMBING AND DRAINAGE INSTITUTE (PDI) STANDARD PD-WH201.
9. CLEANOUTS WITH PLUGS SHALL BE INSTALLED PER THE PLANS, AND AT NO GREATER THAN 100 FT. INTERVALS, AT ALL CHANGES OF DIRECTION AND AT THE BASE OF ALL RISERS.
10. AT A MINIMUM, SHUTOFF VALVES SHALL BE INSTALLED ON COLD AND HOT WATER SUPPLY PIPING AT ALL FIXTURES AND EQUIPMENT.
11. EXISTING PIPING, FIXTURES, AND EQUIPMENT THAT ARE NOT TO BE REUSED SHALL BE REMOVED AND SHALL REMAIN THE PROPERTY OF THE OWNER IF THEY WISH TO RETAIN OWNERSHIP. IF NOT, THE EQUIPMENT SHALL BECOME THE PROPERTY OF THE CONTRACTOR AND DISPOSED OF ACCORDINGLY.
12. COORDINATE WITH AMERICAN WATER FOR ALL EXTERIOR PLUMBING WORK. ALL EXTERIOR PLUMBING MUST BE DONE IN ACCORDANCE WITH AMERICAN WATER STANDARDS.
13. DOMESTIC WATER PIPE SHALL BE COPPER. SANITARY PIPE SHALL BE PVC. REFER TO AMERICAN WATER STANDARDS FOR EXTERIOR PIPE MATERIAL.
14. EYE WASH STATION SHALL PROVIDE TEPID WATER (60°-100°F).

LEGEND

BELOW FLOOR WASTE PIPING
ABOVE FLOOR WASTE PIPING
COLD WATER PIPING
HOT WATER PIPING
VENT PIPING
PIPE DOWN
PIPE UP
PIPE TEE DOWN
PIPE TEE UP
WALL CLEANOUT
FLOOR CLEANOUT
FLOOR DRAIN

ABBREVIATIONS

W	WASTE
VTR	VENT THROUGH ROOF
FD	FLOOR DRAIN
WCO	WALL CLEANOUT
FCO	FLOOR CLEANOUT

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE
1	Addendum #1	3/12/26

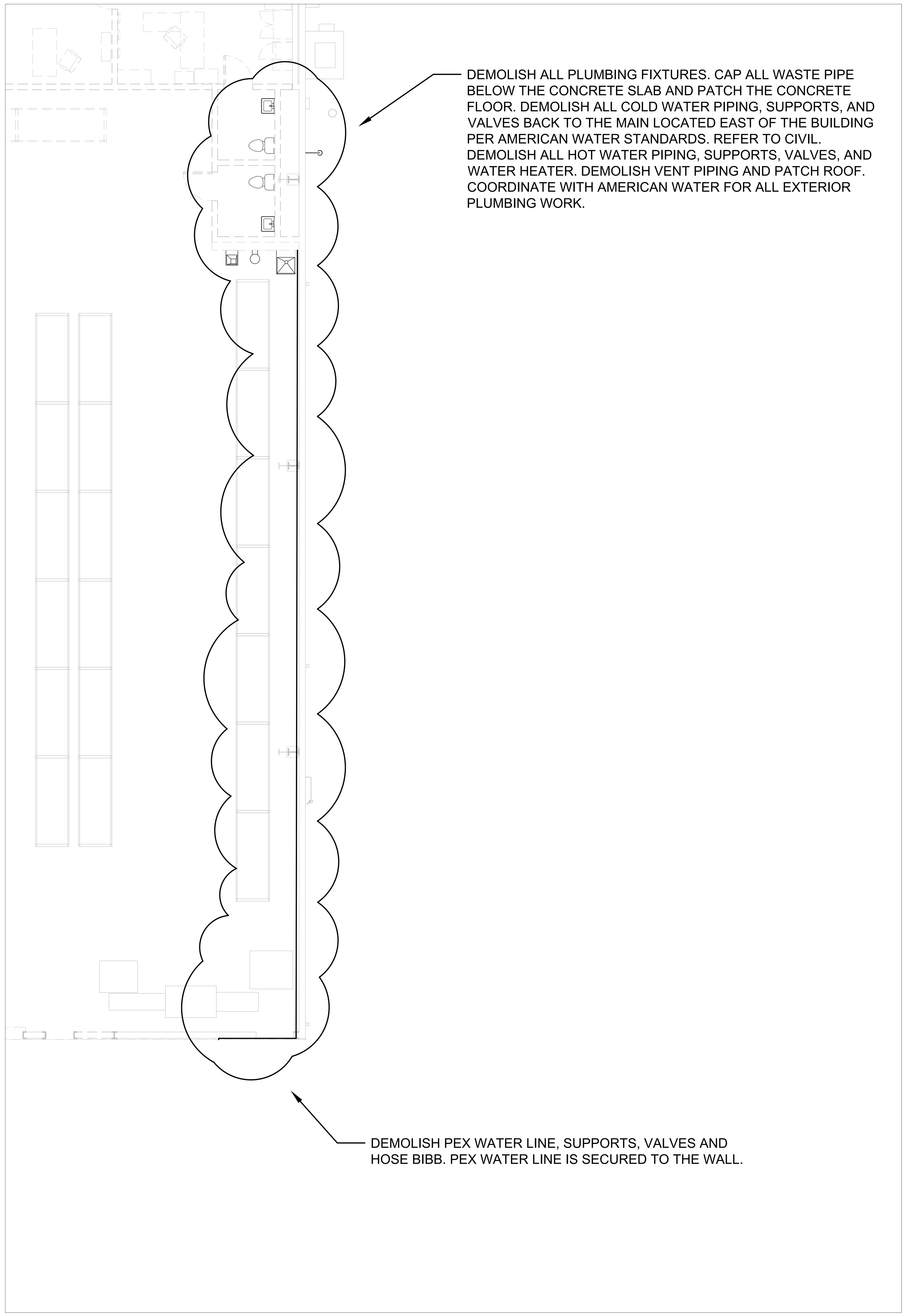
ISSUE DATE: 2/21/2025
PROJECT NO.: 300000331343
DESIGNED BY: NET
DRAWN BY: NET
CHECKED BY: SER/JCM
SIZE: ANSI 'D'

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH
820 McCLELLAN AVE,
FORT LEAVENWORTH, KS
66027-1292

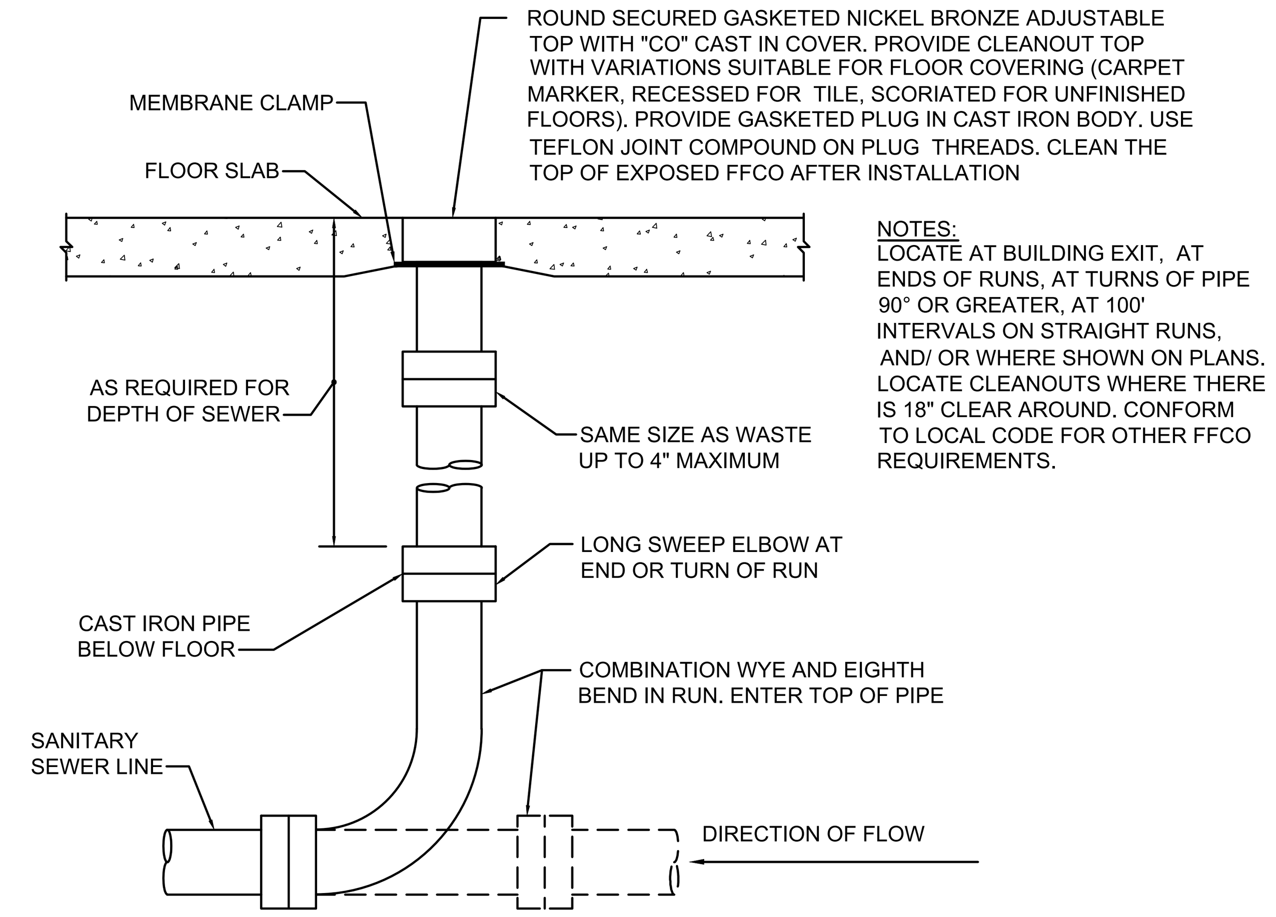
B1146 - Textiles Warehouse - Office Addition
PLUMBING

SHEET ID
P1

RTA SUBMITTAL

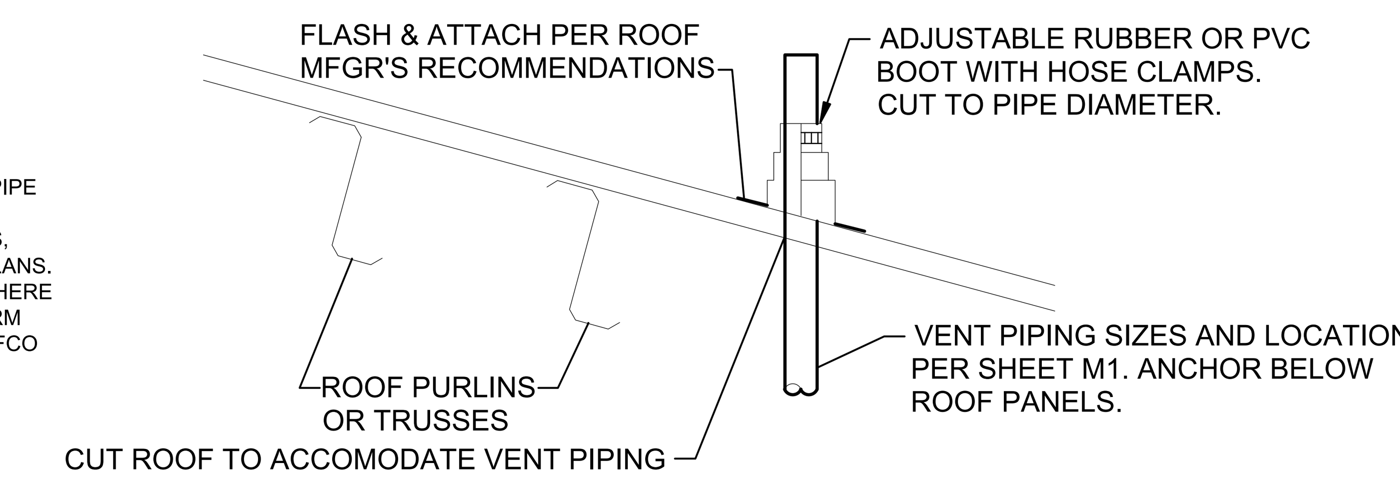


A1 PLUMBING DEMOLITION
P2 1/8" = 1'-0"



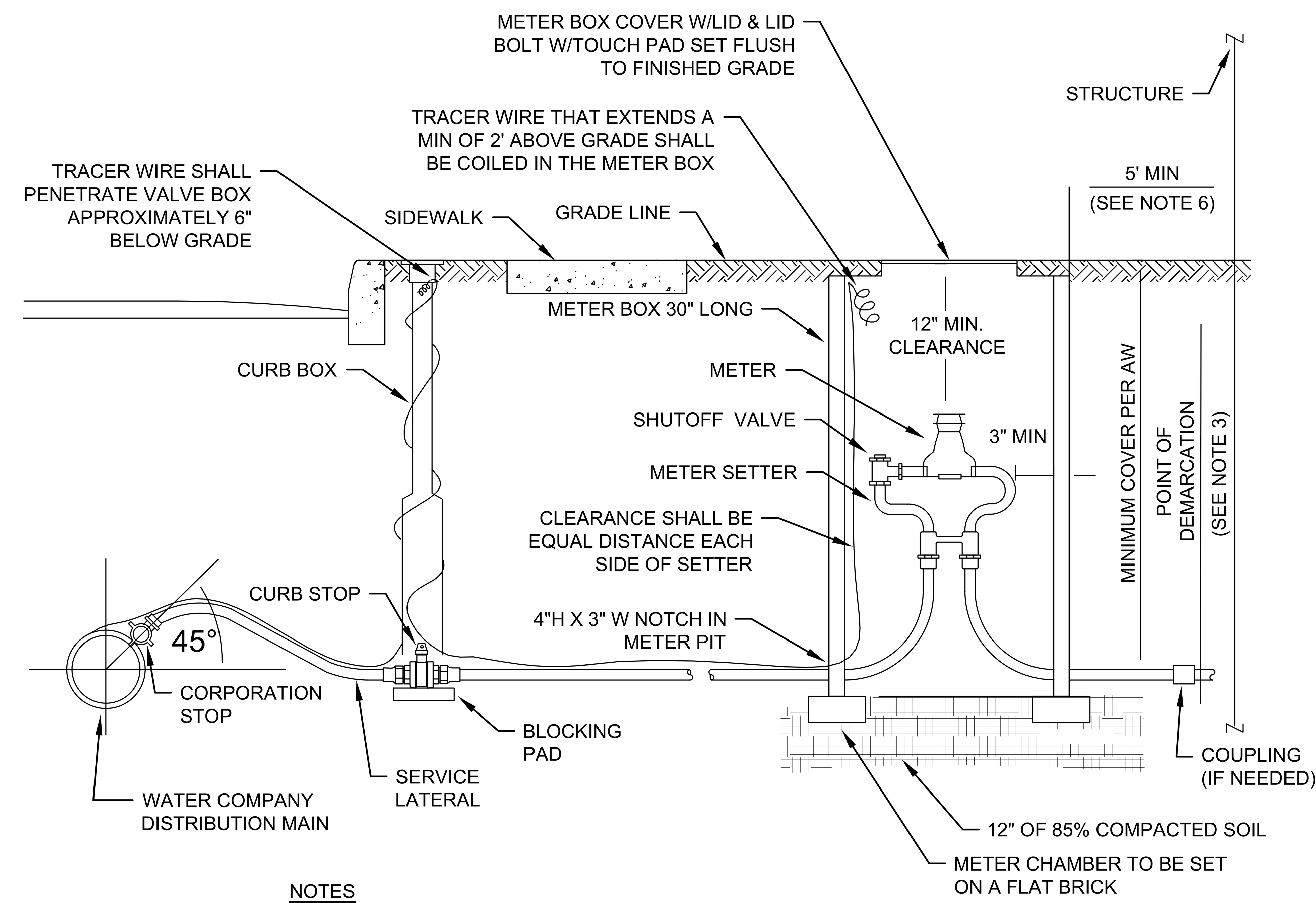
NOTES:
LOCATE AT BUILDING EXIT, AT ENDS OF RUNS, AT TURNS OF PIPE 90° OR GREATER, AT 100' INTERVALS ON STRAIGHT RUNS, AND/ OR WHERE SHOWN ON PLANS. LOCATE CLEANOUTS WHERE THERE IS 18" CLEAR AROUND. CONFORM TO LOCAL CODE FOR OTHER FFCO REQUIREMENTS.

C3 FLOOR CLEANOUT DETAIL
P2 NTS



NOTE: FLASH ALL PENETRATIONS WEATHERTIGHT.

D4 VENT THRU ROOF DETAIL
P2 NTS



NOTES
1. CENTER METER IN BOX, MIN 3" CLEARANCE ON ALL SIDES.
2. TRENCH, INSTALL, AND BACKFILL SERVICE LINE IN ACCORDANCE WITH AW SPECIFICATIONS.
3. POINT OF DEMARCATION FOR SERVICE LINE OWNERSHIP PER PRIME CONTRACT BETWEEN AW AND GOVERNMENT.
4. SERVICE LINE MATERIALS SHALL MEET AW SPECIFICATIONS.
5. PROVIDE TRACE WIRE ALONG WATER SERVICE LINE. TRACE WIRE SHALL EXTEND INTO THE CURB BOX AND METER PIT AND TERMINATE IN THE METER PIT.
6. IF WATER SERVICE BETWEEN MAIN AND WATER METER IS REPLACED WITH NON-METALLIC PIPING, THERE SHALL BE A MINIMUM LENGTH OF TEN (10) FT OF METALLIC PIPING BETWEEN THE BUILDING STRUCTURE AND METER PIT TO MAINTAIN PROPER GROUNDING OF BUILDING'S ELECTRICAL SYSTEM.

A3 WATER METER DETAIL
P2 NTS

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE

DESIGNED BY: NET
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SIZE: ANSI 'D'
ISSUE DATE: 2/21/2025
PROJECT NO.: 300000331343

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

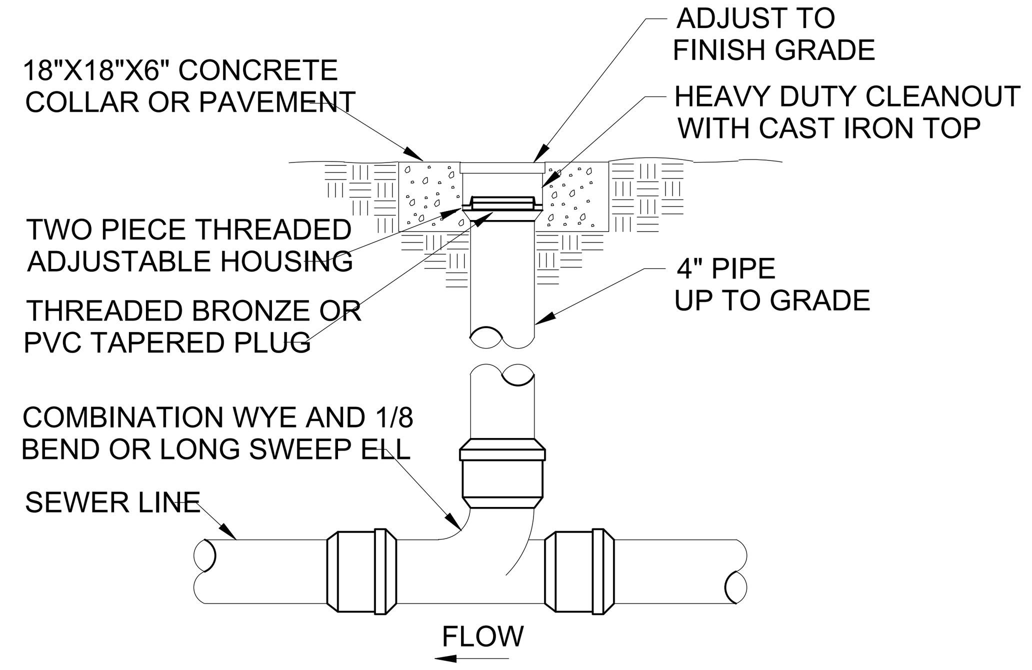
820 McCLELLAN AVE,
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

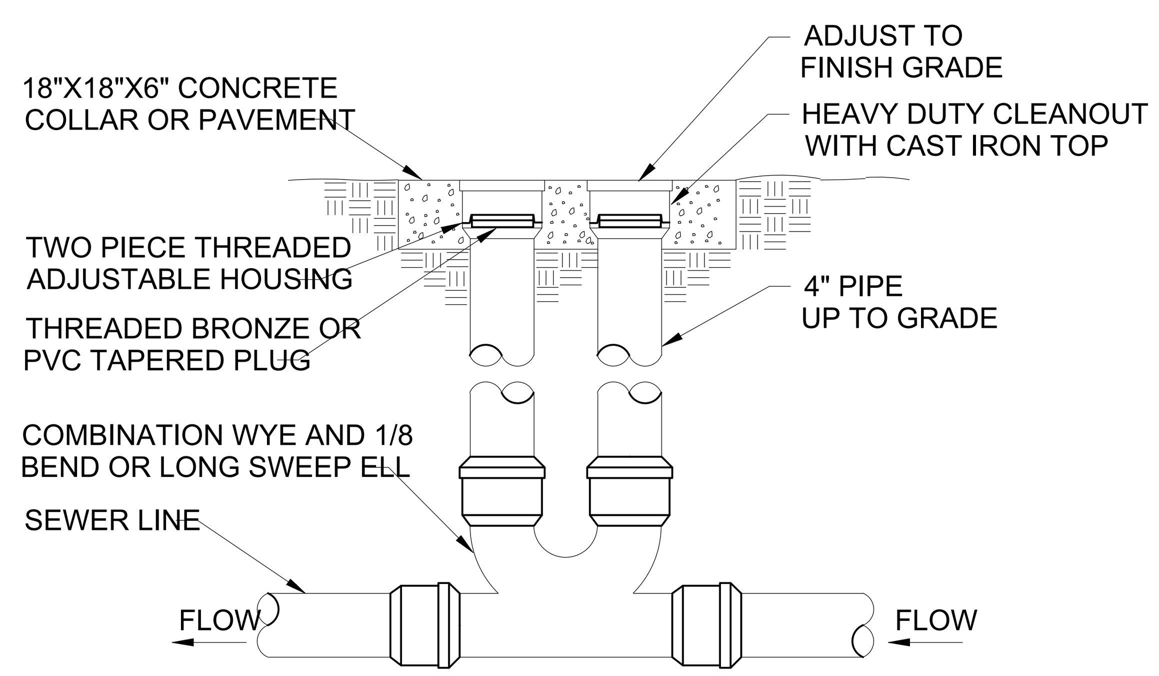
PLUMBING DEMOLITION AND DETAILS

SHEET ID

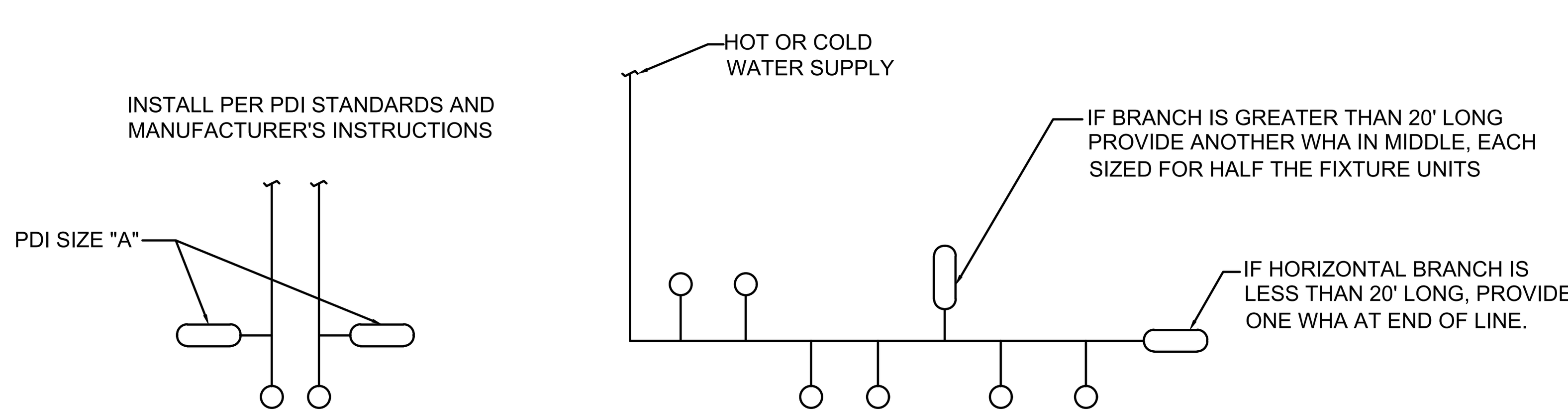
P2



D2	CLEANOUT DETAIL - EXTERIOR
P3	NTS



B2	CLEANOUT DETAIL - TWO-WAY EXTERIOR
P3	NTS

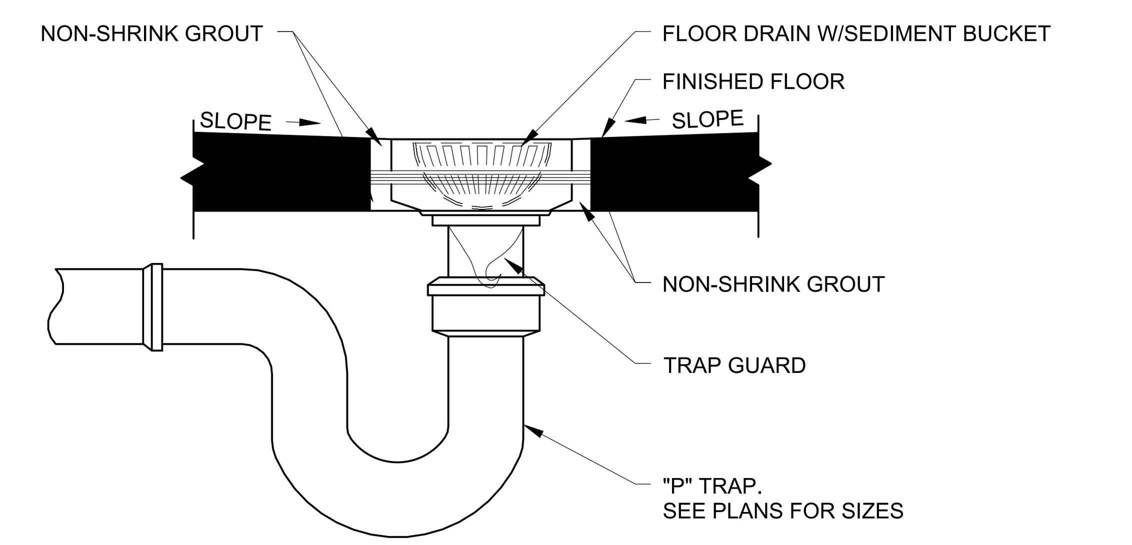


PDI SIZE	PIPE SIZE	FIXTURE UNIT LOAD
A	1/2"	1-11
B	3/4"	12-32
C	1"	33-60
D	1 1/4"	61-113
E	1 1/2"	114-154
F	2"	155-330

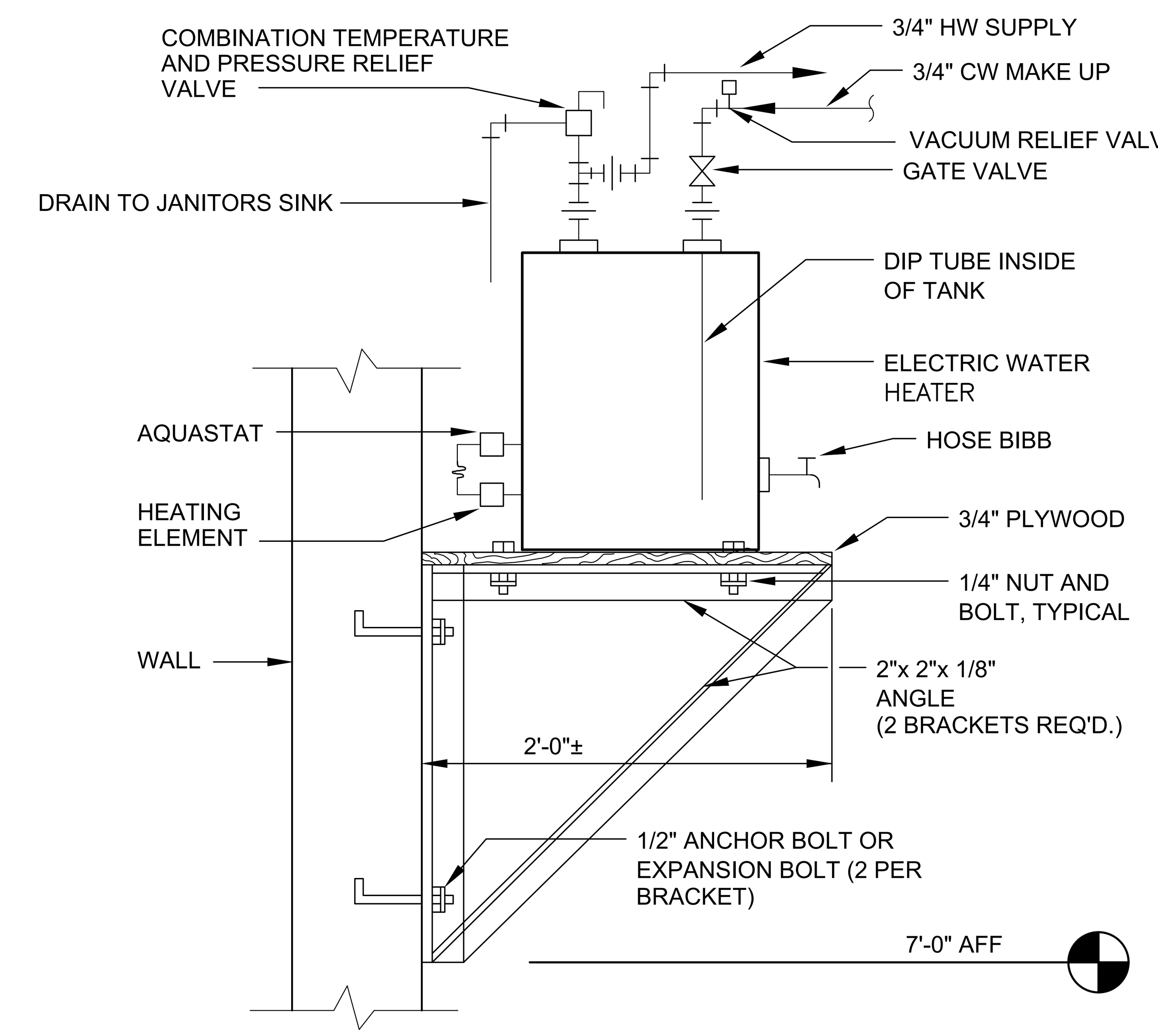
FIXTURE UNIT TABULATION		
FIXTURE	COLD	HOT
VALVE WATER CLOSET	10	--
URINAL	5	--
LAVATORY/SINK	1.5	1.5
JANITOR'S SINK	3	3
SHOWER	2	2

DO NOT PROVIDE AIR CHAMBERS. PROVIDE WATER HAMMER ARRESTERS BY SIOUX CHIEF. PRECISION PLUMBING PRODUCTS, WATTS OR APPROVED EQUIVALENT WITH PISTON AND O-RING CONSTRUCTION, HAVING PDI #WH-201, ASSE # 1010, AND ANSI # A112.26.1M CERTIFICATION. INSTALL IN HORIZONTAL OR VERTICAL POSITION, BUT NEVER UPSIDE DOWN. INSTALL IN LINE WITH WATER FLOW DIRECTION IF POSSIBLE. SIZE THE UNITS PER THE TABLES SHOWN ABOVE.

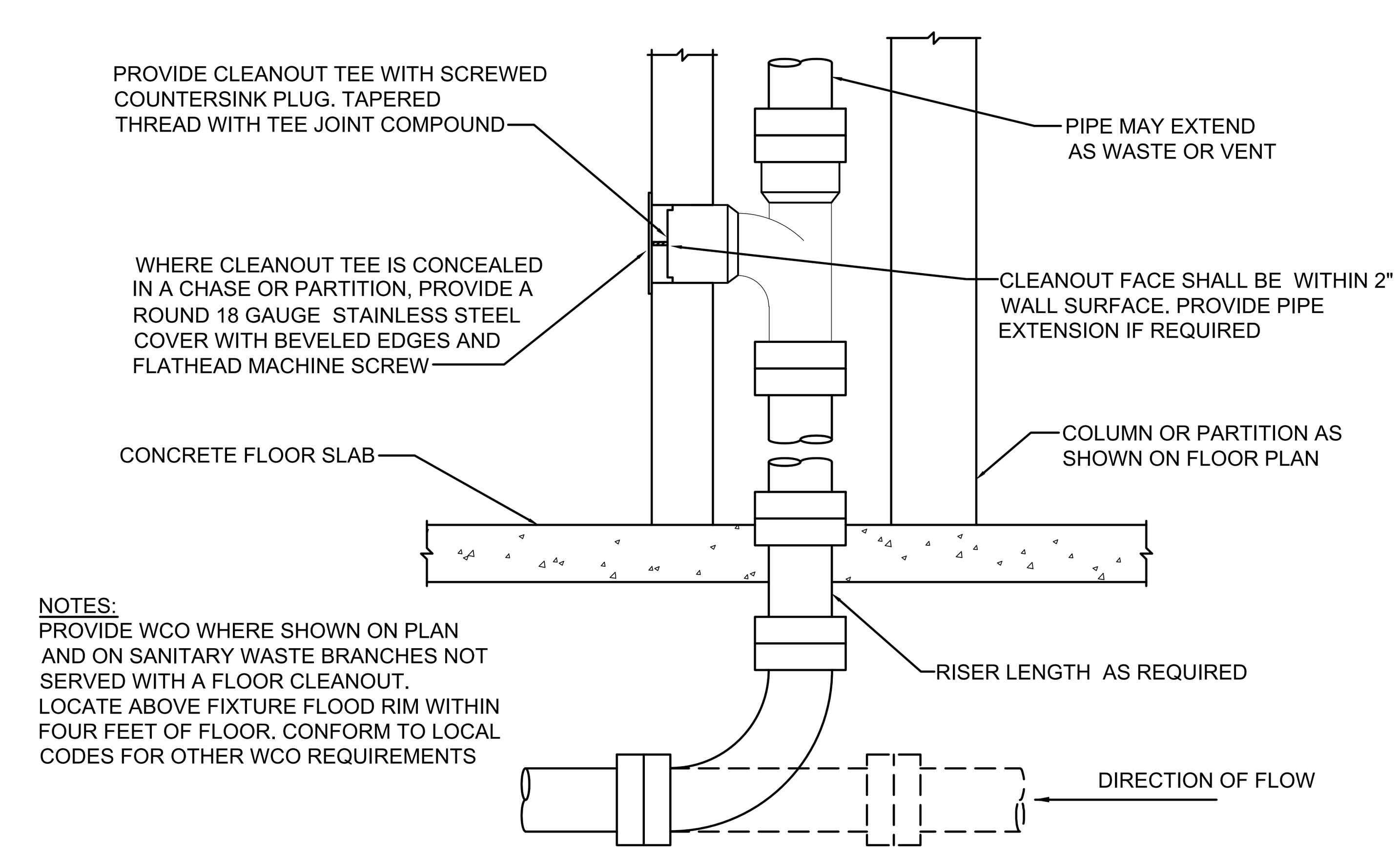
A2	WATER HAMMER ARRESTER DETAIL
P3	NTS



D4	TYPICAL FLOOR DRAIN DETAIL
P3	NTS



B4	WALL MOUNTED ELECTRIC WATER HEATER
P3	NTS



NOTES:
PROVIDE WCO WHERE SHOWN ON PLAN AND ON SANITARY WASTE BRANCHES NOT SERVED WITH A FLOOR CLEANOUT. LOCATE ABOVE FIXTURE FLOOD RIM WITHIN FOUR FEET OF FLOOR. CONFORM TO LOCAL CODES FOR OTHER WCO REQUIREMENTS

A4	WALL CLEANOUT DETAIL
P3	NTS

FORT LEAVENWORTH
DPW

MARK	DESCRIPTION	DATE

ISSUE DATE: 2/21/2025
PROJECT NO.: 300000331343
DESIGNED BY: NET
DRAWN BY: NET
CHECKED BY: SER/JCM
SIZE: ANSI 'D'

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse - Office Addition

PLUMBING DETAILS

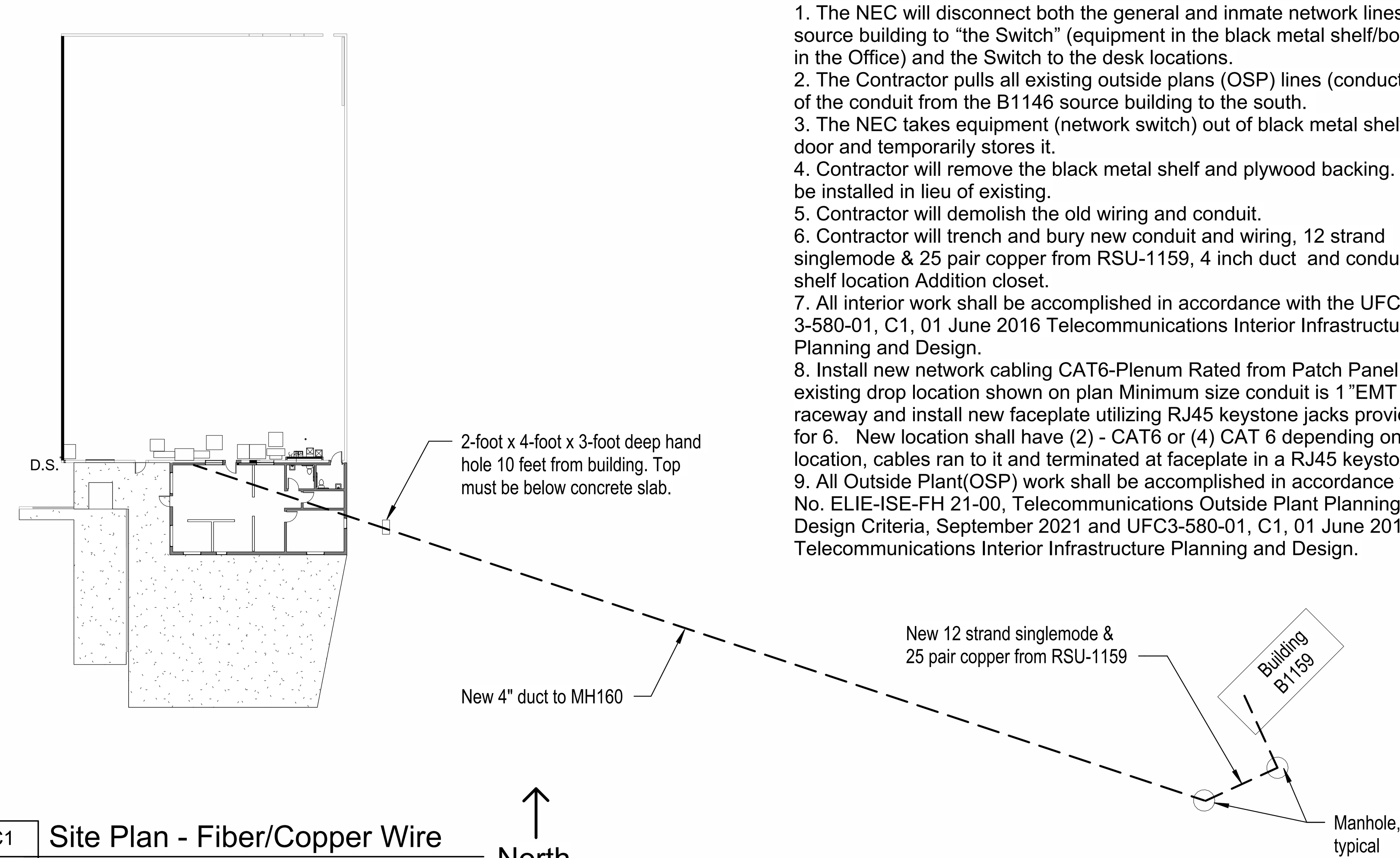
SHEET ID
P3

1 2 3 4 5

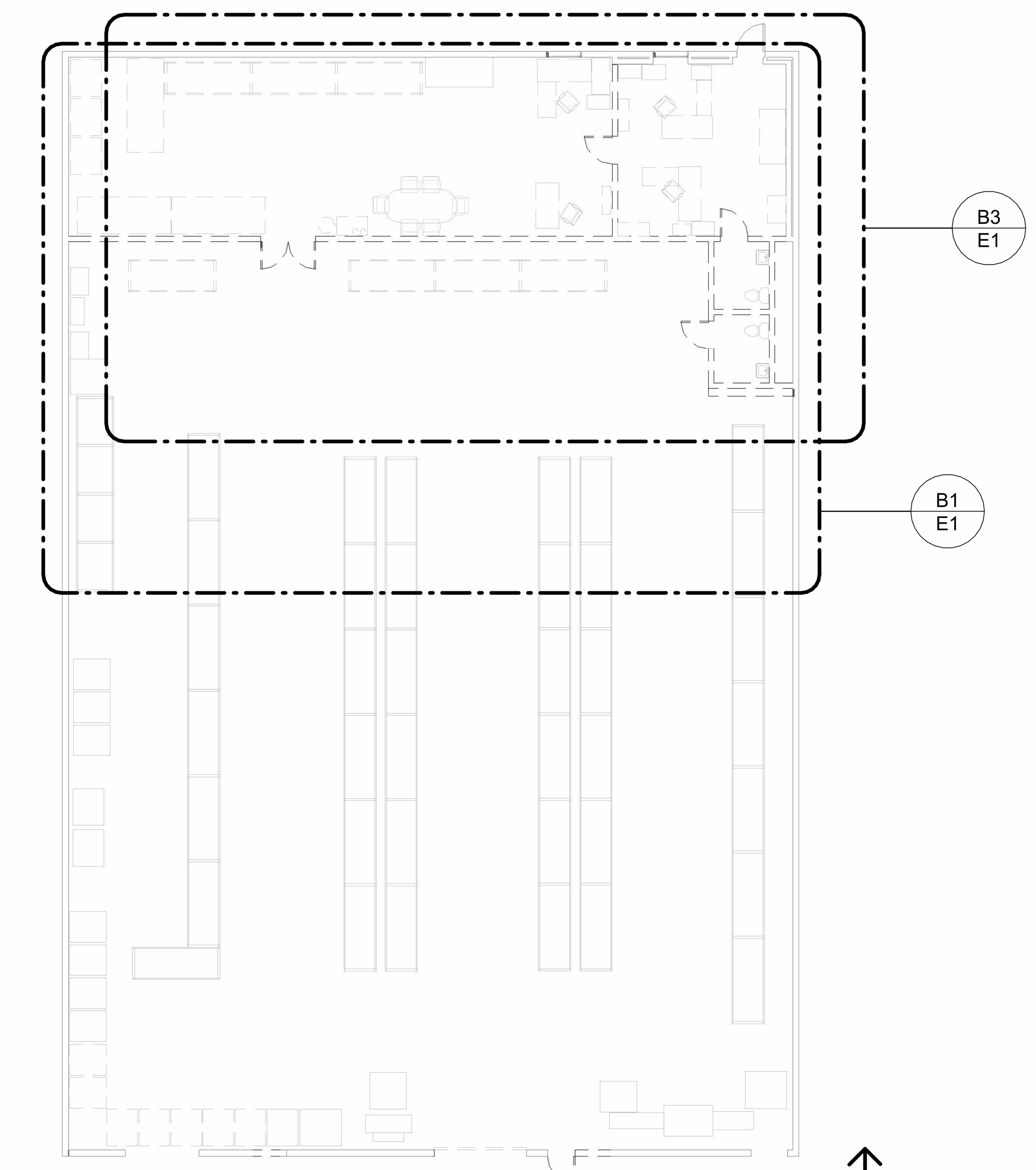
D C B A

Communications Notes

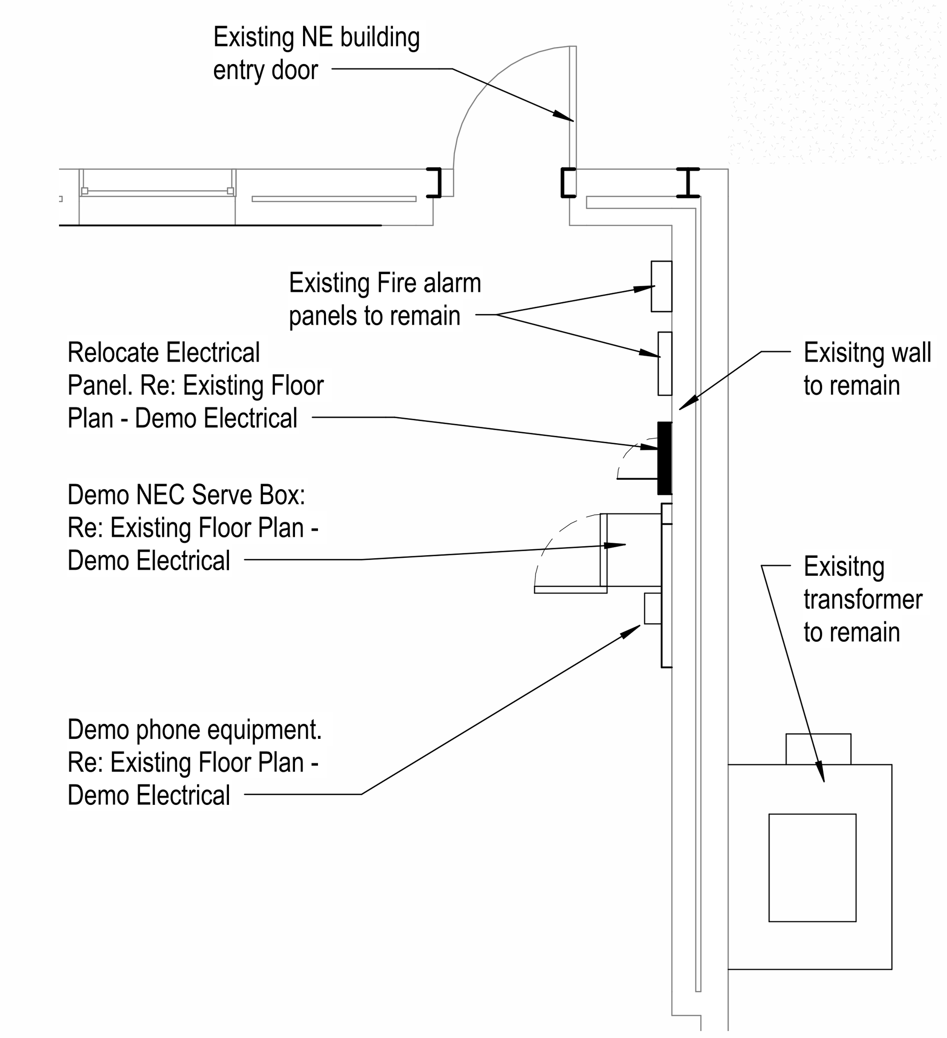
1. The NEC will disconnect both the general and inmate network lines from the source building to "the Switch" (equipment in the black metal shelf/box on wall in the Office) and the Switch to the desk locations.
2. The Contractor pulls all existing outside plans (OSP) lines (conductors) out of the conduit from the B1146 source building to the south.
3. The NEC takes equipment (network switch) out of black metal shelf with door and temporarily stores it.
4. Contractor will remove the black metal shelf and plywood backing. New will be installed in lieu of existing.
5. Contractor will demolish the old wiring and conduit.
6. Contractor will trench and bury new conduit and wiring, 12 strand singlemode & 25 pair copper from RSU-1159, 4 inch duct and conduit to new shelf location Addition closet.
7. All interior work shall be accomplished in accordance with the UFC 3-580-01, C1, 01 June 2016 Telecommunications Interior Infrastructure Planning and Design.
8. Install new network cabling CAT6-Plenum Rated from Patch Panel to existing drop location shown on plan Minimum size conduit is 1"EMT or raceway and install new faceplate utilizing RJ45 keystone jacks provide space for 6. New location shall have (2) - CAT6 or (4) CAT 6 depending on the location, cables ran to it and terminated at faceplate in a RJ45 keystone jack.
9. All Outside Plant(OSP) work shall be accomplished in accordance with TR No. ELIE-ISE-FH 21-00, Telecommunications Outside Plant Planning and Design Criteria, September 2021 and UFC3-580-01, C1, 01 June 2016 Telecommunications Interior Infrastructure Planning and Design.



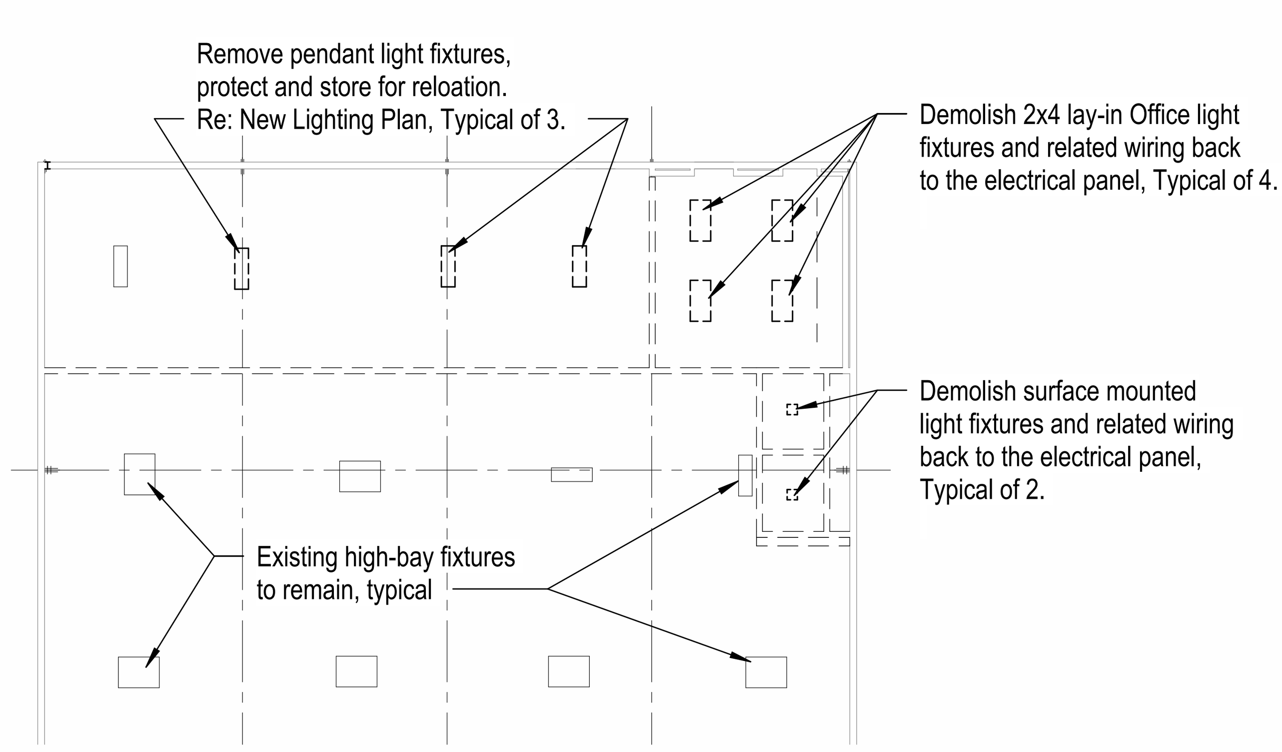
C1 | Site Plan - Fiber/Copper Wire
1/32" = 1'-0"



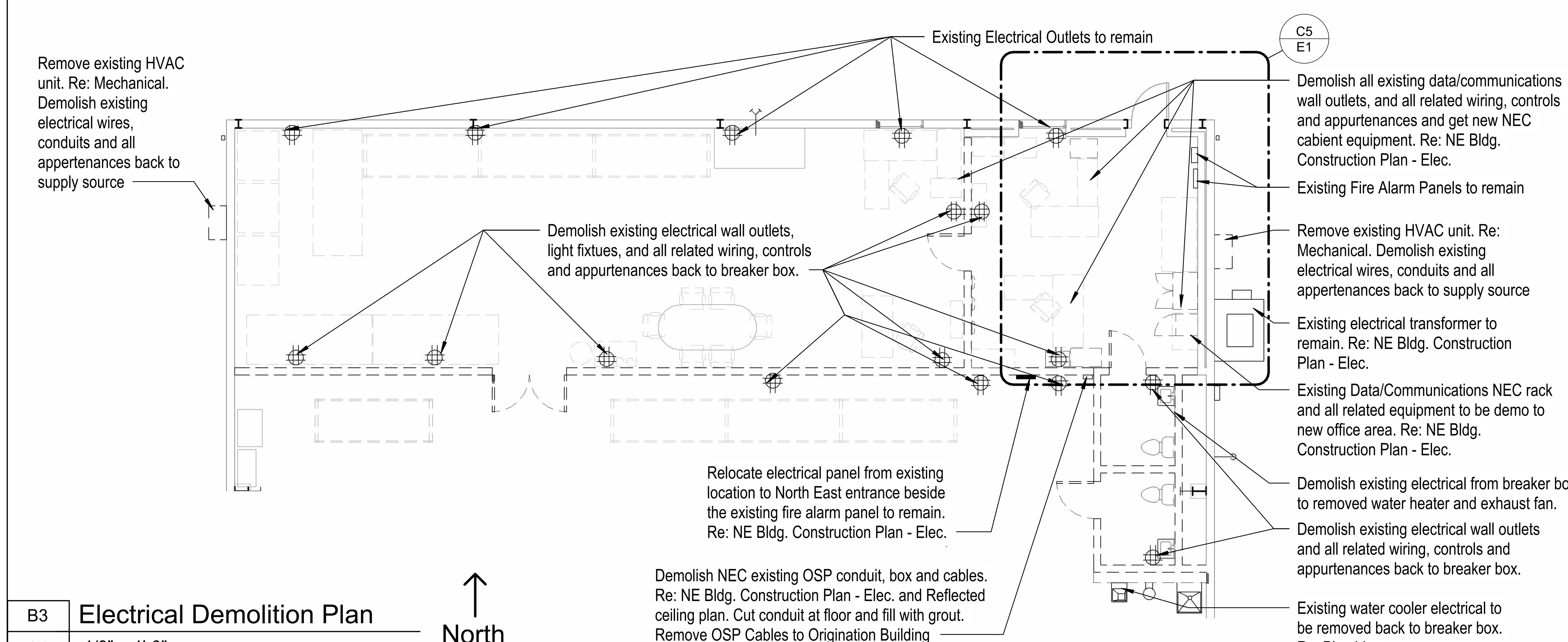
C3 | Electrical Demolition Ref. Plan
A4 | 1/16" = 1'-0"



C5 | NE Building Electrical Demolition Plan
A4 | 1/4" = 1'-0"



B1 | Reflected Ceiling Elec. Demolition Plan
E1 | 1/16" = 1'-0"



B3 | Electrical Demolition Plan
A4 | 1/8" = 1'-0"

Symbol	Description	Symbol	Description	Symbol	Description
	DUPLEX RECEPTACLE - WALL MOUNT 18" AFF	4	FOUR-PLEX EMPLOYEE NETWORK DATA PORT FLOOR-MOUNTED	E	LATERIN SURFACE WALL-MOUNTED LIGHT FIXTURE
GFCI	DUPLEX RECETALE - GROUND-FAULT- CIRCUIT-INTERRUPTER 36" TO CENTERINE IN LATRINE / 16" AT DRINIKING FOUNTAIN	$	LIGHT SWITCH	F	LATERIN EXHUAST FAN
1 GFCI WP	DUPLEX RECEPTALE WITH WEATHERPROOF PLATE, AT LEAST 12" ABOVE THE FINISHED GRADE - GROUND-FAULT CIRCUIT-INTERRUPTER	$ 3W	LIGHT 3-WAY SWITCH	G	EXTERIOR WALL MOUNTED LIGHT FIXTURE
	FOUR -PLEX RECEPTACLE		24-HOUR EMERGENCY FIXT URE	WH-1	WATER HEATER
1 GFCI	FLOOR MOUNTED FOUR-PLEX RECEPTACLES - GROUND-FAULT CIRCUIT-INTERRUPTER	B	CONFERENCE ROOM LIGHTS		EXIT SIXT SIGN, MOUNT JUST AFF OF DOOR FRAME
2	INMATE NETWORK DUPLEX DATA PORT	C	STORAGE ROOM LIGHTS	UH #	UNIT HEATER
4	EMPLOYEE NETWORK FOUR-PLEX DATA PORT	D	LATERINE FIXTURES	HVLS-#	HIGH VOLUME LOW SPEED FAN

General Elec. Demolition Notes

1. Remove necessary electrical wiring, conduit, switches, disconnects, luminaires, lamps, ballasts and associated ancillary equipment shown, items indicated for removal are only suggestive of the amount of demolition work involved. Perform a site investigation to aid in determining the complete extent of demolition work.
2. Coordinated and scheduled all necessary power outages with the owner's representative prior to proceeding with such work so operations in adjacent occupied portions of the building are not interrupted or restricted without prior approval.
3. Do not remove the entire conduit. Splice the existing branch circuits that go to the panel and install a new conduit from the splice part of the conduit to the newly relocated panel. Where an existing panel is removed, provide new wiring and maintain continuity of the existing circuit.

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE
1	Revision 1, Addendum 1	3 / 12 / 2026

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH

820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

DESIGNED BY: Designer
DRAWN BY: DB
CHECKED BY: DB
SIZE: ANSI 'D'
ISSUE DATE: 02/21/2025
PROJECT NO.: 300000241811
100% Drawings

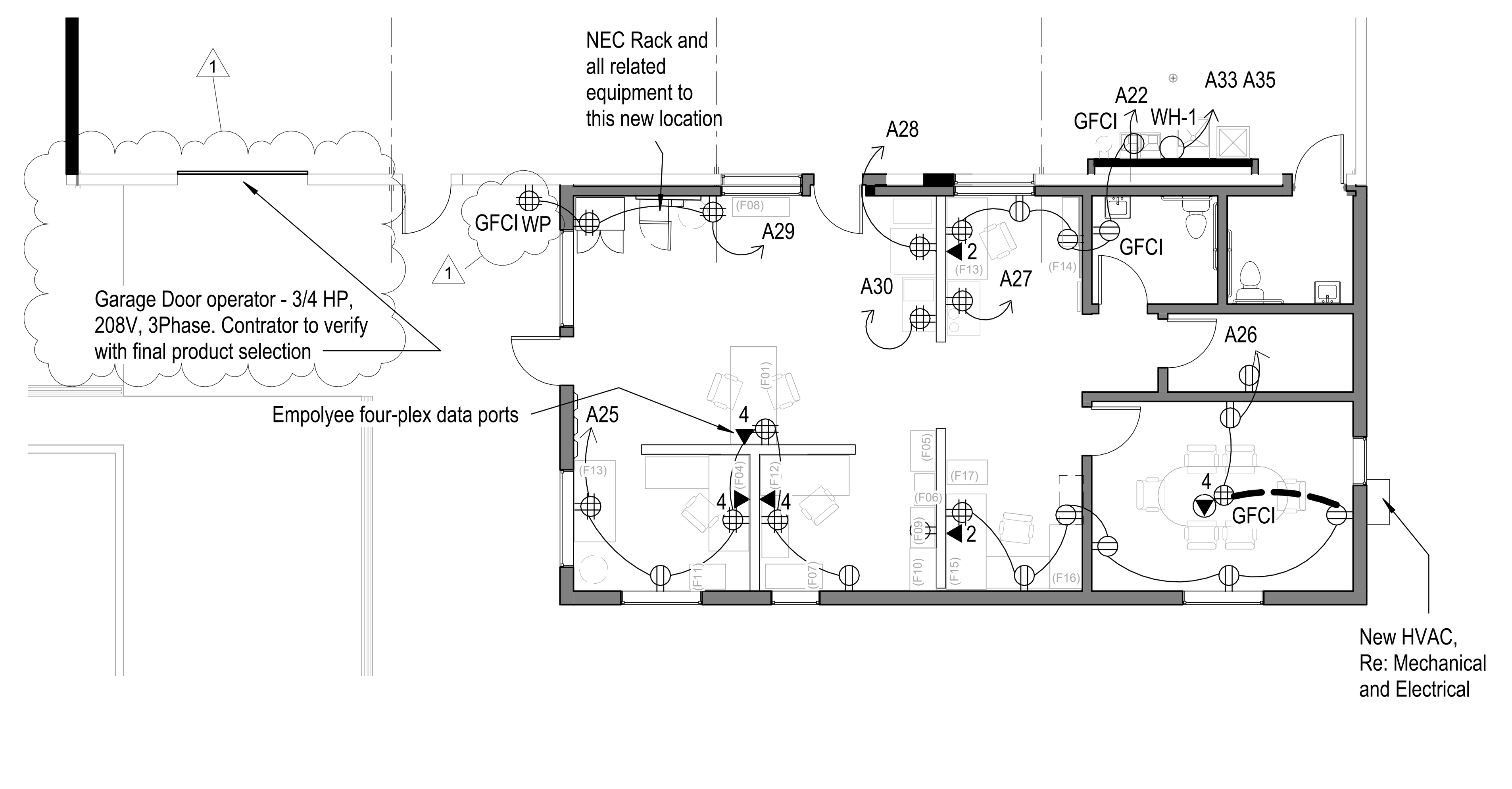
B1146 - Textiles Warehouse

Electrical Demolition and Site Data Plans

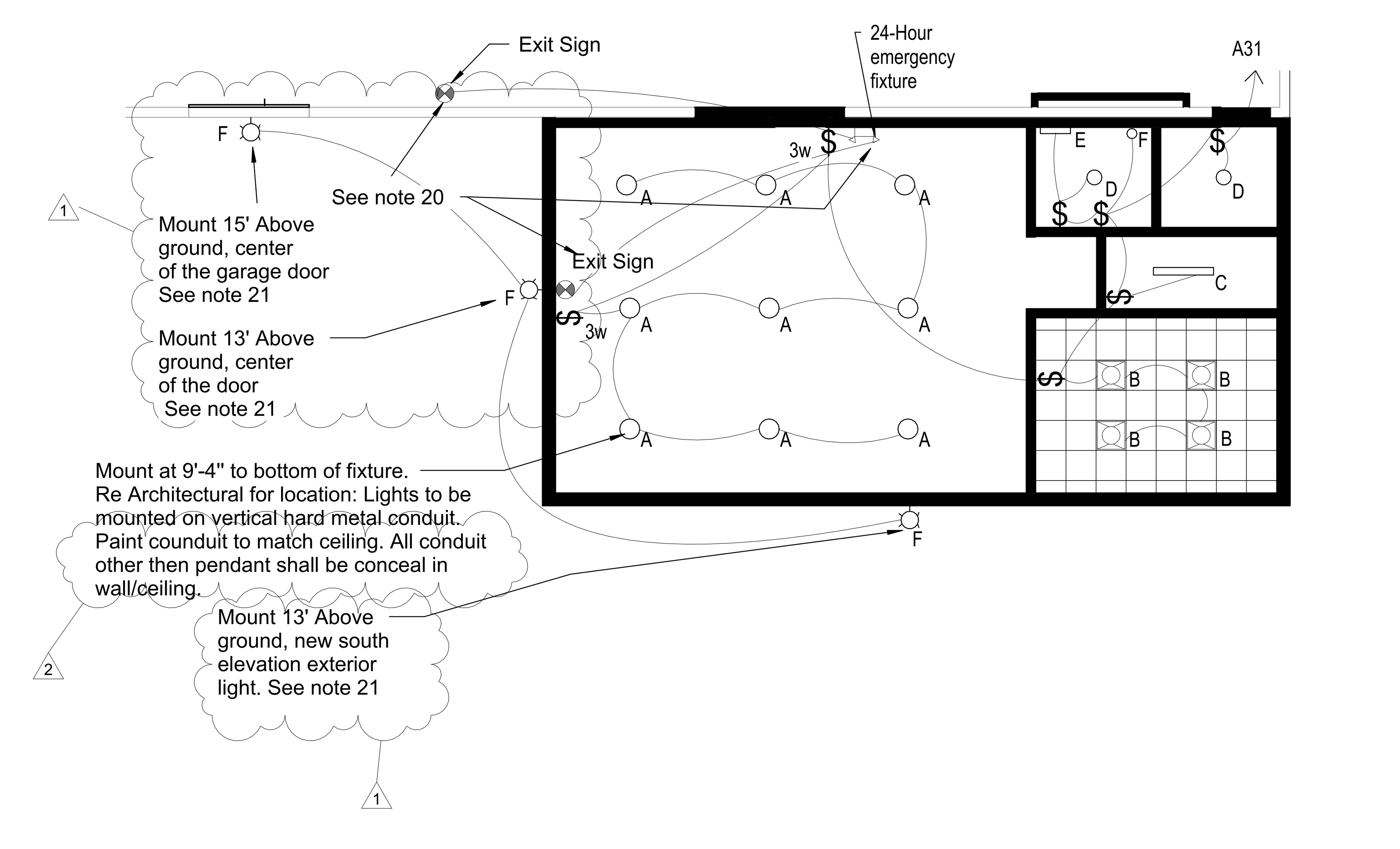
SHEET ID

E1

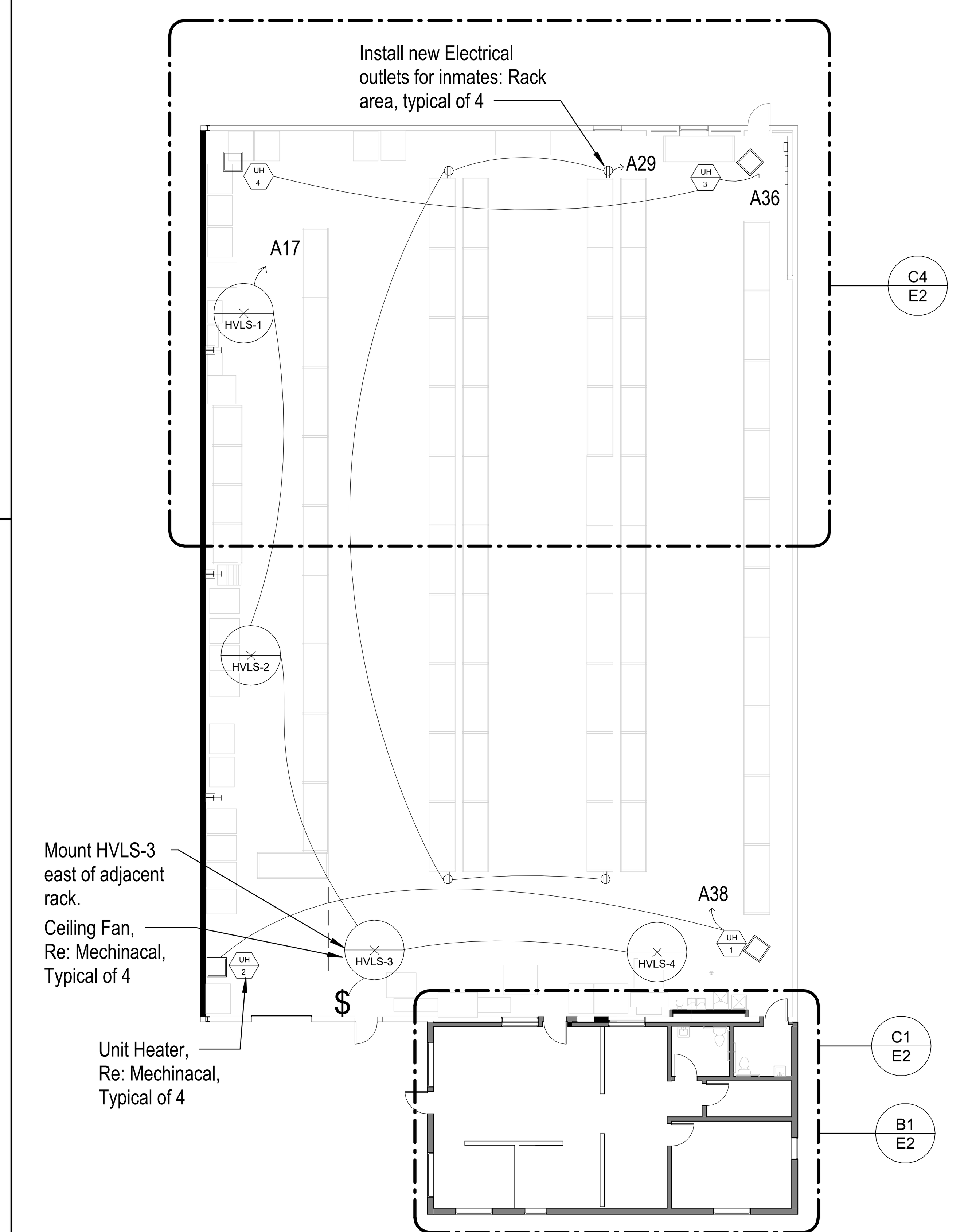
1 2 3 4 5



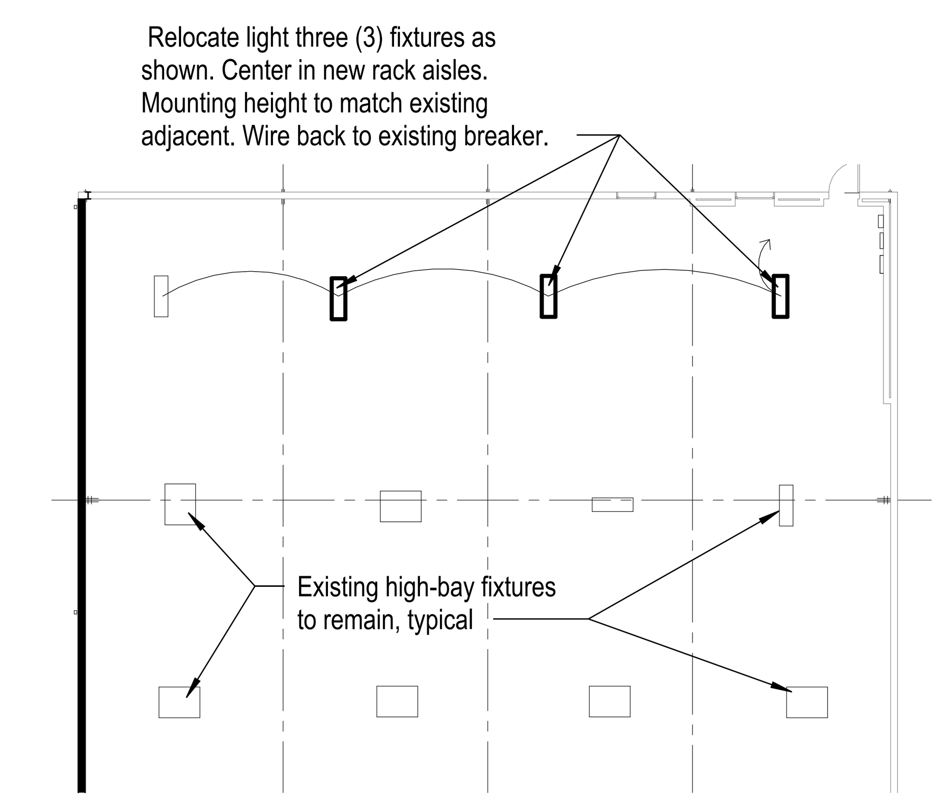
C1 Addition Floor Plan - Power/Data
E2 1/8" = 1'-0"



B1 Addition Reflected Ceiling Plan
A4 1/8" = 1'-0"



B3 Electrical Power Plan
1/16" = 1'-0"



C4 Reflected Ceiling Electrical Plan
E2 1/16" = 1'-0"

LIGHT FIXTURE SCHEDULE (BASIS OF DESIGN)

	ITEM	MANUFACTURE & TYPE	LAMPS NO.	WATTS	LUMENS	TYPE	VOLTS	MINIMUM TEMPERATURE	CONTROL	DIMESMIONS
A	OFFICE ROOM LIGHT	LITHONIA LIGHTING & PENDANT MOUNT - LED - Pendant Mount 10' - 6" above floor Warehouse Lighting - LED CCT Selectable (30/35/40/50K), 0-10V Dimmable, 120-277V, Opal Lens, Sandy White Housing, Gen 2		75	5000	LED	120	4000K	SWITCH	L: .48", W: 3-3/8"
B	CONFERENCE ROOM LIGHT	LITHONIA LIGHTING - LED Recess Lay-in LED Panel Luminaire w/ matte diffuser opal lens Lithonia Lighting - CPX-2x2 200-0LM-80 CRI-40K-SWL-MVolt or Equal		15.06	2000	LED	120	3500K	SWITCH	L: 21.84", W: 21.84"
C	STORAGE ROOM LIGHT	LITHONIA LIGHTING ADJUSTABLE LUMEN & COLOR TEMPERATURE LED COMERICAL - 48" industrial pendant-mount LED industrial fixture. Lithonia Lighting - LED - MNSS L48 AL03 MVolt SWW3 MNSS 48" Adjustable Lumen		36	4000	LED	120	4000K	SWITCH	L: 48", W: 2.25"
D	LATRINE GENERAL LIGHT	LITHONIA LIGHTING & ROUND FLUSH MOUNTED - LED Surface-mounted general restroom on gypsum Lithonia Lighting - LED - FMSATL-13-14840-BN-M4 13" round-shape flush mount		16	1100	LED	120	4000K	SWITCH	Depth/Height: 3.1", Width/Diameter: 13"
E	LATRINE ABOVE MIRROR LIGHT	LITHONIA LIGHTING & CONTEMPORARY CLYINDER WITH ADJUSTABLE LUMEN & COLOR- 24" long LED Wall-mounted light above Employee lavatory Lithonia Lighting - LED - FMVCCLS 36IN MVOLT 30K35K40K 90CRI BN M4		26.39	2700	LED	120	4000K	SWITCH	L: 24", W: 4.44"
F	EXTERIOR LIGHT	LITHONIA LIGHTING & TWR1 LED WALL PACK ADJUSTABLE LUMEN & COLOR TEMPERATURE - LED Wall-pack + Photocell Lithonia Lighting - LED - TWR1-ALO-SWW2-UVOLT-PE-DDBTXD		59	8500	LED	120	4000K	SWITCH	L: 10.8", W: 6.36"

General Electrical Construction Notes

1. The drawings have been prepared utilizing existing drawings and building observations and may not show all conditions.
2. Coordinate installation with existing conditions. Field verify existing conditions.
3. Provide a dedicated neutral conductor for each branch circuit phase conductor. Exception: where an equipment manufacturer requires a multiwire branch circuit for only one utilization equipment and where all ungrounded conductors of that circuit are open simultaneously by the branch circuit overcurrent device.
4. Conceal all conduits in new walls, existing stud walls, or above suspended and expose ceilings.
5. Provide electrical metallic tubing (EMT) sleeves with insulated bushing on each end for all cables passing through walls and floors.
6. All access control cabling shall be installed in conduit. J-hook supports shall not be allowed.
7. The contractor shall coordinate all conduit and cable routing with existing conditions prior to installation.
8. The contractor shall label all cables and connectors (outlets, patch panels, etc) per specifications and in compliance with TIA-606. All cables shall be labeled with a pre-printed stick-on self-adhering labeling system. Label must be applied to both ends of the cable and shall be visible after final installation.
9. All work is to conform to applicable UFC 1-200-01, UFC 1-300-02, UFC 3-120-10, UFC 3-501-01, UFC 3-520-01, UFC 3-540-07, UFC 3-560-01, UFC 4-010-06, NFPA 70, NFPA 70E, NFPA 101.
10. Coordinate all work with other trades prior to installation.
11. Coordinate routing of conduit, conductors, lights, architectural ceiling, and structural elements. Conduit shall rise and drop, jog, or offset as required to avoid conflicts. Rework of installed work to resolve conflicts arising from lack of coordination shall not justify an increase in the contract amount.
12. label panelboard directories, include a description of loads such as lights, receptacles, mechanical unit locations, etc, on typed panelboard directories.
13. Size the conductors and conduit according to their units.
14. Provide mininum conductors 12 AWG conductor with 20 AMP Breaker for lights.
15. NEC server box shall be located to new office area. The new cables shall be CAT 6.
16. The contractor will demo the appurtenances of NEC equiment inculding new plywoodand and will buy new NEC equipment. The contractor will run new lines in conduit, and NEC will connect their NEC equipment to their servers, install new switch, patch connections and drops.
17. All wall outlets, switches & coverplates to be cream colored.
18. The contractor will be provide with NEC standards "21-010 TELE Standard Drawings", "TELECOMMUNCATIONS OUTSIDE PLANT PLANNING AND DESIGN CRITERIA SEP 2021" and B1146 Rack Elev 2" in a PDF format.

19. The contractor shall perform an arc-flash hazard analysis due to relocation of the panel and the addition of new circuits. All work and calculation shall comply with the requirements of UFC 3-560-01.

20. Exit sign/24-hour emergency power: The contractor shall field-verify and identify the source circuit for the nearest existing exit sign.The contractor shall perform a load calculation on the identified circuit to determine its existing load. If sufficient capacity exists (in accordance with nec 210.20 for continuous loads), the contractor shall connect the new exit sign to this existing circuit. If sufficient capacity does not exist, the contractor shall install a new dedicated circuit from an available breaker in panel [specify panel, e.g., a31] to power the new exit sign. In either case, the contractor shall clearly label the final circuit in the panel directory.

21. Exterior lighting power: The contractor shall field-verify and identify the source circuit for the nearest existing exterior lighting. This circuit is presumed to be controlled by an existing photocell. Provided sufficient capacity exists, the contractor shall connect the new exterior wall pack and its associated photocell to this existing exterior lighting circuit. The contractor shall clearly label the final circuit in the panel directory.

FORT LEAVENWORTH
DPW

MARK	DESCRIPTION	DATE
2	Revision 2	Date 2 3 / 12 / 2026
1	Revision 1, Addendum 1	

ISSUE DATE: 02/21/2025
PROJECT NO.: 300000241811
100% Drawings
DESIGNED BY: RK
DRAWN BY: RK
CHECKED BY: JM
SIZE: ANSI 'D'

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH
820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
66027-1292

B1146 - Textiles Warehouse

Electrical Power and Lighting Plans and Schedule

SHEET ID

E2

ELECTRICAL SPECIFICATIONS

26 20 00 INTERIOR DISTRIBUTION SYSTEM

1. PROVIDE EQUIPMENT, MATERIALS, INSTALLATION, AND WORKMANSHIP IN ACCORDANCE WITH NFPA 70 UNLESS MORE STRINGENT REQUIREMENTS ARE SPECIFIED OR INDICATED. NECA NEIS 1 SHALL BE CONSIDERED THE MINIMUM STANDARD FOR WORKMANSHIP.
2. PROVIDE MATERIALS AND EQUIPMENT THAT ARE PRODUCTS OF MANUFACTURERS REGULARLY ENGAGED IN THE PRODUCTION OF SUCH PRODUCTS WHICH ARE OF EQUAL MATERIAL, DESIGN AND WORKMANSHIP AND:
   - A. HAVE BEEN IN SATISFACTORY COMMERCIAL OR INDUSTRIAL USE FOR 2 YEARS PRIOR TO BID OPENING INCLUDING APPLICATIONS OF EQUIPMENT AND MATERIALS UNDER SIMILAR CIRCUMSTANCES AND OF SIMILAR SIZE.
   - B. HAVE BEEN ON SALE ON THE COMMERCIAL MARKET THROUGH ADVERTISEMENTS, MANUFACTURERS' CATALOGS, OR BROCHURES DURING THE 2-YEAR PERIOD.
   - C. WHERE TWO OR MORE ITEMS OF THE SAME CLASS OF EQUIPMENT ARE REQUIRED, PROVIDE PRODUCTS OF A SINGLE MANUFACTURER; HOWEVER, THE COMPONENT PARTS OF THE ITEM NEED NOT BE THE PRODUCTS OF THE SAME MANUFACTURER UNLESS STATED IN THIS SECTION.
3. PRODUCTS MANUFACTURED MORE THAN 3 YEARS PRIOR TO DATE OF DELIVERY TO SITE ARE NOT ACCEPTABLE.
4. AS A MINIMUM, MEET REQUIREMENTS OF UL, WHERE UL STANDARDS ARE ESTABLISHED FOR THOSE ITEMS, AND REQUIREMENTS OF NFPA 70 FOR ALL MATERIALS, EQUIPMENT, AND DEVICES.
5. CONDUIT AND FITTINGS
   - A. CONFORM TO THE FOLLOWING:
     - a. RIGID, THREADED ZINC-COATED STEEL CONDUIT IN ACCORDANCE WITH (IAW) ANSI C80.1, UL 6.
     - b. RIGID NONMETALLIC CONDUIT
       - a. PVC TYPE EPC-40 IAW NEMA TC 2,UL 651.
     - c. ELECTRICAL, ZINC-COATED STEEL METALLIC TUBING (EMT) IAW UL 797, ANSI C80.3.
     - d. FLEXIBLE METAL CONDUIT IAW UL 1, LIMITED TO 6 FEET.
     - e. LIQUID-TIGHT FLEXIBLE METAL CONDUIT, STEEL IAW UL 360, LIMITED TO 6 FEET.
     - f. FITTINGS FOR METAL CONDUIT, EMT, AND FLEXIBLE METAL CONDUIT
       - a. UL 514B. FERROUS FITTINGS: CADMIUM- OR ZINC-COATED IAW UL 514B.
     - g. FITTINGS FOR RIGID METAL CONDUIT AND IMC
       - a. THREADED-TYPE. SPLIT COUPLINGS UNACCEPTABLE.
     - h. FITTINGS FOR EMT
       - a. STEEL COMPRESSION TYPE.
     - i. FITTINGS FOR RIGID NONMETALLIC CONDUIT
       - a. NEMA TC 3 FOR PVC AND UL 514B.
6. OUTLET BOXES AND COVERS: UL 514A, CADMIUM- OR ZINC-COATED, IF FERROUS METAL. UL 514C, IF NONMETALLIC.
7. CABINETS, JUNCTION BOXES, AND PULL BOXES: UL 50; VOLUME GREATER THAN 100 CUBIC INCHES, NEMA TYPE 1 ENCLOSURE; SHEET STEEL, HOT-DIP, ZINC-COATED.
8. WIRES AND CABLES:
   - A. PROVIDE WIRES AND CABLES IN ACCORDANCE APPLICABLE REQUIREMENTS OF NFPA 70 AND UL FOR TYPE OF INSULATION, JACKET, AND CONDUCTOR SPECIFIED OR INDICATED. DO NOT USE WIRES AND CABLES MANUFACTURED MORE THAN 12 MONTHS PRIOR TO DATE OF DELIVERY TO SITE.
   - B. CONDUCTORS
     - a. PROVIDE THE FOLLOWING:
       - CONDUCTOR SIZES AND CAPACITIES SHOWN ARE BASED ON COPPER, UNLESS INDICATED OTHERWISE.
       - CONDUCTORS NO. 8 AWG AND LARGER DIAMETER: STRANDED.
       - CONDUCTORS NO. 10 AWG AND SMALLER DIAMETER: SOLID.
       - CONDUCTORS FOR REMOTE CONTROL, ALARM, AND SIGNAL CIRCUITS, CLASSES 1, 2, AND 3: STRANDED UNLESS SPECIFICALLY INDICATED OTHERWISE.
       - ALL CONDUCTORS: COPPER.
   - C. MINIMUM CONDUCTOR SIZES
     - a. PROVIDE MINIMUM CONDUCTOR SIZE IN ACCORDANCE WITH THE FOLLOWING:
       - BRANCH CIRCUITS: NO. 12 AWG.
       - CLASS 1 REMOTE-CONTROL AND SIGNAL CIRCUITS: NO. 14 AWG.
       - CLASS 2 LOW-ENERGY, REMOTE-CONTROL AND SIGNAL CIRCUITS: NO. 16 AWG.
       - CLASS 3 LOW-ENERGY, REMOTE-CONTROL, ALARM AND SIGNAL CIRCUITS: NO. 22 AWG.
9. SPLICES AND TERMINATION COMPONENTS: UL 486A-486B FOR WIRE CONNECTORS AND UL 510 FOR INSULATING TAPES. CONNECTORS FOR NO. 10 AWG AND SMALLER DIAMETER WIRES: INSULATED, PRESSURE-TYPE IN ACCORDANCE WITH UL 486A-486B OR UL 486C (TWIST-ON SPLICING CONNECTOR). PROVIDE SOLDERLESS TERMINAL LUGS ON STRANDED CONDUCTORS.
10. DEVICE PLATES:
    - A. PROVIDE THE FOLLOWING:
      - a. UL LISTED, ONE-PIECE DEVICE PLATES FOR OUTLETS TO SUIT THE DEVICES INSTALLED.
      - b. FOR METAL OUTLET BOXES, PLATES ON UNFINISHED WALLS: ZINC-COATED SHEET STEEL OR CAST METAL HAVING ROUND OR BEVELED EDGES.
      - c. FOR NONMETALLIC BOXES AND FITTINGS, OTHER SUITABLE PLATES MAY BE PROVIDED.
      - d. PLATES ON FINISHED WALLS: NYLON OR LEXAN, MINIMUM 0.03 INCH WALL THICKNESS AND SAME COLOR AS RECEPTACLE OR TOGGLE SWITCH WITH WHICH THEY ARE MOUNTED.
      - e. SCREWS: MACHINE-TYPE WITH COUNTERSUNK HEADS IN COLOR TO MATCH FINISH OF PLATE.
      - f. SECTIONAL TYPE DEVICE PLATES ARE NOT BE PERMITTED.
      - G. PLATES INSTALLED IN WET LOCATIONS: GASKETED AND UL LISTED FOR "WET LOCATIONS."
11. RECEPTACLES
    - A. PROVIDE THE FOLLOWING:
      - a. UL 498, GENERAL PURPOSE SPECIFICATION GRADE, GROUNDING-TYPE. RESIDENTIAL GRADE RECEPTACLES ARE NOT ACCEPTABLE.
      - b. RATINGS AND CONFIGURATIONS: AS INDICATED.
      - c. BODIES: WHITE AS PER NEMA WD 1.
      - d. FACE AND BODY: THERMOPLASTIC SUPPORTED ON A METAL MOUNTING STRAP.
      - e. DIMENSIONAL REQUIREMENTS: PER NEMA WD 6.
      - f. SCREW-TYPE, SIDE-WIRED WIRING TERMINALS OR OF THE SOLDERLESS PRESSURE TYPE HAVING SUITABLE CONDUCTOR-RELEASE ARRANGEMENT.
      - g. GROUNDING POLE CONNECTED TO MOUNTING STRAP.
      - h. THE RECEPTACLE: CONTAINING TRIPLE-WIPE POWER CONTACTS AND DOUBLE OR TRIPLE-WIPE GROUND CONTACTS.
    - B. GROUND-FAULT CIRCUIT INTERRUPTER RECEPTACLES
      - a. UL 943, DUPLEX TYPE FOR MOUNTING IN STANDARD OUTLET BOX. PROVIDE DEVICE CAPABLE OF DETECTING CURRENT LEAK WHEN THE CURRENT TO GROUND IS 6 MILLIAMPERES OR HIGHER, AND TRIPPING PER REQUIREMENTS OF UL 943 FOR CLASS A GROUND-FAULT CIRCUIT INTERRUPTER DEVICES. PROVIDE SCREW-TYPE, SIDE-WIRED WIRING TERMINALS OR PRE-WIRED (PIGTAIL) LEADS.
12. CIRCUIT BREAKERS
    - A. UL 489, THERMAL MAGNETIC-TYPE HAVING A MINIMUM SHORT-CIRCUIT CURRENT RATING EQUAL TO THE SHORT-CIRCUIT CURRENT RATING OF THE PANELBOARD IN WHICH THE CIRCUIT BREAKER WILL BE MOUNTED. BREAKER TERMINALS: UL LISTED AS SUITABLE FOR TYPE OF CONDUCTOR PROVIDED. WHERE INDICATED ON THE DRAWINGS, PROVIDE CIRCUIT BREAKERS WITH SHUNT TRIP DEVICES. SERIES RATED CIRCUIT BREAKERS AND PLUG-IN CIRCUIT BREAKERS ARE UNACCEPTABLE.
    - B. MULTIPOLE BREAKERS
      - a. PROVIDE COMMON TRIP-TYPE WITH SINGLE OPERATING HANDLE. DESIGN BREAKER SUCH THAT OVERLOAD IN ONE POLE AUTOMATICALLY CAUSES ALL POLES TO OPEN. MAINTAIN PHASE SEQUENCE THROUGHOUT EACH PANEL SO THAT ANY THREE ADJACENT BREAKER POLES ARE CONNECTED TO PHASES A, B, AND C, RESPECTIVELY.
13. TELECOMMUNICATIONS SYSTEM
    - A. PROVIDE SYSTEM OF TELECOMMUNICATIONS WIRE-SUPPORTING STRUCTURES (PATHWAY), INCLUDING: OUTLET BOXES, CONDUITS WITH PULL WIRES CABLE TRAYS, AND OTHER ACCESSORIES FOR TELECOMMUNICATIONS OUTLETS AND PATHWAY IN ACCORDANCE WITH TIA-569 AND AS SPECIFIED HEREIN. ADDITIONAL TELECOMMUNICATIONS REQUIREMENTS ARE SPECIFIED IN SECTION 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.
14. MANUFACTURER'S NAMEPLATE
    - A. PROVIDE ON EACH ITEM OF EQUIPMENT A NAMEPLATE BEARING THE MANUFACTURER'S NAME, ADDRESS, MODEL NUMBER, AND SERIAL NUMBER SECURELY AFFIXED IN A CONSPICUOUS PLACE; THE NAMEPLATE OF THE DISTRIBUTING AGENT WILL NOT BE ACCEPTABLE.
15. INSTALLATION
    - A. ELECTRICAL INSTALLATIONS, INCLUDING WEATHERPROOF AND HAZARDOUS LOCATIONS AND DUCTS, PLENUMS AND OTHER AIR-HANDLING SPACES: CONFORM TO REQUIREMENTS OF NFPA 70 AND IEEE C2 AND TO REQUIREMENTS SPECIFIED HEREIN.
    - B. WIRING METHODS
      - a. PROVIDE INSULATED CONDUCTORS INSTALLED IN RIGID STEEL CONDUIT, IMC, RIGID NONMETALLIC CONDUIT, OR EMT, EXCEPT WHERE SPECIFICALLY INDICATED OR SPECIFIED OTHERWISE OR REQUIRED BY NFPA 70 TO BE INSTALLED OTHERWISE. GROUNDING CONDUCTOR: SEPARATE FROM ELECTRICAL SYSTEM NEUTRAL CONDUCTOR. PROVIDE INSULATED GREEN EQUIPMENT GROUNDING CONDUCTOR FOR CIRCUIT(S) INSTALLED IN CONDUIT AND RACEWAYS. MINIMUM CONDUIT SIZE: 1/2 INCH IN DIAMETER FOR LOW VOLTAGE LIGHTING AND POWER CIRCUITS. VERTICAL DISTRIBUTION IN MULTIPLE STORY BUILDINGS: MADE WITH METAL CONDUIT IN FIRE-RATED SHAFTS, WITH METAL CONDUIT EXTENDING THROUGH SHAFTS FOR MINIMUM DISTANCE OF 6 INCHES. FIRESTOP CONDUIT WHICH PENETRATES FIRE-RATED WALLS, FIRE-RATED PARTITIONS, OR FIRE-RATED FLOORS IN ACCORDANCE WITH NFPA 70.
      - b. INSTALL PULL WIRES IN EMPTY CONDUITS. PULL WIRE: PLASTIC HAVING MINIMUM 200-POUND FORCE TENSILE STRENGTH. LEAVE MINIMUM 36 INCHES OF SLACK AT EACH END OF PULL WIRE.
    - C. CONDUIT INSTALLATION
      - a. UNLESS INDICATED OTHERWISE, CONCEAL CONDUIT WITHIN FINISHED WALLS, CEILINGS, AND FLOORS. KEEP CONDUIT MINIMUM 6 INCHES AWAY FROM PARALLEL RUNS OF FLUES AND STEAM OR HOT WATER PIPES. INSTALL CONDUIT PARALLEL WITH OR AT RIGHT ANGLES TO CEILINGS, WALLS, AND STRUCTURAL MEMBERS WHERE LOCATED ABOVE ACCESSIBLE CEILINGS AND WHERE CONDUIT WILL BE VISIBLE AFTER COMPLETION OF PROJECT. RUN CONDUITS IN CRAWL SPACE AS IF EXPOSED.
      - b. FOLLOW NFPA 70 FOR RESTRICTIONS APPLICABLE TO EMT, NONMETALLIC CONDUIT, FLEXIBLE CONDUIT, ETC.
      - c. STUB-UPS
        - PROVIDE CONDUITS STUBBED UP THROUGH CONCRETE FLOOR FOR CONNECTION TO FREE-STANDING EQUIPMENT WITH ADJUSTABLE TOP OR COUPLING THREADED INSIDE FOR PLUGS, SET FLUSH WITH FINISHED FLOOR. EXTEND CONDUCTORS TO EQUIPMENT IN RIGID STEEL CONDUIT, EXCEPT THAT FLEXIBLE METAL CONDUIT MAY BE USED 6 INCHES ABOVE FLOOR. WHERE NO EQUIPMENT CONNECTIONS ARE MADE, INSTALL SCREWDRIVER-OPERATED THREADED FLUSH PLUGS IN CONDUIT END.
      - d. CONDUIT SUPPORT
        - SUPPORT CONDUIT BY PIPE STRAPS, WALL BRACKETS, THREADED ROD CONDUIT HANGERS, OR CEILING TRAPEZE. PLASTIC CABLE TIES ARE NOT ACCEPTABLE. FASTEN BY WOOD SCREWS TO WOOD; BY TOGGLE BOLTS ON HOLLOW MASONRY UNITS; BY CONCRETE INSERTS OR EXPANSION BOLTS ON CONCRETE OR BRICK; AND BY MACHINE SCREWS, WELDED THREADED STUDS, OR SPRING-TENSION CLAMPS ON STEEL WORK. THREADED C-CLAMPS MAY BE USED ON RIGID STEEL CONDUIT ONLY. DO NOT WELD CONDUITS OR PIPE STRAPS TO STEEL STRUCTURES. FILL UNUSED HOLES. IN PARTITIONS OF LIGHT STEEL CONSTRUCTION, USE SHEET METAL SCREWS. IN SUSPENDED-CEILING CONSTRUCTION, RUN CONDUIT ABOVE CEILING. DO NOT SUPPORT CONDUIT BY CEILING SUPPORT SYSTEM. CONDUIT AND BOX SYSTEMS: SUPPORTED INDEPENDENTLY OF BOTH (A) TIE WIRES SUPPORTING CEILING GRID SYSTEM, AND (B) CEILING GRID SYSTEM INTO WHICH CEILING PANELS ARE PLACED. DO NOT SHARE SUPPORTING MEANS BETWEEN ELECTRICAL RACEWAYS AND MECHANICAL PIPING OR DUCTS. COORDINATE INSTALLATION WITH ABOVE-CEILING MECHANICAL SYSTEMS TO ASSURE MAXIMUM ACCESSIBILITY TO ALL SYSTEMS. SPRING-STEEL FASTENERS MAY BE USED FOR LIGHTING BRANCH CIRCUIT CONDUIT SUPPORTS IN SUSPENDED CEILINGS IN DRY LOCATIONS. WHERE CONDUIT CROSSES BUILDING EXPANSION JOINTS, PROVIDE SUITABLE WATERTIGHT EXPANSION FITTING THAT MAINTAINS CONDUIT ELECTRICAL CONTINUITY BY BONDING JUMPERS OR OTHER MEANS. FOR CONDUITS GREATER THAN 2 1/2 INCHES INSIDE DIAMETER, PROVIDE SUPPORTS TO RESIST FORCES OF 0.5 TIMES THE EQUIPMENT WEIGHT IN ANY DIRECTION AND 1.5 TIMES THE EQUIPMENT WEIGHT IN THE DOWNWARD DIRECTION.
    - E. DIRECTIONAL CHANGES IN CONDUIT RUNS
      - MAKE CHANGES IN DIRECTION OF RUNS WITH SYMMETRICAL BENDS OR CAST-METAL FITTINGS. MAKE FIELD-MADE BENDS AND OFFSETS WITH HICKEY OR CONDUIT-BENDING MACHINE. DO NOT INSTALL CRUSHED OR DEFORMED CONDUITS. AVOID TRAPPED CONDUITS. PREVENT PLASTER, DIRT, OR TRASH FROM LODGING IN CONDUITS, BOXES, FITTINGS, AND EQUIPMENT DURING CONSTRUCTION. FREE CLOGGED CONDUITS OF OBSTRUCTIONS.
    - F. LOCKNUTS AND BUSHINGS
      - FASTEN CONDUITS TO SHEET METAL BOXES AND CABINETS WITH TWO LOCKNUTS WHERE REQUIRED BY NFPA 70, WHERE INSULATED BUSHINGS ARE USED, AND WHERE BUSHINGS CANNOT BE BROUGHT INTO FIRM CONTACT WITH THE BOX; OTHERWISE, USE AT LEAST MINIMUM SINGLE LOCKNUT AND BUSHING. PROVIDE LOCKNUTS WITH SHARP EDGES FOR DIGGING INTO WALL OF METAL ENCLOSURES. INSTALL BUSHINGS ON ENDS OF CONDUITS, AND PROVIDE INSULATING TYPE WHERE REQUIRED BY NFPA 70.
    - g. FLEXIBLE CONNECTIONS
      - PROVIDE FLEXIBLE STEEL CONDUIT BETWEEN 3 AND 6 FEET IN LENGTH FOR RECESSED AND SEMIRECESSED LIGHTING FIXTURES. INSTALL FLEXIBLE CONDUIT TO ALLOW 20 PERCENT SLACK. MINIMUM FLEXIBLE STEEL CONDUIT SIZE: 1/2 INCH DIAMETER. PROVIDE LIQUID TIGHT FLEXIBLE METAL CONDUIT IN WET AND DAMP LOCATIONS FOR EQUIPMENT SUBJECT TO VIBRATION, NOISE TRANSMISSION, MOVEMENT OR MOTORS. PROVIDE SEPARATE GROUND CONDUCTOR ACROSS FLEXIBLE CONNECTIONS. PLASTIC CABLE TIES ARE NOT ACCEPTABLE AS A SUPPORT METHOD.
    - D. BOXES, OUTLETS, AND SUPPORTS
      - a. PROVIDE BOXES IN WIRING AND RACEWAY SYSTEMS WHEREVER REQUIRED FOR PULLING OF WIRES, MAKING CONNECTIONS, AND MOUNTING OF DEVICES OR FIXTURES. PROVIDE EACH BOX WITH VOLUME REQUIRED BY NFPA 70 FOR NUMBER OF CONDUCTORS ENCLOSED IN BOX. BOXES FOR MOUNTING LIGHTING FIXTURES: MINIMUM 4 INCHES SQUARE, OR OCTAGONAL, EXCEPT THAT SMALLER BOXES MAY BE INSTALLED AS REQUIRED BY FIXTURE CONFIGURATIONS, AS APPROVED. BOXES FOR USE IN MASONRY-BLOCK OR TILE WALLS: SQUARE-CORNERED, TILE-TYPE, OR STANDARD BOXES HAVING SQUARE-CORNERED, TILE-TYPE COVERS. PROVIDE GASKETS FOR CAST-METAL BOXES INSTALLED IN WET LOCATIONS. PROVIDE SEPARATE BOXES FOR FLUSH OR RECESSED FIXTURES WHEN REQUIRED BY FIXTURE TERMINAL OPERATING TEMPERATURE; PROVIDE READILY REMOVABLE FIXTURES FOR ACCESS TO BOXES UNLESS CEILING ACCESS PANELS ARE PROVIDED. SUPPORT BOXES AND PENDANTS FOR SURFACE-MOUNTED FIXTURES ON SUSPENDED CEILINGS INDEPENDENTLY OF CEILING SUPPORTS. FASTEN BOXES AND SUPPORTS WITH WOOD SCREWS ON WOOD, WITH BOLTS AND EXPANSION SHIELDS ON CONCRETE OR BRICK, WITH TOGGLE BOLTS ON HOLLOW MASONRY UNITS, AND WITH MACHINE SCREWS OR WELDED STUDS ON STEEL.IN OPEN OVERHEAD SPACES, CAST BOXES THREADED TO RACEWAYS NEED NOT BE SEPARATELY SUPPORTED EXCEPT WHERE USED FOR FIXTURE SUPPORT; SUPPORT SHEET METAL BOXES DIRECTLY FROM BUILDING STRUCTURE OR BY BAR HANGERS. WHERE BAR HANGERS ARE USED, ATTACH BAR TO RACEWAYS ON OPPOSITE SIDES OF BOX, AND SUPPORT RACEWAY WITH APPROVED-TYPE FASTENER MAXIMUM 24 INCHES FROM BOX. WHEN PENETRATING REINFORCED CONCRETE MEMBERS, AVOID CUTTING REINFORCING STEEL.
    - E. BOXES
      - a. BOXES FOR USE WITH RACEWAY SYSTEMS: MINIMUM 1 1/2 INCHES DEEP, EXCEPT WHERE SHALLOWER BOXES REQUIRED BY STRUCTURAL CONDITIONS ARE APPROVED BOXES FOR OTHER THAN LIGHTING FIXTURE OUTLETS: MINIMUM 4 INCHES SQUARE, EXCEPT THAT 4 BY 2 INCH BOXES MAY BE USED WHERE ONLY ONE RACEWAY ENTERS OUTLET. TELECOMMUNICATIONS OUTLETS: A MINIMUM OF 4 11/16 INCHES SQUARE BY 2 1/8 INCHES DEEP. MOUNT OUTLET BOXES FLUSH IN FINISHED WALLS.
16. CONDUCTOR IDENTIFICATION
    - A. PROVIDE CONDUCTOR IDENTIFICATION WITHIN EACH ENCLOSURE WHERE TAP, SPLICE, OR TERMINATION IS MADE. FOR CONDUCTORS NO. 6 AWG AND SMALLER DIAMETER, PROVIDE COLOR CODING BY FACTORY-APPLIED, COLOR-IMPREGNATED INSULATION.
17. COVERS AND DEVICE PLATES
    - A. INSTALL WITH EDGES IN CONTINUOUS CONTACT WITH FINISHED WALL SURFACES WITHOUT USE OF MATS OR SIMILAR DEVICES. PLASTER FILLINGS ARE NOT PERMITTED. INSTALL PLATES WITH ALIGNMENT TOLERANCE OF 1/16 INCH. USE OF SECTIONAL-TYPE DEVICE PLATES ARE NOT PERMITTED. PROVIDE GASKET FOR PLATES INSTALLED IN WET LOCATIONS.
18. ELECTRICAL PENETRATIONS
    - A. SEAL OPENINGS AROUND ELECTRICAL PENETRATIONS THROUGH FIRE RESISTANCE-RATED WALLS, PARTITIONS, FLOORS, OR CEILINGS IN ACCORDANCE WITH NFPA 70.
19. REPAIR OF EXISTING WORK
    - A. PERFORM REPAIR OF EXISTING WORK, DEMOLITION, AND MODIFICATION OF EXISTING ELECTRICAL DISTRIBUTION SYSTEMS AS FOLLOWS:
    - B. WORKMANSHIP
      - a. LAY OUT WORK IN ADVANCE. EXERCISE CARE WHERE CUTTING, CHANNELING, CHASING, OR DRILLING OF FLOORS, WALLS, PARTITIONS, CEILINGS, OR OTHER SURFACES IS NECESSARY FOR PROPER INSTALLATION, SUPPORT, OR ANCHORAGE OF CONDUIT, RACEWAYS, OR OTHER ELECTRICAL WORK. REPAIR DAMAGE TO BUILDINGS, PIPING, AND EQUIPMENT USING SKILLED CRAFTSMEN OF TRADES INVOLVED.
    - C. EXISTING CONCEALED WIRING TO BE REMOVED
      - a. DISCONNECT EXISTING CONCEALED WIRING TO BE REMOVED FROM ITS SOURCE. REMOVE CONDUCTORS; CUT CONDUIT FLUSH WITH FLOOR, UNDERSIDE OF FLOOR, AND THROUGH WALLS; AND SEAL OPENINGS.
    - D. REMOVAL OF EXISTING ELECTRICAL DISTRIBUTION SYSTEM
      - a. REMOVAL OF EXISTING ELECTRICAL DISTRIBUTION SYSTEM EQUIPMENT INCLUDES EQUIPMENT'S ASSOCIATED WIRING, INCLUDING CONDUCTORS, CABLES, EXPOSED CONDUIT, SURFACE METAL RACEWAYS, BOXES, AND FITTINGS, BACK TO EQUIPMENT'S POWER SOURCE AS INDICATED.
    - E. CONTINUATION OF SERVICE
      - a. MAINTAIN CONTINUITY OF EXISTING CIRCUITS OF EQUIPMENT TO REMAIN. MAINTAIN EXISTING CIRCUITS OF EQUIPMENT ENERGIZED. RESTORE CIRCUITS WIRING AND POWER WHICH ARE TO REMAIN BUT WERE DISTURBED DURING DEMOLITION BACK TO ORIGINAL CONDITION.
20. FIELD QUALITY CONTROL
    - A. FURNISH TEST EQUIPMENT AND PERSONNEL AND SUBMIT WRITTEN COPIES OF TEST RESULTS. GIVE CONTRACTING OFFICER 5 WORKING DAYS NOTICE PRIOR TO EACH TEST. WHERE APPLICABLE, TEST ELECTRICAL EQUIPMENT IN ACCORDANCE WITH NETA ATS.
21. DEVICES SUBJECT TO MANUAL OPERATION
    - A. OPERATE EACH DEVICE SUBJECT TO MANUAL OPERATION AT LEAST FIVE TIMES, DEMONSTRATING SATISFACTORY OPERATION EACH TIME.
22. 600-VOLT WIRING TEST
    - A. TEST WIRING RATED 600 VOLT AND LESS TO VERIFY THAT NO SHORT CIRCUITS OR ACCIDENTAL GROUNDS EXIST. ALL EXISTING WIRING TO BE REUSED MUST ALSO BE TESTED.
23. GROUND-FAULT RECEPTACLE TEST
    - A. TEST GROUND-FAULT RECEPTACLES WITH A "LOAD" (SUCH AS A PLUG IN LIGHT) TO VERIFY THAT THE "LINE" AND "LOAD" LEADS ARE NOT REVERSED. PRESS THE TEST BUTTON AND THEN THE RESET BUTTON TO VERIFY BY LED STATUS THAT THE DEVICE IS A SELF-TEST MODEL AS SPECIFIED IN UL 943.

FORT LEAVENWORTH DPW

MARK	DESCRIPTION	DATE

ISSUE DATE: 02/21/2025
PROJECT NO.: 300000241811

DESIGNED BY: Designer
DRAWN BY: Author
CHECKED BY: Checker
SIZE: ANSI 'D'
100% Drawings

DIRECTORATE OF PUBLIC WORKS
ENGINEERING DIVISION
DESIGN BRANCH
820 McCLELLAN AVE.
FORT LEAVENWORTH, KS
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B1146 - Textiles Warehouse
Electrical Specifications

SHEET ID
E3